All Final Term Exams Solve Current Papers BT102

27:02:2021

Solved By: **Amir Ahmadani**

دعاوں میں یاد رکھیے گا

23:28

23:26 0.13 KB/S 48%
Amir Ahmadan online
23:26
سب رابطوں میں رہا کرو یارو لاپرواہی عزیز چھین لیتی ہے عامر انظر

Current All Papers
27:02:2021

Bt102. 6 30Pm😆

1.Reverse transcription.
2.How adoptive immunity was discovered?
3. What are the target sites of antibiotics in a prokaryotic cells.
4. what is epidemiology and discuss methods of epidemiology.
5. clonal selection theory.
6. write mode of action of chloramphenicol, streptomycin and tetracycline.
7. Explain the following terms
phaeophayta
Rhodophyta
chlorophyta
Bacillariophyta
Dinoglagellata
8. Natural and aquired active and passive immunity.
Cause diagnosis and treatment of anophylaxis.

Amir Ahmadani

What is plaque?2
Drugs which resist cell wall?3
Describe forms of algae? (names were already written)

Type a message

سب رابطوں میں رہا کرو یارو لاپرواہی عزیز چھین لیتی ہے عامر انظر

Slime and capsule?2
Spores are stained by?3
Hypersensitivity and types?5
Primary and secondary immune system? 5

Amir Ahmadani

Disk diffusion
Viral replication
Anaphylaxis
Inflamation
Antigen presenting cell
Revers transcripties

Amir Ahmadani

Functio of capsule
Diff bw epitope nd antigen
Bacterial transmission
Disk staining
Primary and secondry immunity
Hypersensitivity types
Bactericidal and bacteriostatic.

Amir Ahmadani

Virus, viroid, viron 2
Cytokines 2
Bactericidal and bacteriostatic drugs 2

سب رابطوں میں رہا کرو یارو لاپرواہی عزیز چھین لیتی ہے عامر انظر

Amir Ahmadani

Virus, viroid, viron 2
Cytokines 2
Bactericidal and bacteriostatic drugs 2
Composition and distribution of microbes 3
Antigen and epitopes 3
Type 3 hypersensitivity 3
Infectious diseases by contact, vehicle, and vector 5
Hypersensitivity and types 5
Primary nd secondary immune response 5

Amir Ahmadani

BT 102. at 9:30 am
1.incidence and prevalence
2.function of capsule and name the bacteria which form capsule
3.antifungal drugs
4.inflammation and its consequences 5.hypersensitivity and its types.
6.Algae and its structure.
7.antigen presenting cells.
8.how microbs become resistant against antimicrobial.

AMIR AHMADANI

23:26 0.00 KB/S 48%

سب رابطوں میں رہا کرو یارو لاپرواہی عزیز چھین لیتی ہے عامر انظر

AMIR AHMADANI

Why heptin is a weak antigen 3
Hypersensitivity and its type 5
Disk method 3
Acid fast staining 3
How disease can be transmitted all types 5
Primary and secondary immune response 5
Mortality pathogenicity and morbidity 3
Mcqs from ppts.

Amir Ahmadani

Bohat bura MCqs tha
New MCQs the
1.Incidence prevelnace
2.Spore staining
3.Microorganism affecting body
4.protein inhibiting antibiotics names
5.hypersensitivity III mechanism and diagnosis
6.Hypersensitivity define and four types
7. Pendamic, Endemic, Epidemic
8. Acquired immunity's nd natural for active and pasive.

Amir Ahmadani

Type a message

23:26 0.00 KB/S 48%

سب رابطوں میں رہا کرو یارو لاپرواہی عزیز چھین لیتی ہے عامر انظر

Amir Ahmadani

Why heptin is a weak antigen 3
Hypersensitivity and its type 5
Disk method 3
Acid fast staining 3
How disease can be transmitted all types 5
Primary and secondary immune response 5
Mortality pathogenicity and morbidity 3
Mcqs from ppts.

Amir Ahmadani

Bohat bura MCqs tha
New MCQs the
1.Incidence prevelnace
2.Spore staining
3.Microorganism affecting body
4.protein inhibiting antibiotics names
5.hypersensitivity III mechanism and diagnosis
6.Hypersensitivity define and four types
7. Pendamic, Endemic, Epidemic
8. Acquired immunity's nd natural for active and pasive.

Amir Ahmadani 21:01



Type a message

22:40 8.00 KB/S 55%

reverse 1 of 3

VIROIDS: They are composed only of circular single stranded (ss) RNA.

34. Write entry point of microorganism 2 marks

Mucus membranes: These include linings of digestive tract, respiratory tract, urogenital system, and conjunctiva

Skin: Surface area wise, skin is the largest organ of the body. Unbroken skin is impermeable to organisms

Parenteral Route: Injury to the skin or mucous membrane leads to easy access for the organisms in the body. Any cut received on the skin for example becomes a portal of entry

35. Role of thin clonal selection 2 marks

According to this theory, B cells are generated in the bone marrow randomly without seeing the antigen. When such a cell encounters its specific antigen in the body, it starts proliferating into a clone which ultimately becomes plasma cells and start secreting antibodies. A beautiful example to understand clonal selection theory is like going to a shoe store and select a shoe size that fits you very well. Note, that the shoe that you selected was made randomly without any regard to your foot size.

35.Antigen presenting cell 3 marks

These cells are able to present antigens to B cells for making antibodies by B cells.

There are three cells that fall into this category: Dendritic cells, macrophages, and B cells

36.How antibodies provide defense against microbes 5 marks

37.Reverse transcription 2 marks

A reverse transcriptase is an enzyme used to generate complementary DNA from an RNA template, a process termed reverse transcription

38.Bactericidal and bacteriostatic drug 2 marks

Bactericidal: Those antibiotics that kill and lyse the cells are called bactericidal.

Bacteriostatic: Such antibiotics stop the growth of the organisms, they do not kill or lyse the cells. Cells of the immun system then clear them from the body

39. Define plasmogamy karyogamy and meiosis 3 marks

1.Plasmogamy: A haploid nucleus of a donor cell (+) penetrates the cytoplasm of a recipient cell (-). Both cells are haploid (one set of chromosomes)

2. **Karyogamy** The (+) and (-) nuclei fuse to form a diploid zygote nucleus.

3. **Meiosis**: The diploid nucleus gives rise to haploid nuclei (sexual spores).

40. Mechanism of hypersensitivity iv 5 marks

It is called as delayed type because its bad effect becomes visible 2-3 days after exposure to the antigen. This is the onl hypersensitivity that involves cells rather than antibodies. Robert Koch discovered this when he injected tuberculin in patients. Contact dermatitis is the most common example of this kind of hypersensitivity. This is basically an interactio between an antigen, T cells and macrophages. Transplant rejection is another good example of delayed type IV hypersensitivity. Reaction to hair dyes is another example. Poison Ivy is another example in which contact with this pla initiate type IV hypersensitivity response. Chemicals from poison ivy interact with skin proteins and change them into foreign-looking protein molecules. Body's immune system mounts an immune response against these proteins resultin in a damaging type of intense inflammatory response

41. Cytokines 2 marks

Cytokines are released as a result of defense cell stimulation by PAMPs and TLRs interactions. Cytokines recruits more immune cells and also stimulate adaptive immune response

42. Note on infectious disease transmission with vector vehicle and contacts?

Direct Transmission: person to person physical contact is a direct contact. Influenza spread is a typical example.

following: o Bone marrow o Lymph nodes o Spleen o Tonsils o Thymus o Mucosa associated o Bursa of Fabricius in birds

5. Disk diffusion method

A standardized inoculum of the organisms is spread on agar plate. Filter paper disks soaked with anti bacterials are then placed on the lawn of the organism. Plates are incubated at 37oC and results are recorded. Drugs diffuse from the paper disks radially and if the organisms are sensitive to the drug, a clear area with no growth around the disks develops. The size of the zone directly relates to the degree of sensitivity of the drug. This zone is compared to a standard table for recording the results of sensitivity test. Results are reported as sensitive or resistant or intermediate (between the two extremes).

6. How microbes develop resistance in antimicrobial?

7. Define immunity and difference between innate immunity and adaptive immunity. 10 marks

What is immunity?

Immunity: Ability of the body to protect itself from harmful effects of diseases.

Susceptibility: lack of immunity which leads to easy establishment of disease.

Types of immunity

Innate (Natural):

Present at birth: One is born with these defenses, so it is present without any exposure to the microbes. That is the reason, it is non-specific.

Rapid to respond: Because it is already there, it interacts with the microbes as soon as they enter the body.

Nonspecific: It means that it does not differentiate if it is an E. coli or a Staph or Salmonella. Body defense system against these entire pathogens act in a similar fashion. In other words, the same macrophage can phagocytize E. coli, Staph or Salmonella at the same time (if these pathogens are present there).

Does not increase in response with repeated exposures to the same agent. Therefore, it has no memory.

First line of defense: Physical barriers (skin, mucus membrane) provide the first line of defense as they prevent the entry of the organisms by providing a physical barrier which if broken will let the microbes go inside.

Second line of defense: Phagocytic cells and their secretions provide a second line of barriers for microbes. So, if microbes succeed in breaching or breaking the first barrier, macrophages, neutrophils and such other cells phagocytize these microbes and kill them before these can establish infection in the body.

Adaptive Immunity: This will be studied in detail in a separate lecture soon.

Adaptive (Acquired): It is the third line of defense. Acquired by experience, specific to an organism.

Specific: It means that antibodies made against E. coli will destroy E. coli only, and will not do any harm to a Staph for example.

Have memory: It means if antibodies have once been made against E. coli, this exposure will be remembered by the body, and if a second encounter is made with the same type of E. coli again, the body would mount a heightened immune response against E. coli because it has memory cells developed in the body which upon second stimulation proliferate quickly and mount a greater defensive response.

O Because of memory, body's immune response increase with every repeated exposure. It has a limit also!

Slower to respond: For the first time exposure, the body's response is slower and takes a long time compared with the innate immunity which is immediate.

Adaptive Immune Response can further be divided into:

Humoral immune response (antibody response by B cells or lymphocytes)

Cellular immune response (T cells are made which kills microbial harboring cells).

8. Agglutination and Neutralization 2 marks

Agglutination: Antibodies can agglutinate antigens which are then cleared as a group by macrophages

Neutralization: Toxins bind with antibodies and they become incapable of binding to their target receptors thus the toxins become harmless.

9. What is plaque 2 marks

Plaques are equivalent to bacterial colonies. Basically a plaque is an area where the virus infection ha
These plaques are formed on cultured cells in a culture dish. A monolayer of the cells is infected with
enumerated and then the culture is overlaid with the semisolid agar so that infected and lysed cells do
from the site in the dish. Plaques can be counted easily much like bacterial colonies

10. Difference between capsule and slime

22:43 6.00 KB/S 55%
google.com/search?q=
Google
target sites of antibiotics
All Images News Videos Books

In principal, there are three main antibiotic targets in bacteria:

- The cell wall or membranes that surrounds the bacterial cell.
- The machineries that make the nucleic acids DNA and RNA.
- The machinery that produce proteins (the ribosome and associated proteins)

https://www.reactgroup.org › toolbox

How do antibiotics work? – Antibiotics – ReAct

About featured snippets

Is this helpful? Yes · No

People also ask

What are the 4 targets of antimicrobial agents?

Why is the cell wall a good target for antibiotics?

22:43 9.00 KB/S 55%
google.com/search?q=
Google
epidemiology
All Books Images News Videos Sea

See results about

Epidemiology
Peer-reviewed journal

Dictionary

Search for a word

epidemiology

/ˌɛpɪdiːmɪˈɒlədʒi/

noun

the branch of medicine which deals with the incidence, distribution, and possible control of diseases and other factors relating to health.

Definitions from Oxford Languages *Feedback*

Translations and more definitions

People also ask

What does a Epidemiologist do?

22:44 2.00 KB/S 55%

sectional and ecolog- ical study types.

https://www.neurologic.theclinics.com › ... PDF

Research Methods in Epidemiology - Neurologic Clinics

About featured snippets Feedback

People also ask

What are the four methods of epidemiology?

Epidemiological studies generally fall into four broad categories:

- cross-sectional studies.
- case-control studies.
- cohort studies.
- intervention studies.

https://www.open.edu › health › con...

Epidemiology: An introduction: 3.2 Types of epidemiological studies ...

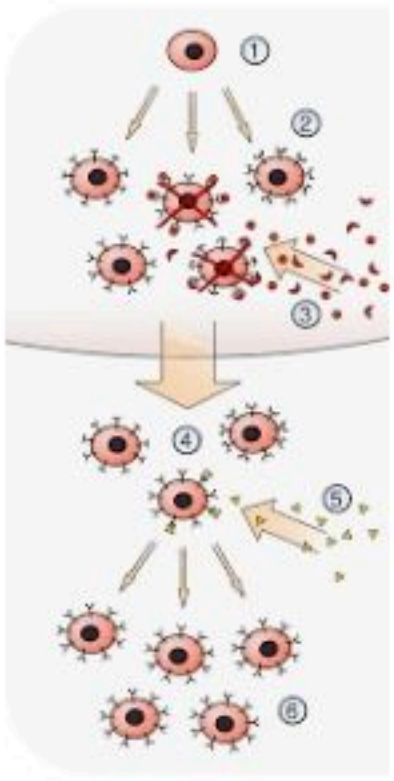
More results

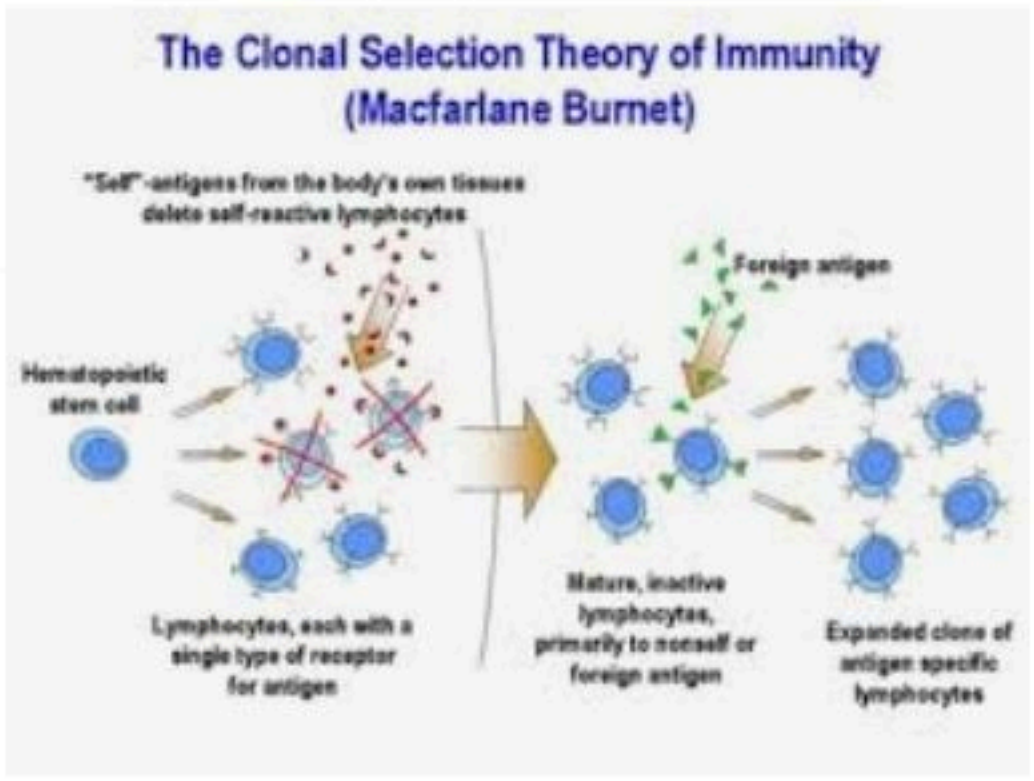
What are the 3 types of epidemiological studies?

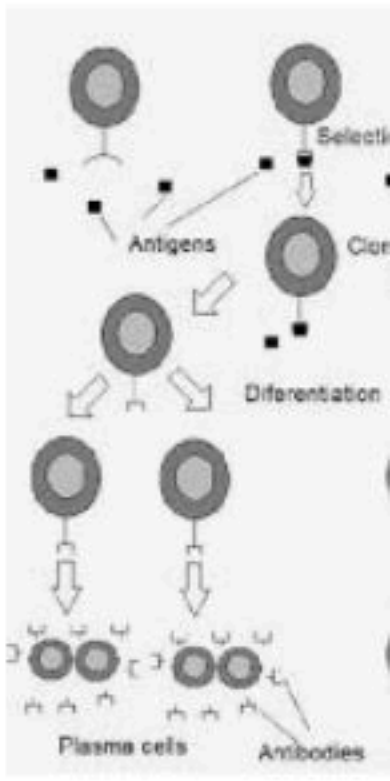
What are the 5 W's of epidemiology?

22:44 127 KB/S 55%
google.com/search?cli
5
Google
clonal selection theory
All Images Videos News Books Sea

Clonal selection







Clonal selection theory is a scientific **theory** in immunology that explains the functions of cells of the immune system (lymphocytes) in response to specific antigens invading the body. ... In short, the **theory** is an explanation of the mechanism for the generation of diversity of antibody specificity.

https://en.m.wikipedia.org › wiki

Clonal selection - Wikipedia

Videos

YouTube · WEHImovies

YouTube · The A

22:45 40.0 KB/S 55%
google.com/search?q=
Google
mode of action of antibiotics
All Images Videos News Books

Antibacterial **action** generally falls within one of four **mechanisms**, three of which involve the inhibition or regulation of enzymes involved in cell wall biosynthesis, nucleic acid metabolism and repair, or protein synthesis, respectively. The fourth **mechanism** involves the disruption of membrane structure.

https://www.sigmaaldrich.com › bio...

Antibiotics by Mechanism of Action - Antibiotics | Sigma-Aldrich

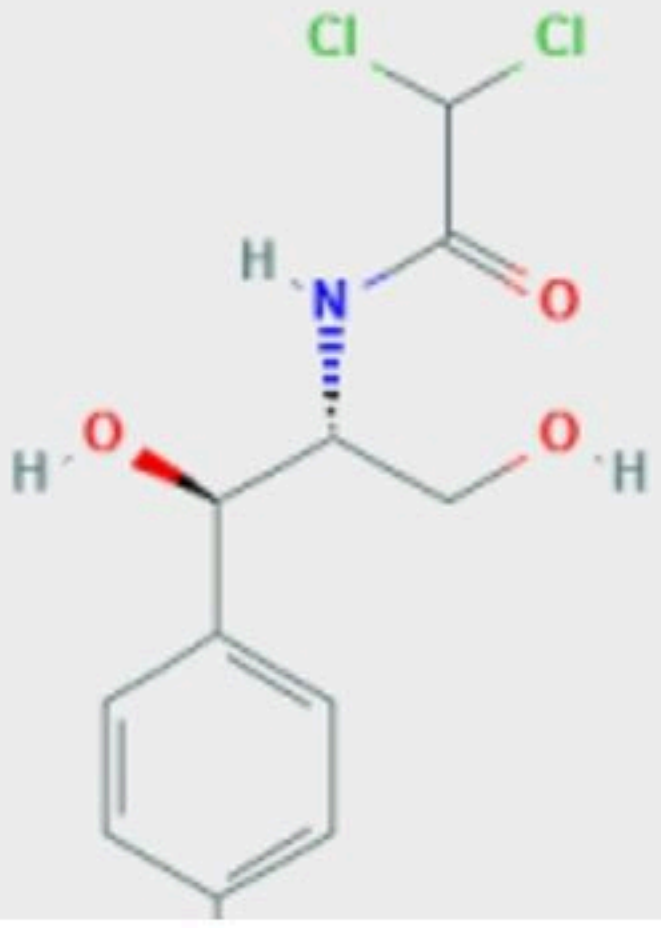
About featured snippets Feedback

People also ask

What are the 5 mechanisms of action of antibiotics?

What are the 4 classes of antibiotics based on mode of action?

22:45 6.00 KB/S 54%
google.com/search?cli
Google
mode of action of chloramphenic
All Images News Videos Books Sea



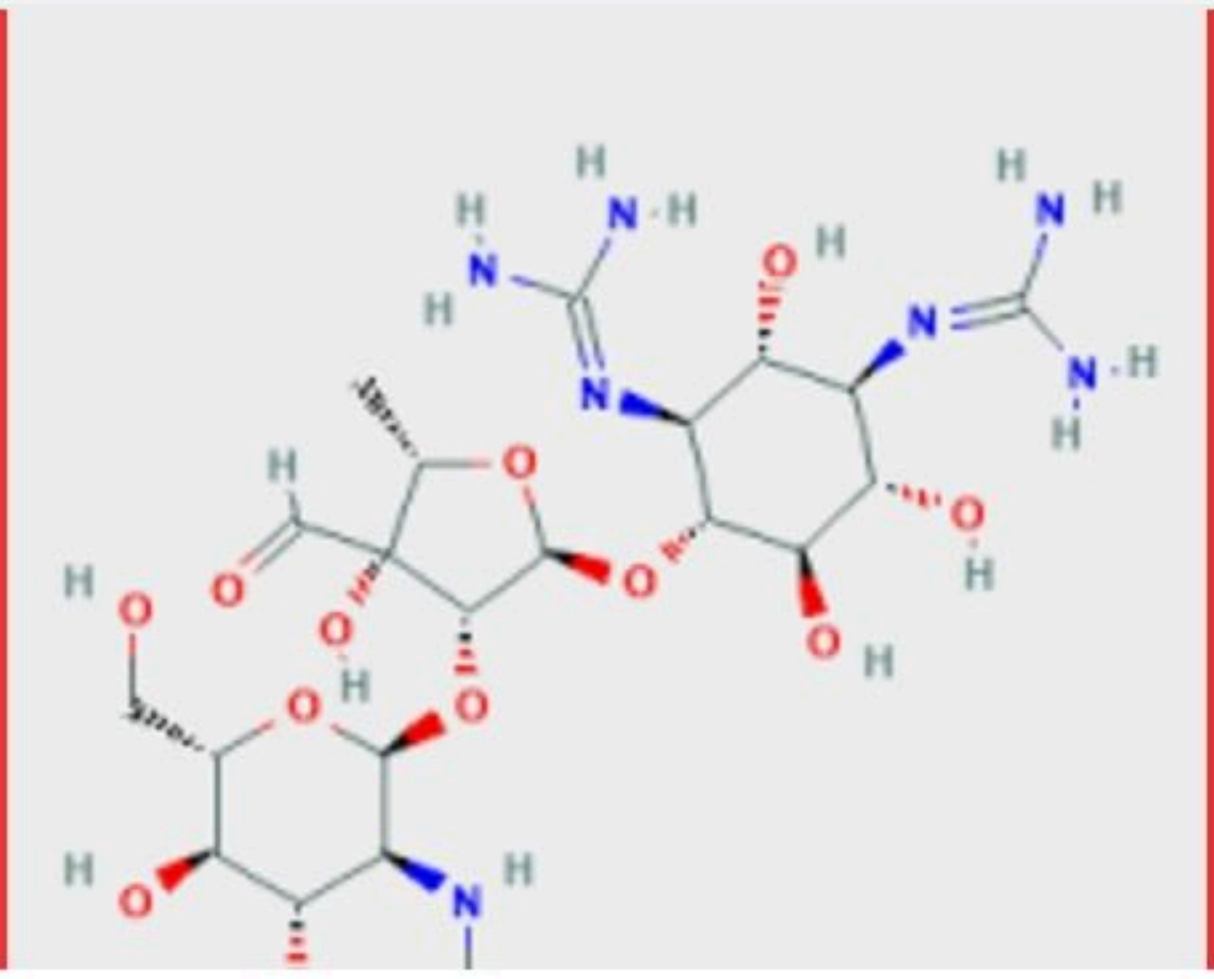
Chloramphenicol diffuses through the bacterial cell wall and reversibly binds to the bacterial 50S ribosomal subunit. The binding interferes with peptidyl transferase activity, thereby prevents transfer of amino acids to the growing peptide chains and blocks peptide bond formation.

https://pubchem.ncbi.nlm.nih.gov › ...

Chloramphenicol | C11H12Cl2N2O5 - PubChem

About featured snippets Feedback

22:46 3.00 KB/S 54%
google.com/search?cli
Google
mode of action of streptomycin
All Images News Videos Books



Streptomycin irreversibly binds to the 16S rRNA and S12 protein within the bacterial 30S ribosomal subunit. As a result, this agent interferes with the assembly of initiation complex between mRNA and the bacterial ribosome, thereby inhibiting the initiation of protein synthesis.

https://pubchem.ncbi.nlm.nih.gov › ...

Streptomycin | C21H39N7O12 - PubChem

About featured snippets Feedback

22:46 18.0 KB/S Vo LTE2 4G 54%
google.com/search?cli 5
Google
mode of action of tetracycline
All Images Videos News Books Sea

Mode of action: Inhibition of protein synthesis. Once **tetracyclines** have been transported into the cell, this class of antibiotic reversibly binds to receptors on the 30S ribosomal subunit of the bacteria, preventing attachment of aminoacyl-tRNA to the RNA-ribosome complex.

Mode of action: Inhibition of protein synthesis. Once tetracyclines have been transpo...

Spectrum of activity: Broad spectrum. Exhibits activity against a wide range of Gram-...

Example: Chlortetracycline, oxytetracycline, demethylchlortetracycline, rolitetracycline, ...

http://amrls.cvm.msu.edu › tetracycl...

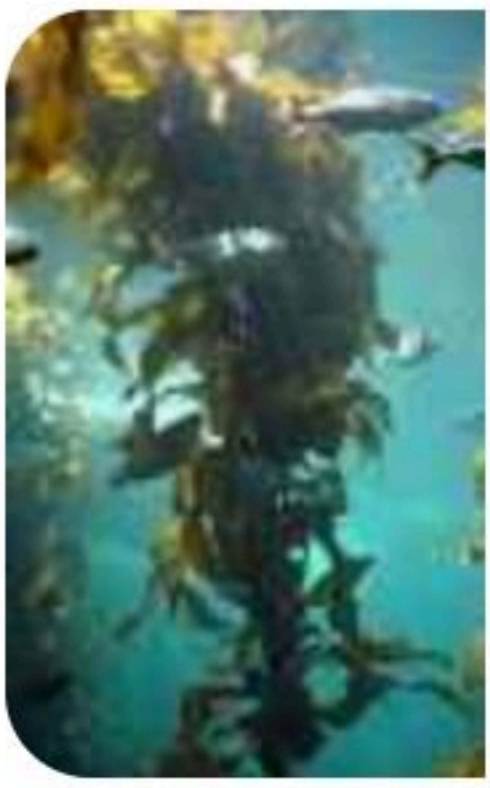
Tetracyclines — Antimicrobial Resistance Learning Site For ...

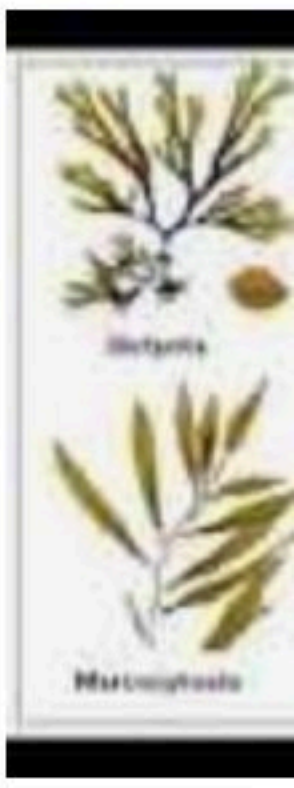
About featured snippets Feedback

22:47 17.0 KB/S 54%
google.com/search?cli
5
phaeophyta
All Images Books Videos News Sea

Brown algae

Overview Lower classifications Videos





The brown algae, comprising the class Phaeophyceae, are a large group of multicellular algae, including many seaweeds located in colder waters within the Northern Hemisphere. Most brown algae live in marine environments, where they play an important role both as food and as a potential habitat. Wikipedia

Scientific name: Phaeophyceae

Class: Phaeophyceae; Kjellman, 1891

Phylum: Ochrophyta

Higher classification: Fucistia

Clade: SAR

Rank: Class

22:47 7.00 KB/S Vo LTE2 4G 54%
google.com/search?cli 5
rhodophyta
All Images Books Videos News Sea

Red algae

Overview | Lower classifications | Videos





Red algae, or Rhodophyta, are one of the oldest groups of eukaryotic algae. The Rhodophyta also comprises one of the largest phyla of algae, containing over 7,000 currently recognized species with taxonomic revisions ongoing. Wikipedia

Scientific name: Rhodophyta

Higher classification: Archaeplastida

Rank: Phylum

Division: Rhodophyta; Wettstein, 1922 Classification is currently disputed. See Taxonomy

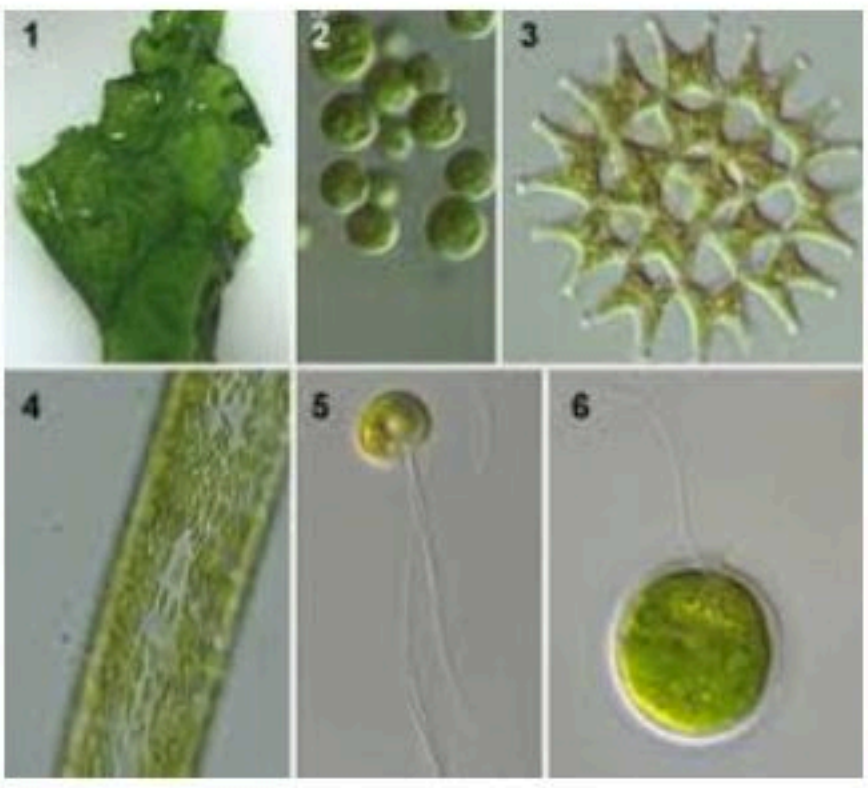
Domain: Eukaryota

Types

22:47 159 KB/S 54%
google.com/search?cli
5
All Images Books Videos News Sea

Green algae

Overview | Lower classifications | Videos



Chlorophyta or Prasinophyta is a taxon of green algae informally called chlorophytes. The name is used in two very different senses, so care is needed to determine the use by a particular author.
Wikipedia

Scientific name: Chlorophyta

Higher classification: Viridiplantae

Phylum: Chlorophyta; Reichenbach, 1828, emend. Pascher, 1914, emend. Lewis & McCourt, 2004

Clade: Chloroplastida

Domain: Eukaryota

Rank: Phylum

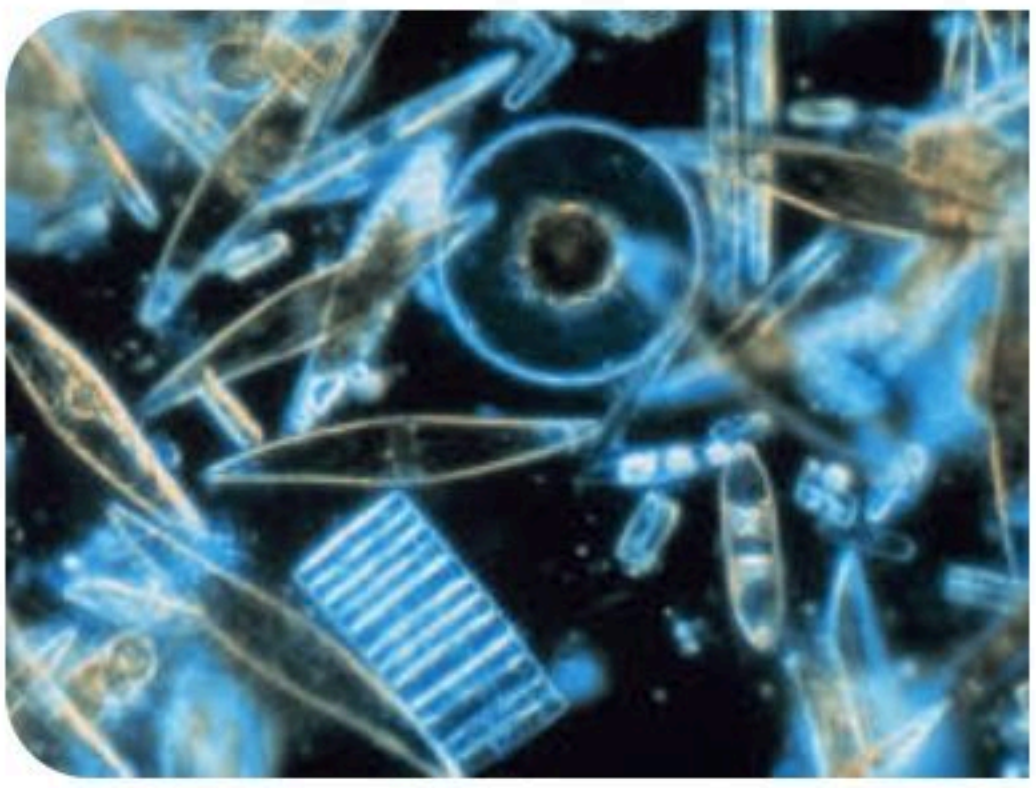
Feedback

22:48 98.0 KB/S 54%
google.com/search?cli
5

Diatom

Algae

Overview | Videos | People also search for





Diatoms are a major group of algae, specifically microalgae, found in the oceans, waterways and soils of the world. Wikipedia

Scientific name: Bacillariophyceae

Rank: Class

Higher classification: Heterokont

Phylum: Ochrophyta

Class: Bacillariophyceae; Dangeard, 1933

Clade: SAR

Ecological role

Life cycle

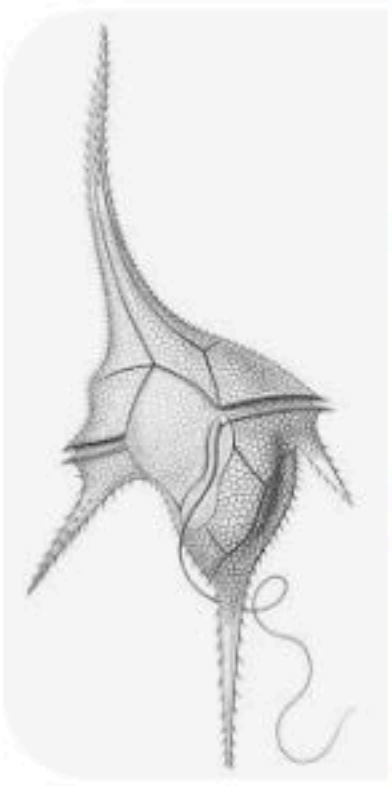
Size

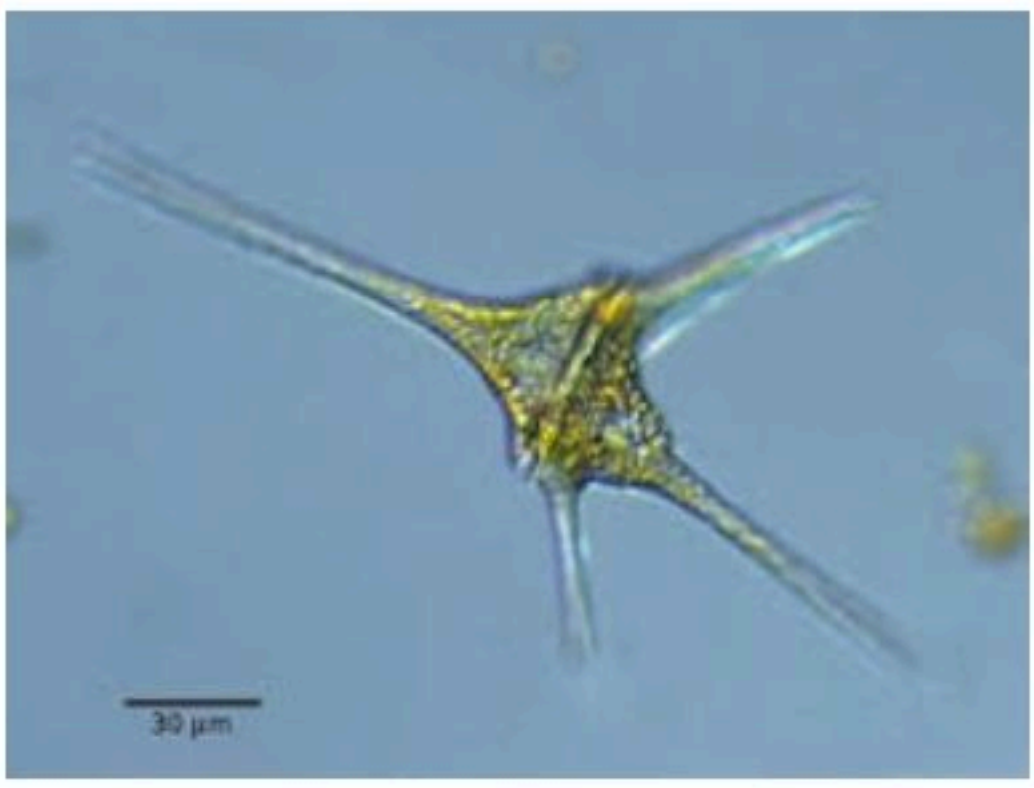
22:49 85.0 KB/S Vo LTE2 4G 54%

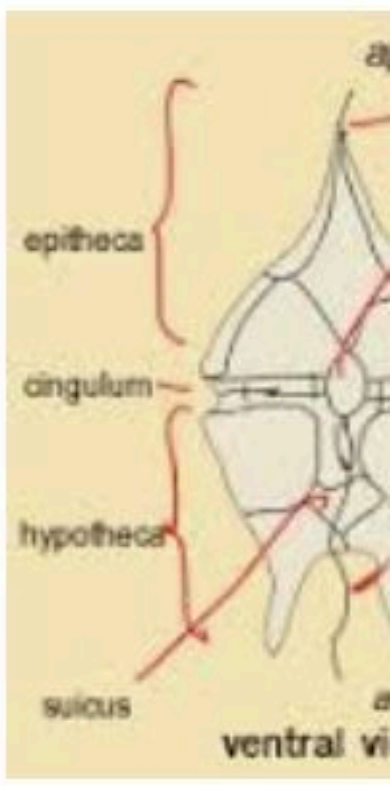
Dinoflagellates

Algae

Overview | Lower classifications | Videos







The dinoflagellates are single-celled eukaryotes constituting the phylum Dinoflagellata. Usually considered algae, dinoflagellates are mostly marine plankton, but they also are common in freshwater habitats. Their populations are distributed depending on sea surface temperature, salinity, or depth.

Wikipedia

Scientific name: Dinoflagellata

Higher classification: Alveolate

Rank: Phylum

Phylum: Myzozoa

Clade: SAR

Subphylum: Dinozoa

Characteristics

22:50 24.0 KB/S 54%

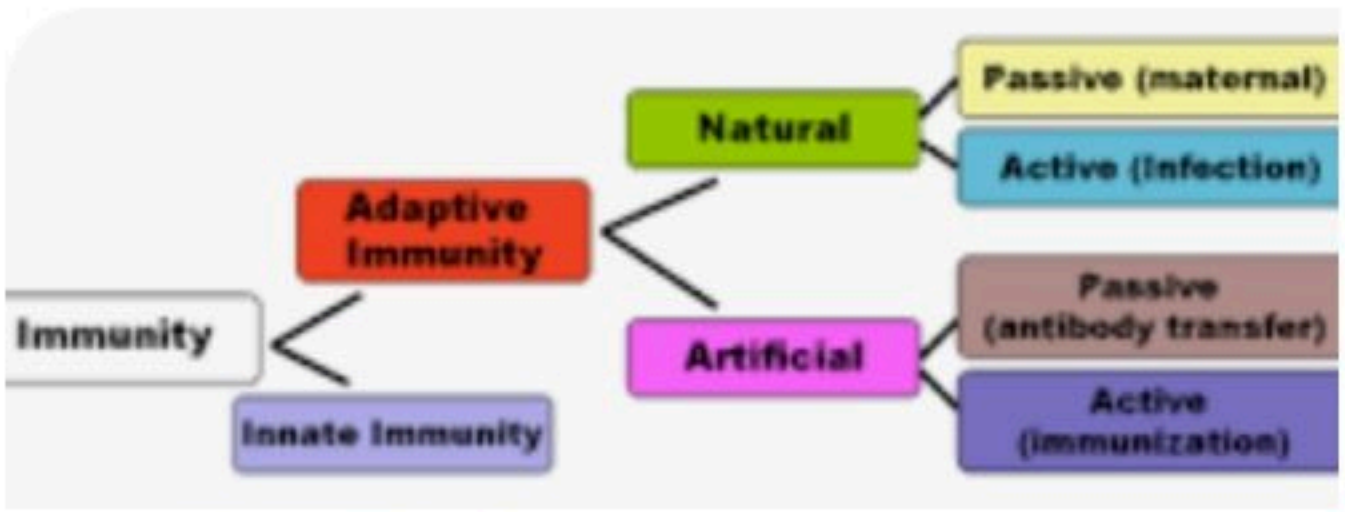
google.com/search?cli

Google

define natural immunity

All Images News Videos Books Sea

Innate immune system definition











Natural immunity: **Immunity** that is **naturally** existing, **Natural immunity** does not require prior sensitization to an antigen. See: **Innate immunity**.

https://www.rxlist.com › definition

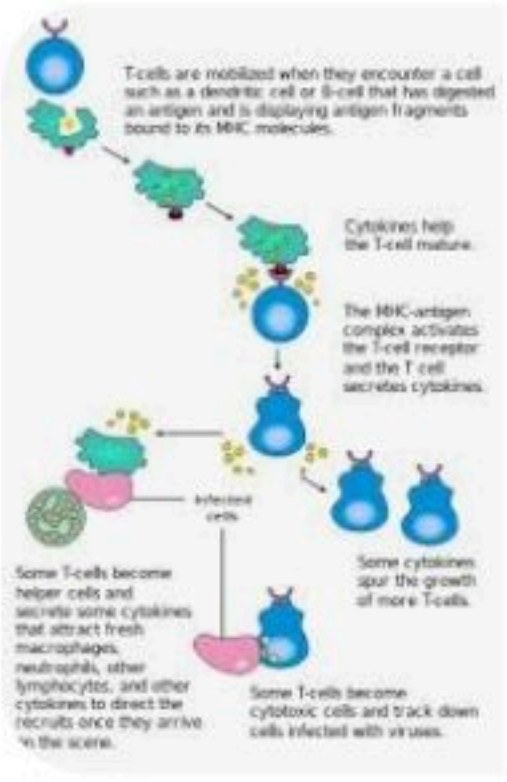
Definition of Natural immunity - RxList

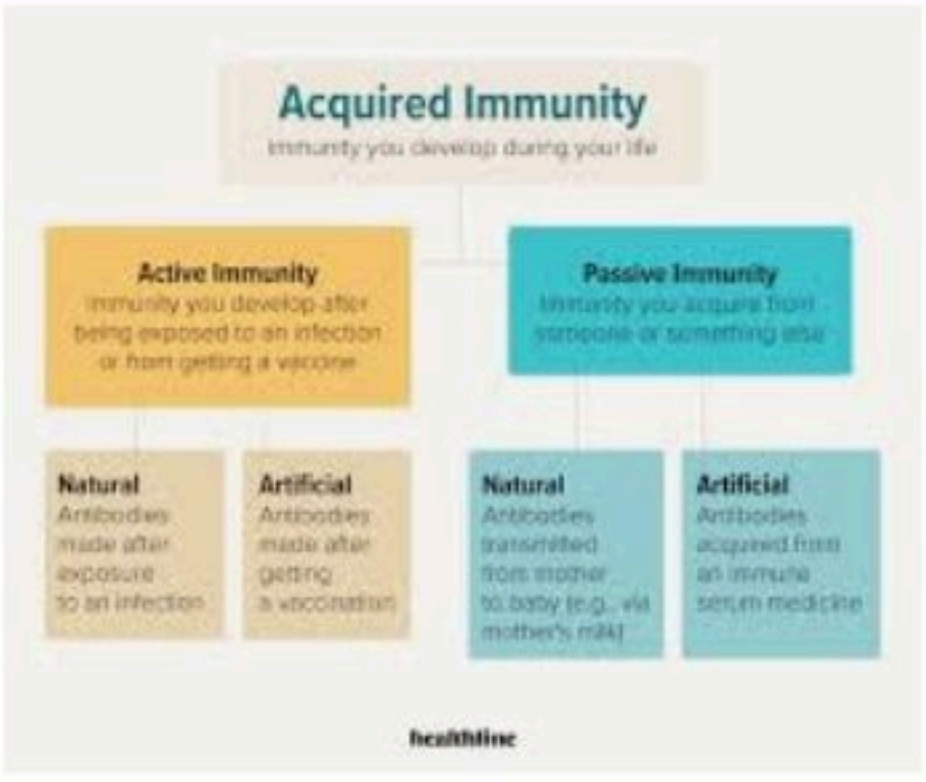
About featured snippets Feedback

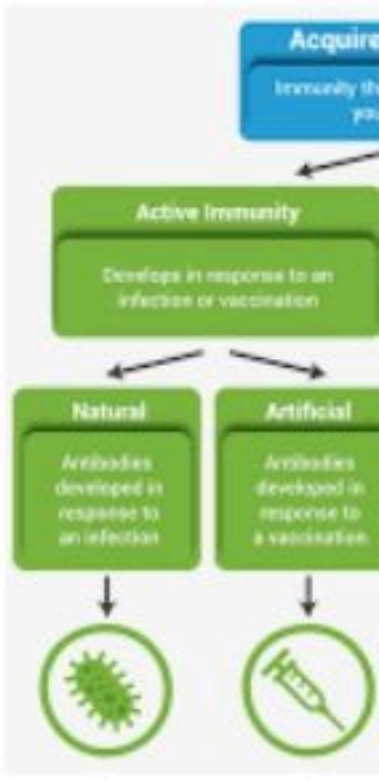
People also ask

22:50 27.0 KB/S 54%
google.com/search?cli
5
Google
define acquired immunity
All Images News Videos Books Sea

Adaptive immune system







The adaptive immune system, also referred as the acquired immune system, is a subsystem of the immune system that is composed of specialized, systemic cells and processes that eliminates pathogens by preventing their growth. The acquired immune system is one of the two main immunity strategies found in vertebrates. Wikipedia

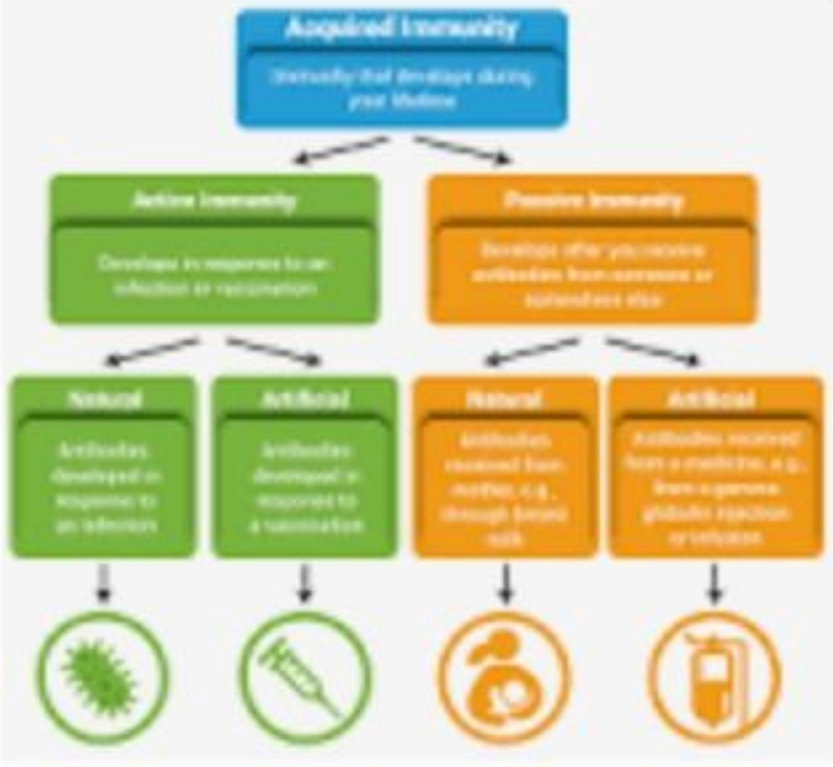
Advantages

Stages

Basics

22:50 4.00 KB/S 54%
google.com/search?cli
5
Google
define acquired active immunity
All News Images Videos Books





Naturally **acquired active immunity** occurs when a person is exposed to a live pathogen and develops a primary **immune** response, which leads to immunological memory. This type of **immunity** is "natural" because deliberate exposure does not induce it.

https://en.m.wikipedia.org › wiki › I...

Immunity (medical) - Wikipedia

About featured snippets Feedback

People also ask

22:51 0.00 KB/S 54%

define passive immunity

All Images Videos News Books Sea

Dictionary

Search for a word

passive immunity

noun PHYSIOLOGY

the short-term immunity which results from the introduction of antibodies from another person or animal.

Definitions from Oxford Languages *Feedback*

Translations and more definitions

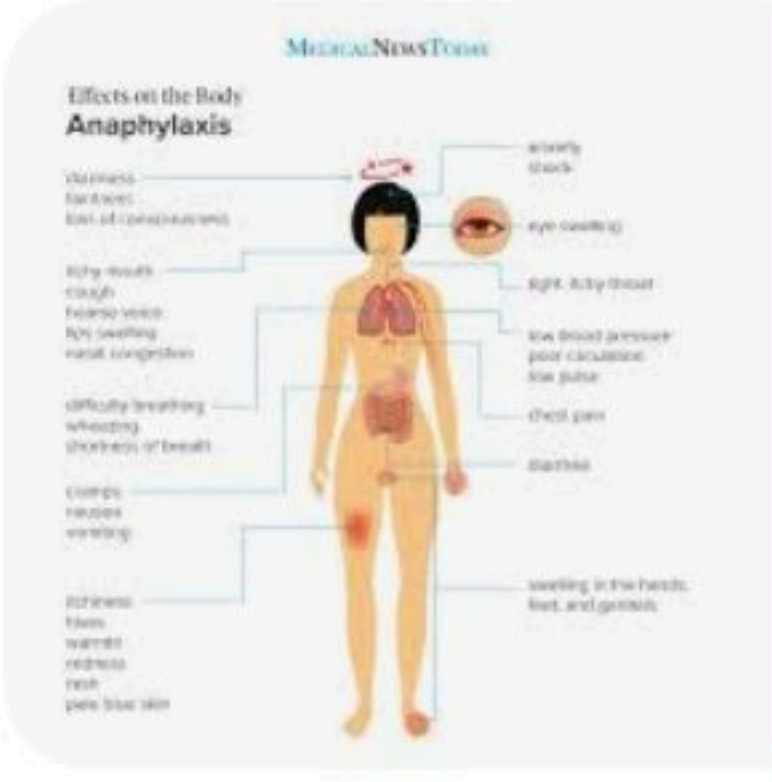
Passive immunity

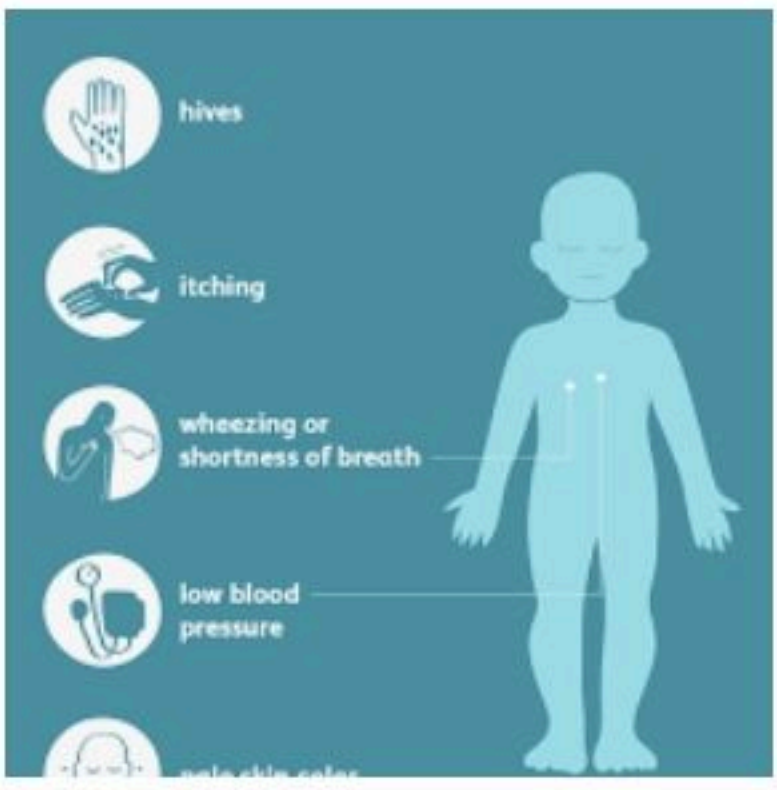
Passive immunity is the transfer of active humoral immunity of ready-made antibodies. Passive immunity can occur naturally, when maternal antibodies are transferred to the fetus through the placenta, ... Wikipedia

People also search for: Vaccine, Therapy, Immunosuppression, Enzyme inhibitor, **more**

People also ask

22:51 19.0 KB/S 54%
google.com/search?cli
Google
anaphylaxis
All Images News Books Videos Sea





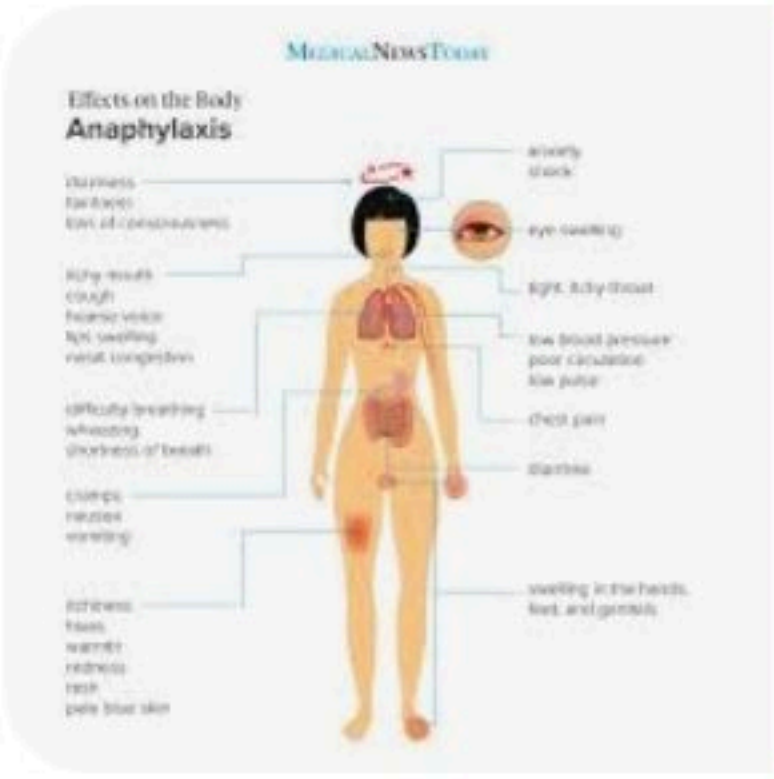
Anaphylaxis is a severe allergic reaction to venom, food, or medication. Most cases are caused by a bee sting or eating foods that are known to cause allergies, such as peanuts or tree nuts. **Anaphylaxis** causes a series of symptoms, including a rash, low pulse, and shock, which is known as **anaphylactic** shock.

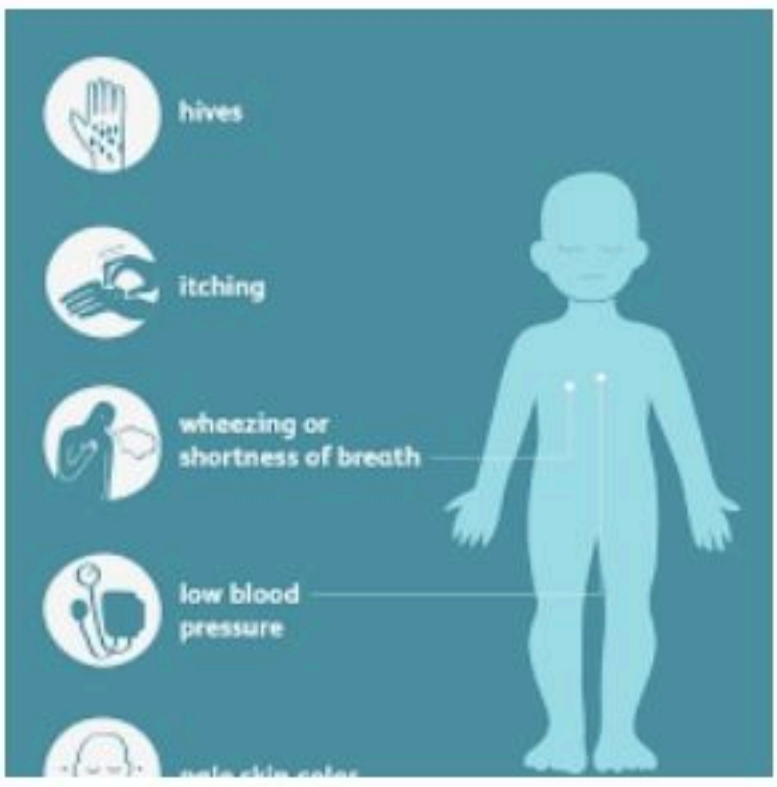
https://www.healthline.com › health

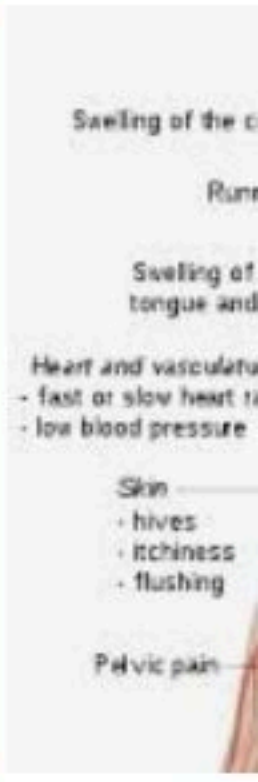
Anaphylaxis: Causes, Symptoms & Diagnosis - Healthline

About featured snippets Feedback

22:51 29.0 KB/S 54%
google.com/search?cli
anaphylaxis symptoms
All Images Videos Books News







Symptoms of anaphylaxis

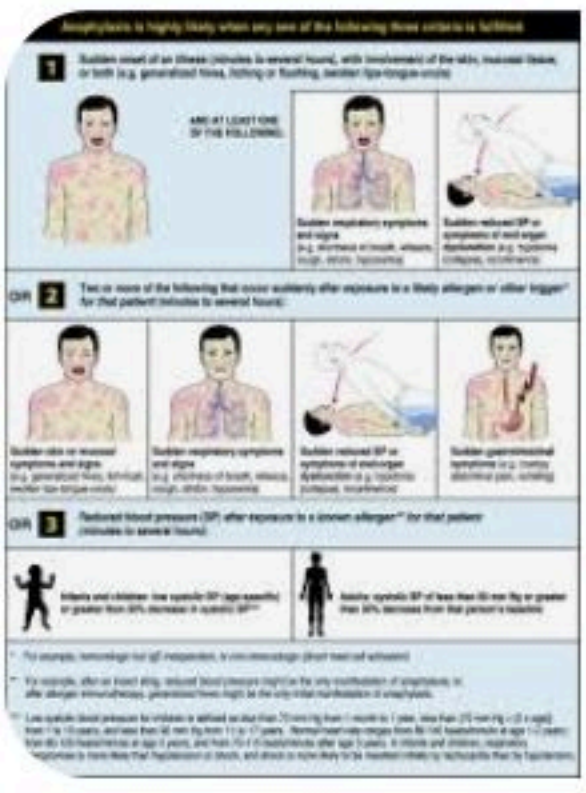
- feeling lightheaded or faint.
- **breathing** difficulties – such as fast, **shallow breathing**.
- **wheezing**.
- a fast heartbeat.
- clammy **skin**.
- **confusion** and anxiety.
- collapsing or losing consciousness.

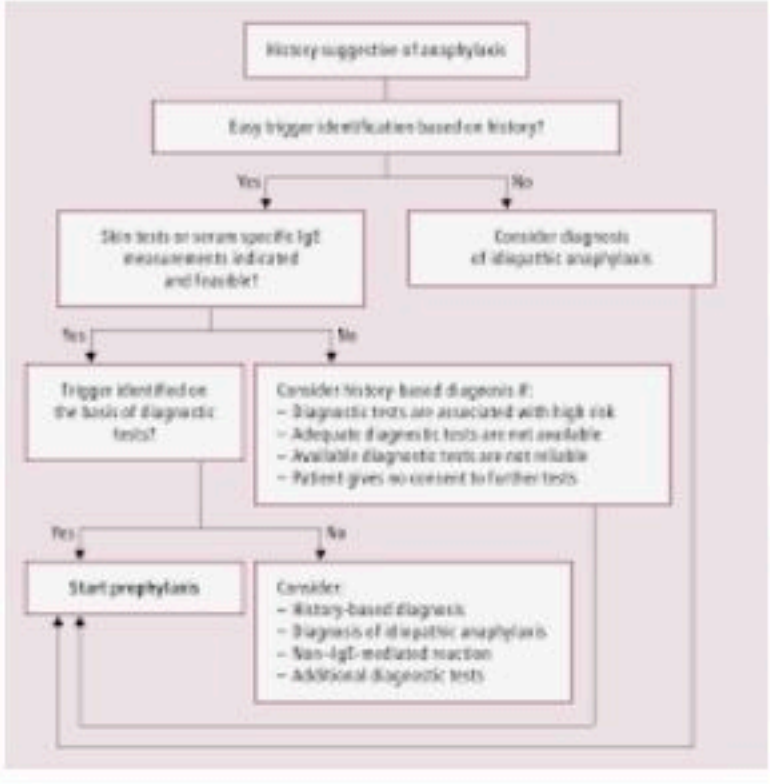
https://www.nhs.uk › conditions › a...

Anaphylaxis - NHS

About featured snippets Feedback

22:52 8.00 KB/S 53%
google.com/search?cli
Google
anaphylaxis diagnosis
All Images News Videos Books Sea







To help confirm the **diagnosis**: You might be given a blood test to measure the amount of a certain enzyme (tryptase) that can be elevated up to three hours after **anaphylaxis**. You might be tested for allergies with skin tests or blood tests to help determine your trigger.

14-Sep-2019

https://www.mayoclinic.org › drc-2...

Anaphylaxis - Diagnosis and treatment - Mayo Clinic

About featured snippets Feedback

22:52 20.0 KB/S 53%
google.com/search?cli
Google
anaphylaxis treatment
All Images Videos Books News

Treatment

1. **Epinephrine** (adrenaline) to reduce your body's allergic response.
2. **Oxygen**, to help you breathe.
3. **Intravenous** (IV) **antihistamines** and cortisone to reduce **inflammation** of your air passages and improve breathing.
4. A **beta**-agonist (such as albuterol) to relieve breathing **symptoms**.

14-Sep-2019

https://www.mayoclinic.org › drc-2...

Anaphylaxis - Diagnosis and treatment - Mayo Clinic

About featured snippets Feedback

People also ask

What is the protocol for the treatment of anaphylaxis?

22:53 65.0 KB/S 53%
google.com/search?cli
5
Google
define plague
All Images News Books Videos Sea

Dictionary

Search for a word

plague

/pleɪg/

noun

1. a contagious bacterial disease characterized by fever and delirium, typically with the formation of buboes (*bubonic plague*) and sometimes infection of the lungs (*pneumonic plague*).
2. an unusually large number of insects or animals infesting a place and causing damage.
   "a **plague of** locusts"

Similar: huge number infestation

verb

cause continual trouble or distress to.
"he has been plagued by ill health"

Similar: afflict bedevil

22:53 32.0 KB/S 53%
google.com/search?cli
Google
drugs which resist cell wall
All Images News Videos Books

Significant glycopeptide **antibiotics** include vancomycin, teicoplanin, telavancin, bleomycin, ramoplanin, and decaplanin. This class of drugs inhibit the synthesis of cell walls in susceptible microbes by inhibiting peptidoglycan synthesis. peptidoglycan: A polymer of glycan and peptides found in bacterial cell walls.

https://courses.lumenlearning.com › ...

Functions of Antimicrobial Drugs | Boundless Microbiology

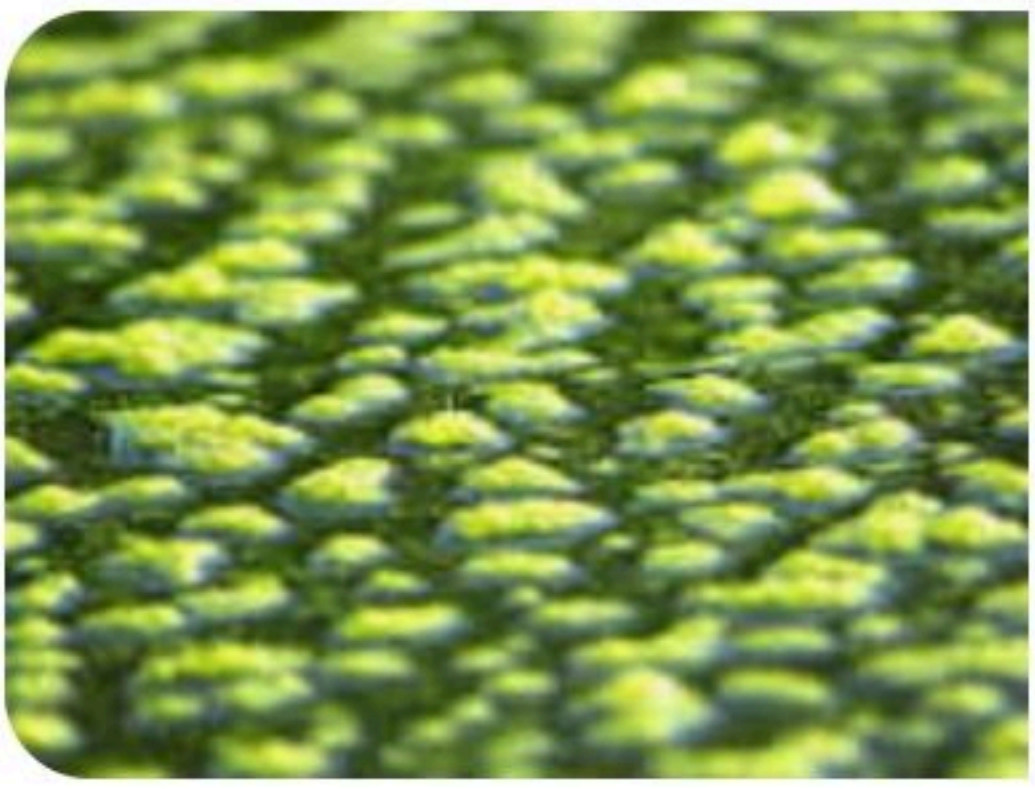
About featured snippets Feedback

People also ask

Which drugs inhibit cell wall synthesis?

What drugs target the cell wall?

22:53 74.0 KB/S 53%
google.com/search?cli
forms of algae
All Images News Videos Books





The different divisions include:

- Euglenophyta (Euglenoids)
- Chrysophyta (Golden-brown **algae** and Diatoms)
- Pyrrophyta (Fire **algae**)
- Chlorophyta (Green **algae**)
- Rhodophyta (Red **algae**)
- Paeophyta (Brown **algae**)
- Xanthophyta (Yellow-green **algae**)

12-Sep-2018

https://www.thoughtco.com › major...

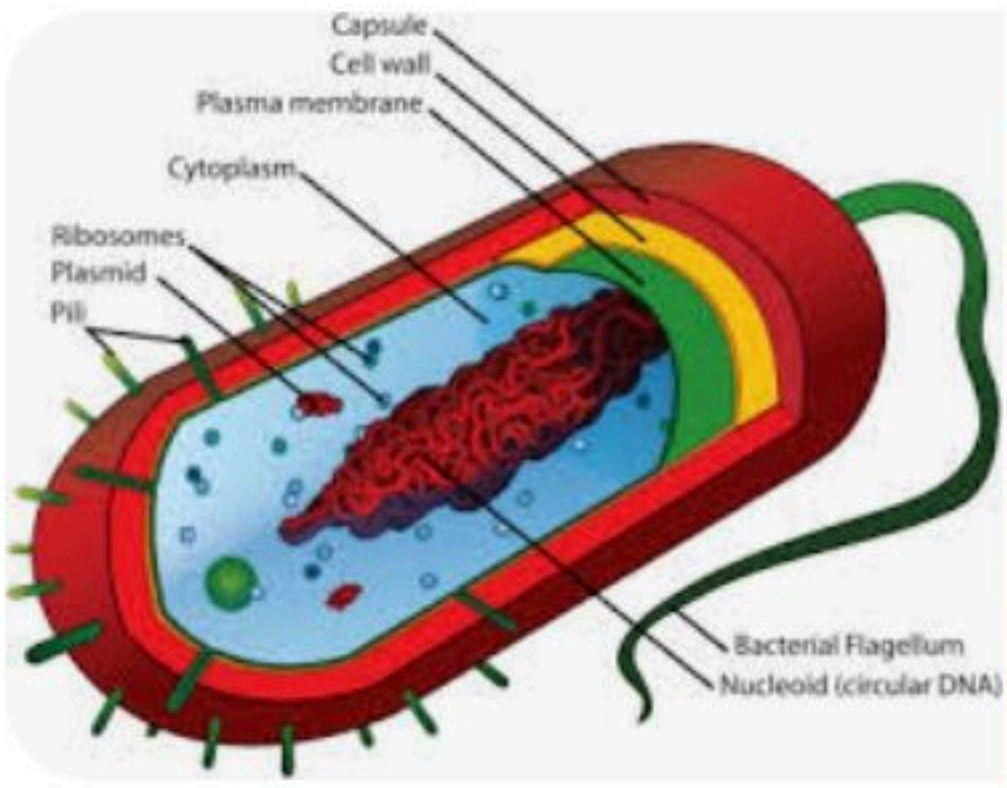
7 Major Types of Algae - ThoughtCo

About featured snippets Feedback

People also ask

22:54 39.0 KB/S 53%
google.com/search?cli
Google
slime and capsule
All Images Videos News Books Sea

Slime layer



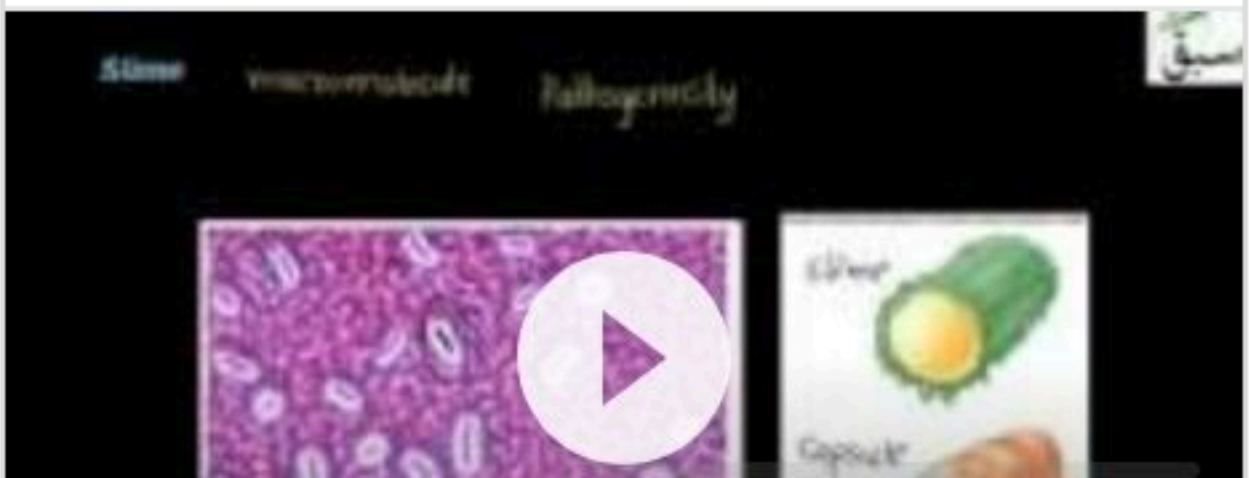




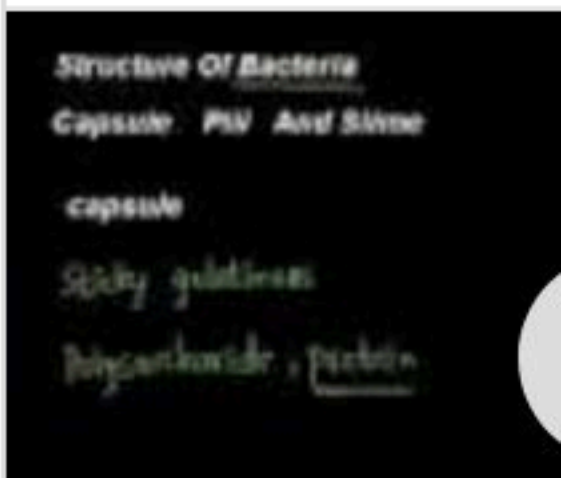
A slime layer in bacteria is an easily removable, unorganized layer of extracellular material that surrounds bacteria cells. Specifically, this consists mostly of exopolysaccharides, glycoproteins, and glycolipids. Therefore, the slime layer is considered as a subset of glycocalyx. Wikipedia

Videos

YouTube · Sabaq Foundation - Free...



YouTube · SWAP

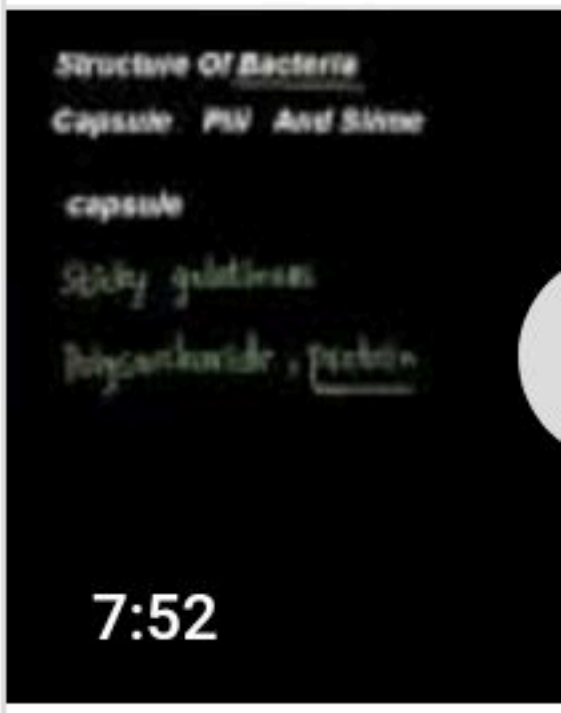


22:54 3.00 KB/S 53%



Structure of Bacteria Capsule, Slime and Pilli, Biology Lecture | Sabaq ...

21-Nov-2016



Structure of (Capsule Pill year Biology

22-Dec-2017

People also ask

What is the difference between a capsule and a slime layer?

Many bacterial cells secrete some extracellular material **in the** form of a **capsule** or a **slime layer**. A **slime layer** is loosely associated with the bacterium and can be easily washed off, whereas a **capsule** is attached tightly to the bacterium and has definite boundaries.

https://www.britannica.com › science

Bacteria - Capsules and slime layers | Britannica

More results

What does a slime capsule do?

22:55 16.0 KB/S 53%

https://microbenotes.com › spore-sta...

Spore Stain- Principle, Procedure and Result Interpretation

05-Aug-2017 — Procedure of Spore Stain (Schaeffer-Fulton Method) · Prepare smears in the usual manner using aseptic technique. · Allow ...

Principle of Spore Stain | Procedure of Spore Stain...

People also ask

What is Spore staining?

The **endospore stain** is a differential **stain** used to visualize bacterial endospores. Endospores are formed by a few genera of bacteria, such as Bacillus . By forming **spores**, bacteria can survive in hostile conditions. ... Endospores can form within different areas of the vegetative cell.

https://www.austincc.edu › microbugz

Welcome to Microbugz - Endospore Stain

More results

How does spore staining work?

Are endospores easily stained?

22:55 11.0 KB/S 53%

What are the 4 types of hypersensitivity?

The four types of hypersensitivity are:

- **Type** I: reaction mediated by IgE antibodies.
- **Type** II: cytotoxic reaction mediated by IgG or IgM antibodies.
- **Type** III: reaction mediated by immune complexes.
- **Type** IV: delayed reaction mediated by cellular response.

10-Sep-2020

https://www.ncbi.nlm.nih.gov › books

Type IV Hypersensitivity Reaction - StatPearls - NCBI Bookshelf

More results

What is a hypersensitivity?

Hypersensitivity (also called **hypersensitivity** reaction or intolerance) refers to undesirable reactions produced by the normal immune system, including allergies and autoimmunity.

https://en.m.wikipedia.org › wiki

Hypersensitivity - Wikipedia

22:56 2.00 KB/S 53%
google.com/search?cli
Google
primary immune system
All Images News Videos Books

The **primary** parts of the **immune system** include the bone marrow and thymus. The bone marrow is extremely important to the **immune system** because all the body's blood cells (including T and B lymphocytes) originate in the bone marrow. B lymphocytes remain in the marrow to mature, while T lymphocytes travel to the thymus.

https://www.healio.com › componen...

Components of the Immune System - Healio

About featured snippets Feedback

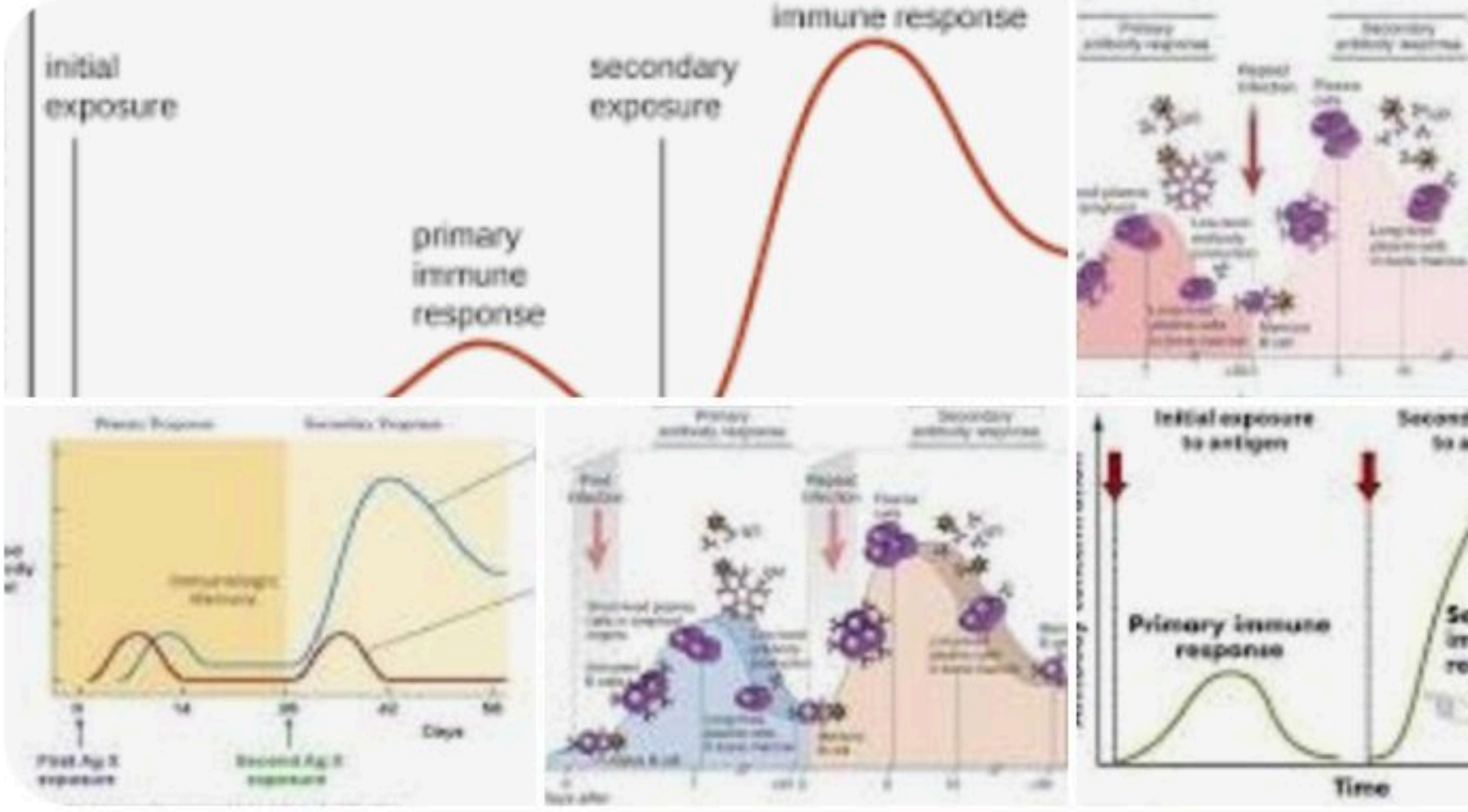
People also ask

What are the primary immune organs?

What are the 3 types of immune systems?

22:56 49.0 KB/S 53%
google.com/search?cli
secondary immune system
All Images Videos News Books

Secondary immune system



Secondary Immune Response is the reaction of the **immune system** when it contacts an antigen for the second and subsequent times. Appears mainly in the lymph nodes and spleen. Appears mainly in the bone marrow and then, in the spleen and lymph nodes. This occurs in **response** to the primary contact of the antigen. 29-Apr-2018

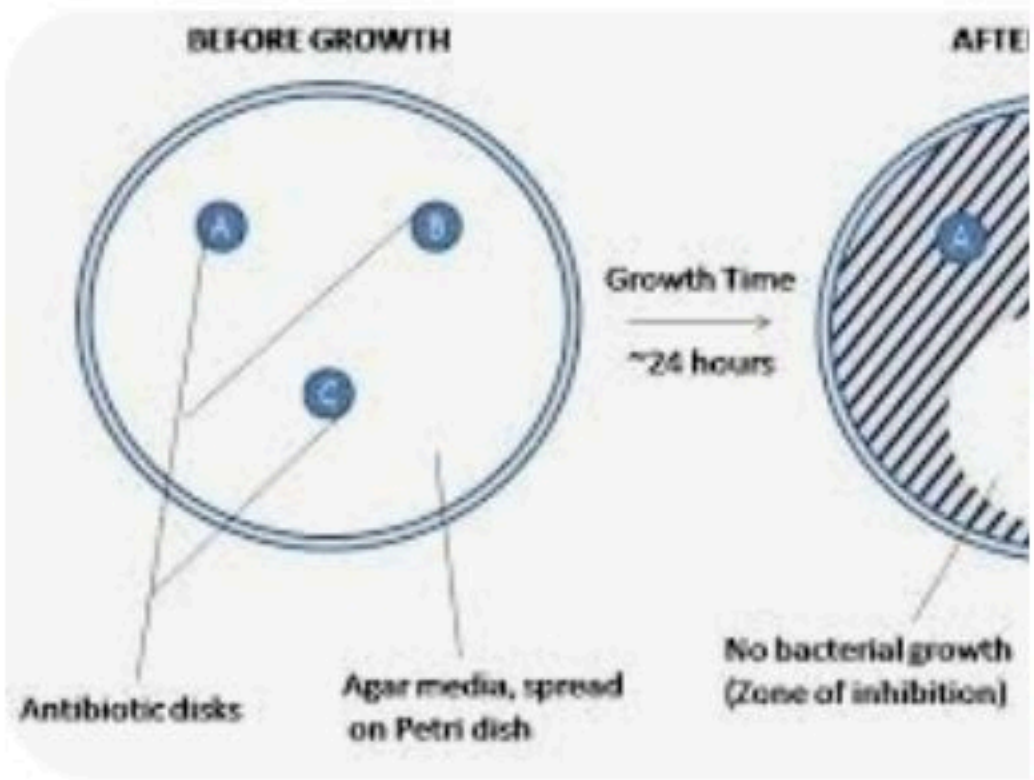
https://microbenotes.com › differenc...

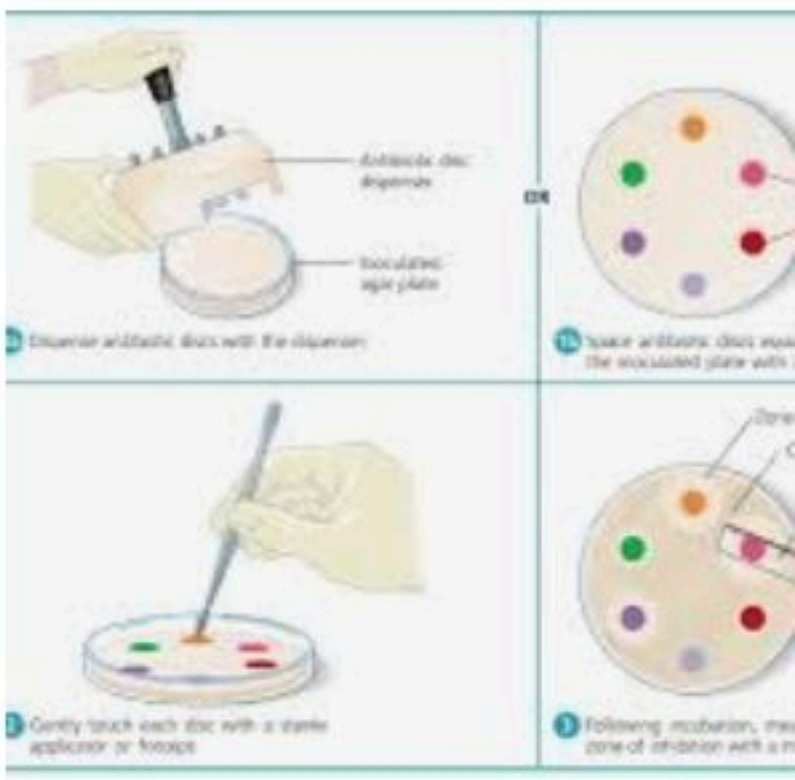
12 Differences between Primary and Secondary Immune Response

About featured snippets Feedback

22:56 22.0 KB/S 53%
google.com/search?cli
5
Google
disc diffusion
All Images Videos News Books

Disk diffusion test

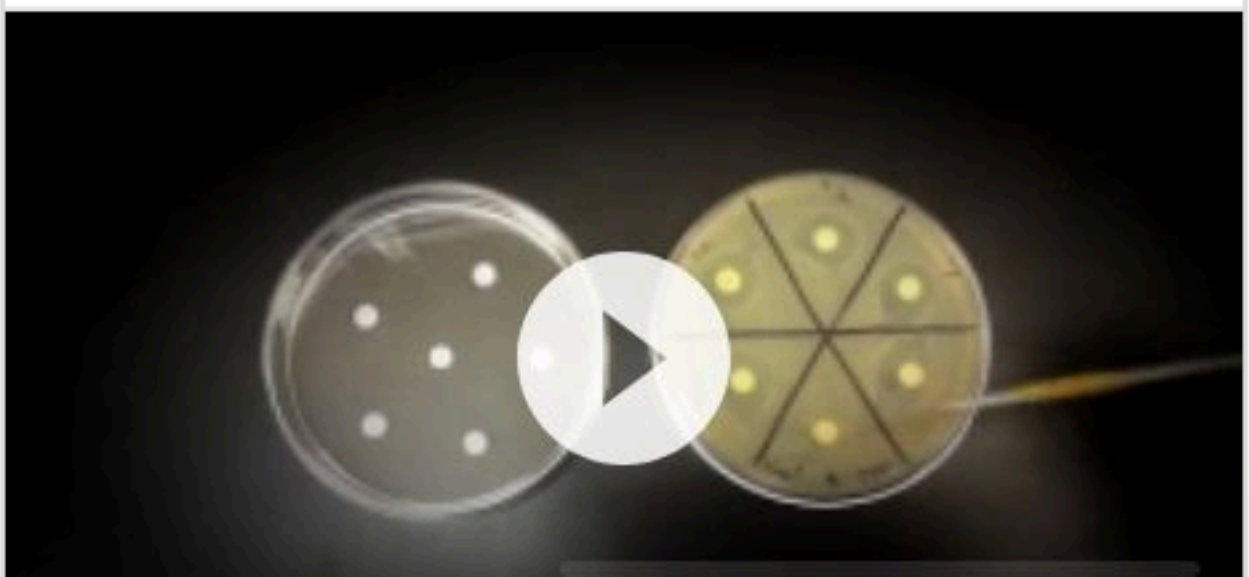




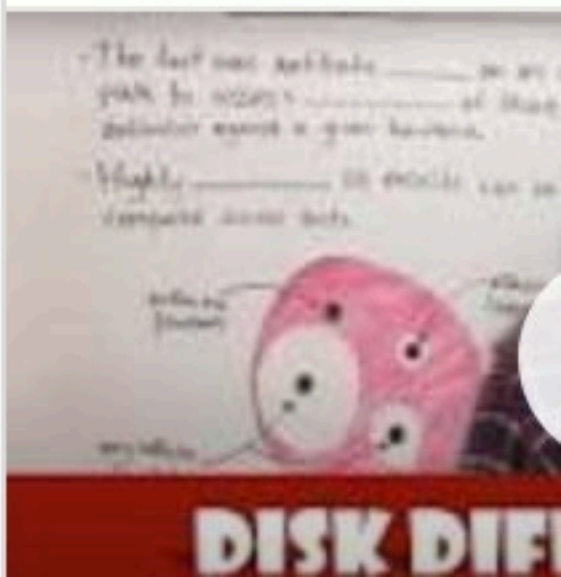
The disk diffusion test, or agar diffusion test, or Kirby–Bauer test, is an antibiotic susceptibility test. It uses antibiotic discs to test the extent to which bacteria are affected by those antibiotics. Wikipedia

Videos

YouTube · Gold Biotechnology, Inc.



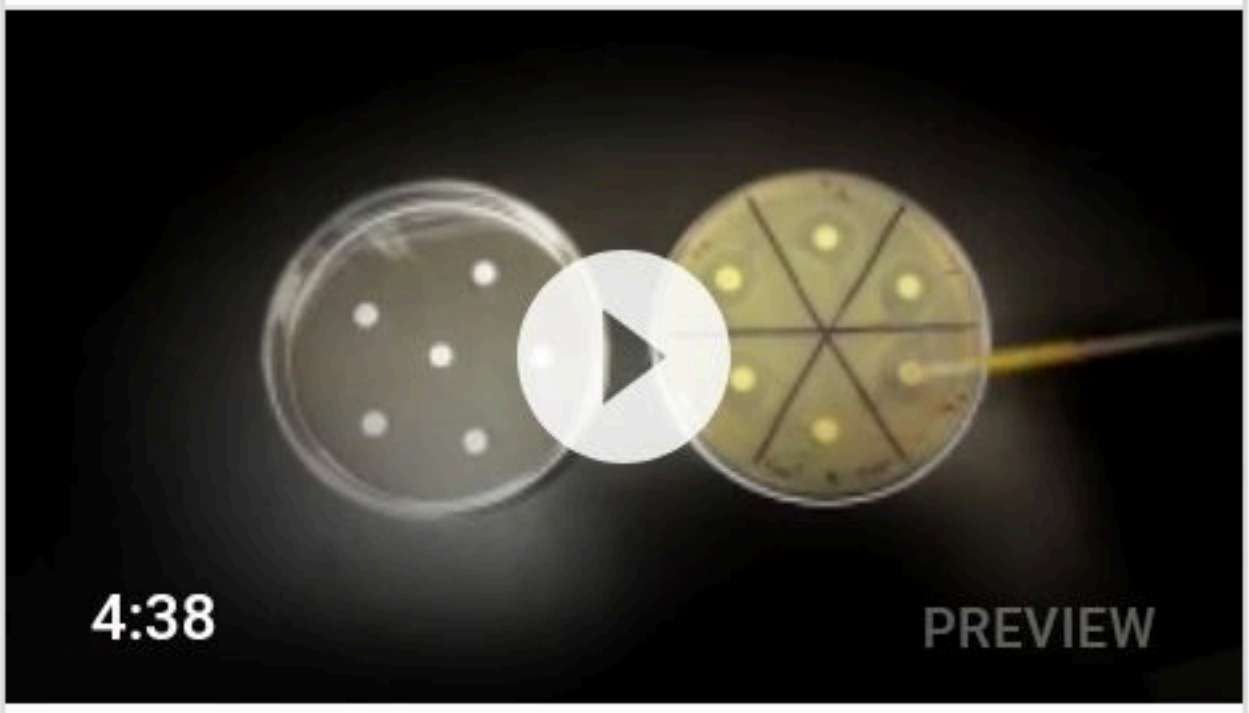
YouTube · Biolo



22:56 2.00 KB/S 53%

Videos

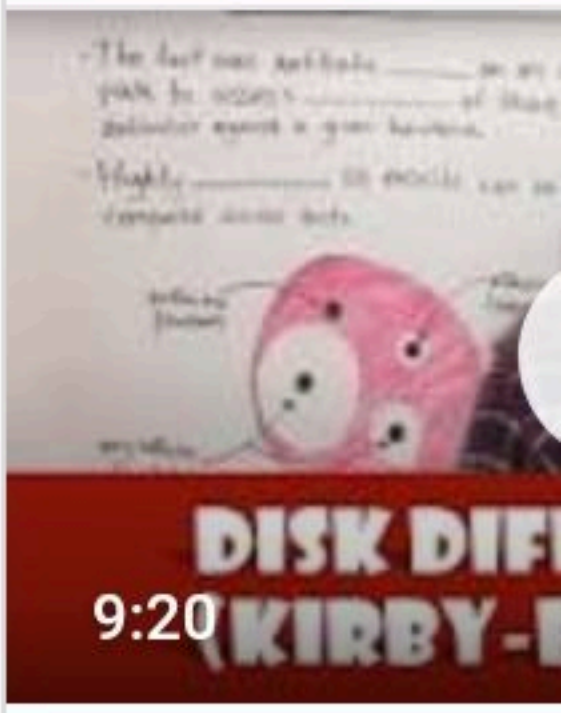
YouTube · Gold Biotechnology, Inc.



Testing an Antibiotic Using a Disk Diffusion Assay - Kirby Bauer ...

15-Aug-2013

YouTube · Biolo



Disk Diffusio Bauer Test)

29-Oct-2018

People also ask

How does disc diffusion work?

The **disk-diffusion** agar method tests the effectiveness of antibiotics on a specific microorganism. An agar plate is first spread with bacteria, then paper disks of antibiotics are placed atop of it. The bacteria is then allowed to grow on the agar media and then observed for growth and effect of the antibiotic on it.

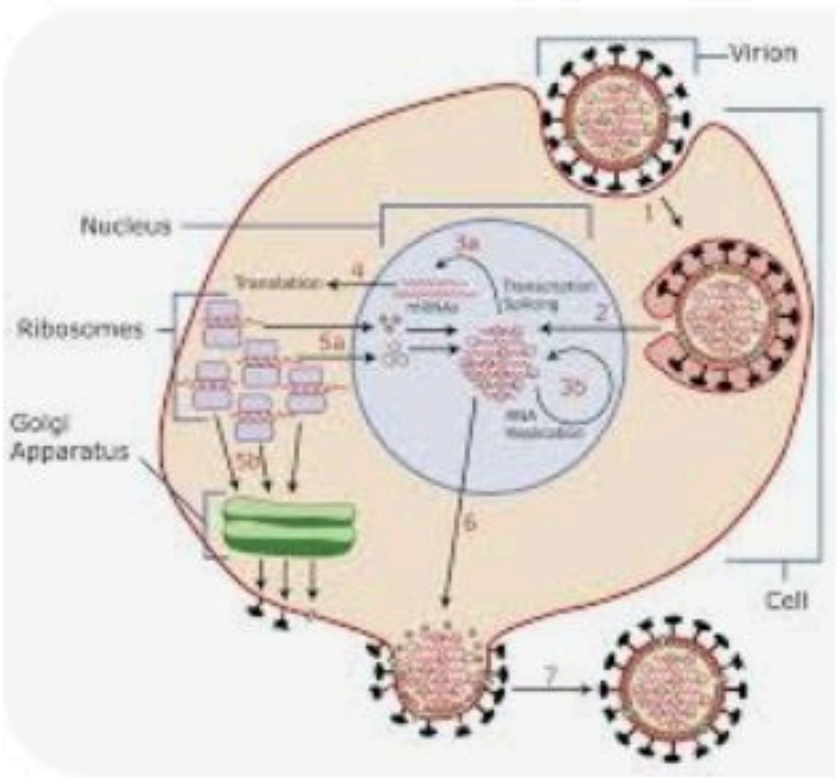
https://en.m.wikipedia.org › wiki

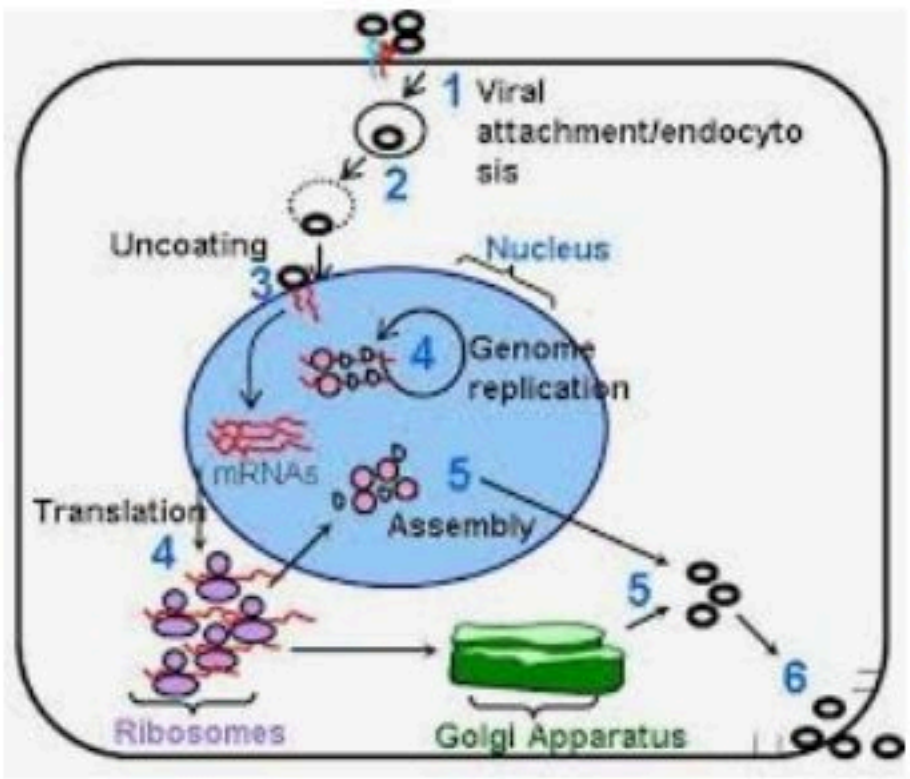
Disk diffusion test - Wikipedia

More results

22:57 29.0 KB/S 53%
google.com/search?cli
5
Google
viral replication
All Images Videos News Books Sea

Viral replication

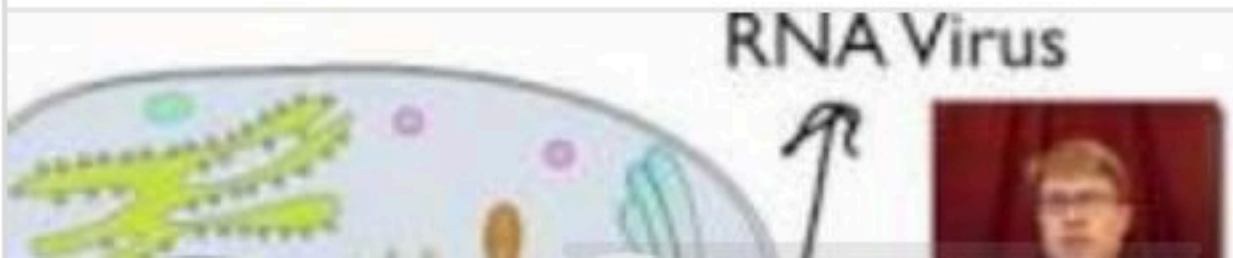




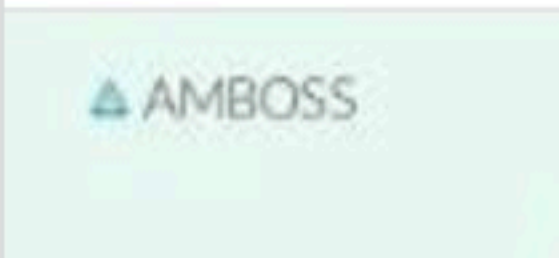
Viral replication is the formation of biological viruses during the infection process in the target host cells. Viruses must first get into the cell before viral replication can occur. Through the generation of abundant copies of its genome and packaging these copies, the virus continues infecting new hosts.
Wikipedia

Videos

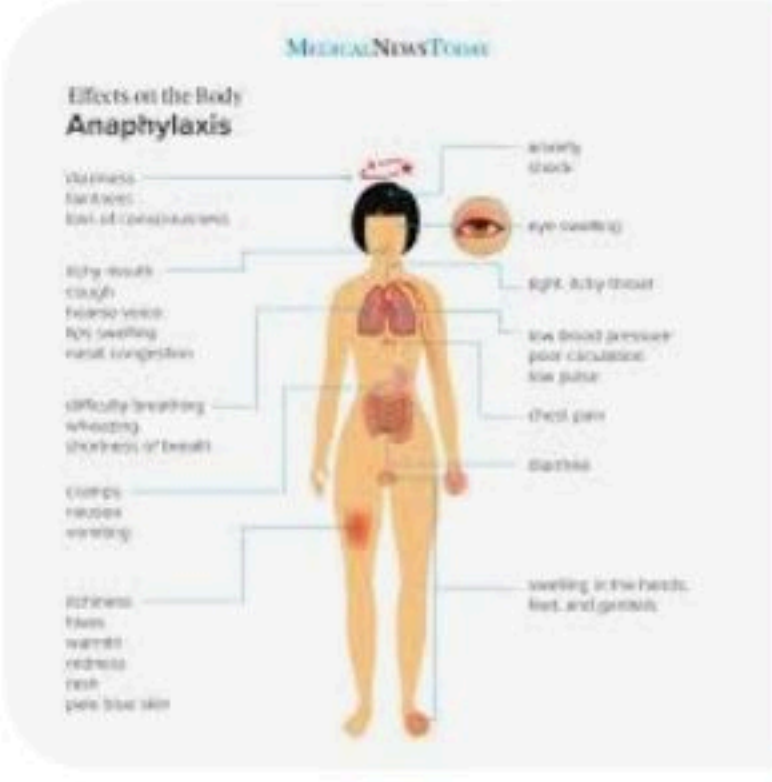
YouTube · Bozeman Science

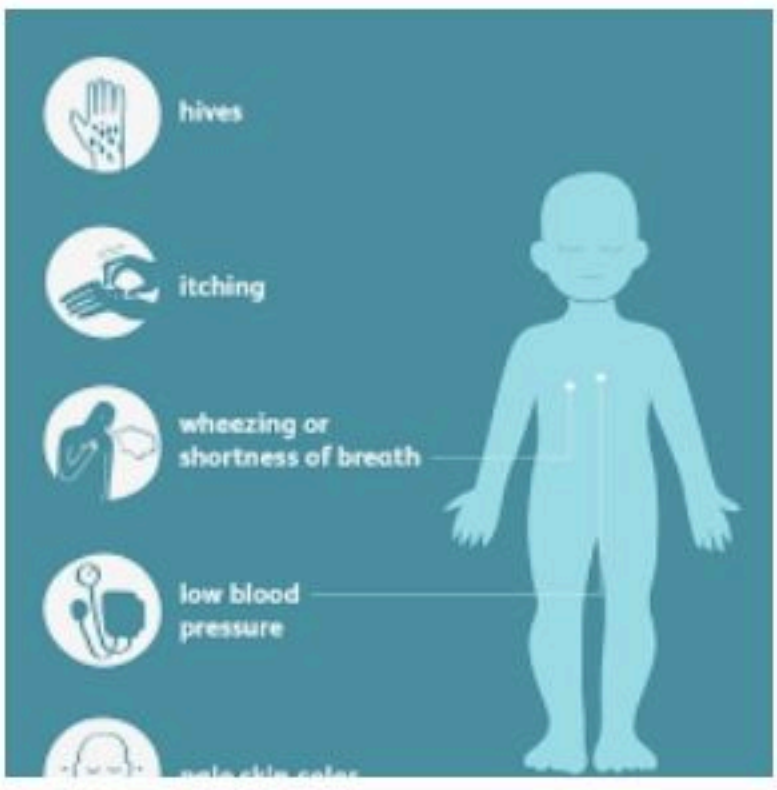


YouTube · AMB



22:57 8.00 KB/S 52%
google.com/search?cli
Google
anaphylaxis
All Images News Books Videos Sea





Anaphylaxis is a severe allergic reaction to venom, food, or medication. Most cases are caused by a bee sting or eating foods that are known to cause allergies, such as peanuts or tree nuts. **Anaphylaxis** causes a series of symptoms, including a rash, low pulse, and shock, which is known as **anaphylactic** shock.

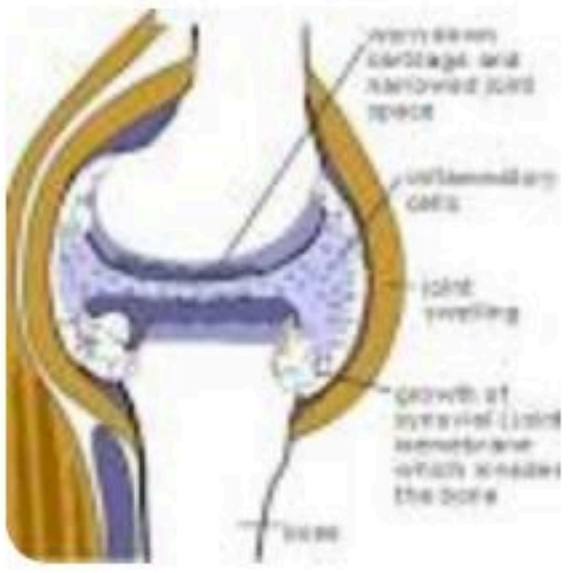
https://www.healthline.com › health

Anaphylaxis: Causes, Symptoms & Diagnosis - Healthline

About featured snippets Feedback

22:57 3.00 KB/S 52%
google.com/search?cli
5
Google
inflammation definition
All Books Images News Videos Sea

Inflammation is a process by which your body's white blood cells and the things they make protect you from infection from outside invaders, such as bacteria and viruses. 03-Aug-2017



https://www.webmd.com › arthritis

Inflammation: Definition, Diseases, Types, and Treatment - WebMD

About featured snippets Feedback

WORD COACH Learn new words

Which word is *similar* to inflammation?

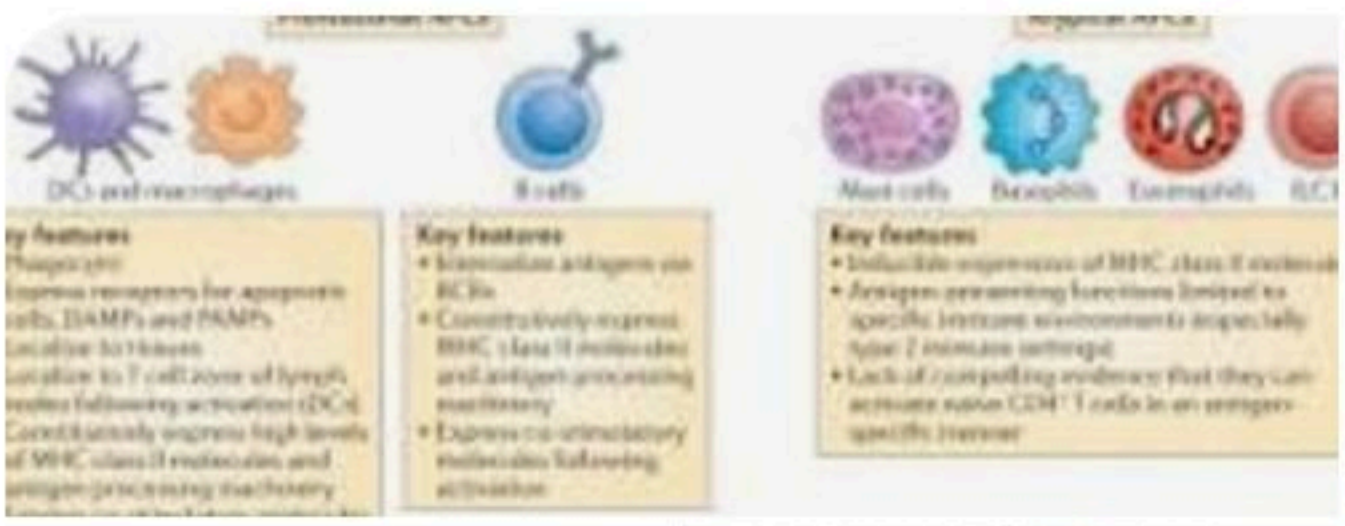
Styling

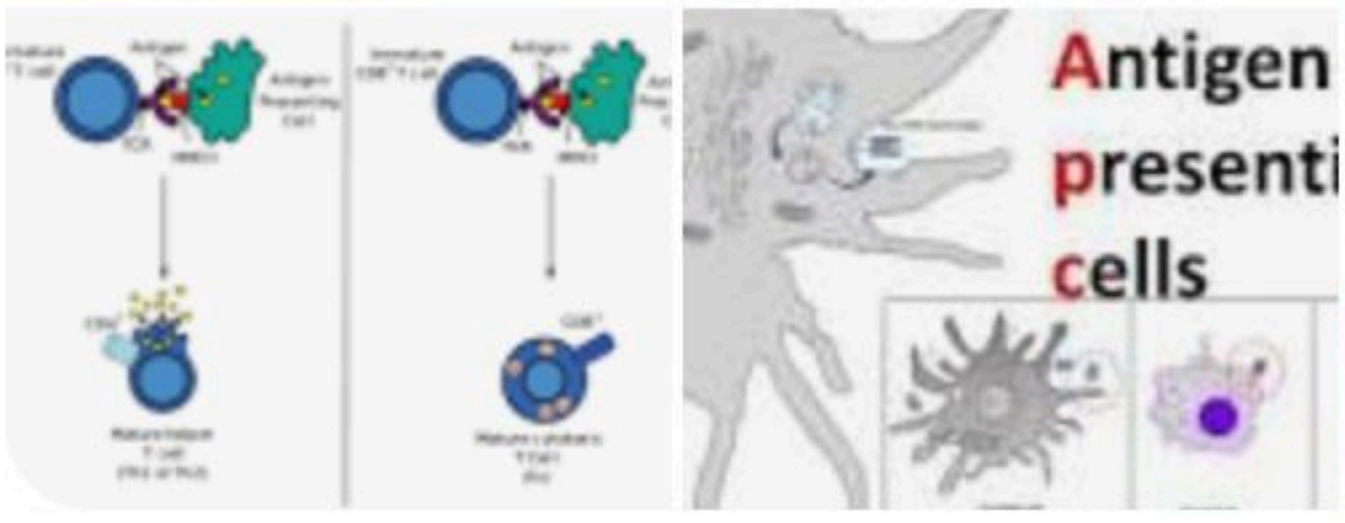
or

Swelling

22:58 113 KB/S 52%
google.com/search?cli
5
antigen presenting cells
All Images Books Videos News Sea

Antigen presenting cells





Antigen-presenting cells (APCs) are a heterogeneous group of immune **cells** that mediate the **cellular** immune response by processing and **presenting antigens** for recognition by certain lymphocytes such as T **cells**. Classical APCs include dendritic **cells**, macrophages, Langerhans **cells** and B **cells**.

https://www.nature.com › subjects

Antigen-presenting cells - Latest research and news | Nature

22:59 41.0 KB/S 52%
google.com/search?cli
5
Google
define reverse transcriptase
All Images Videos News Books Sea

Dictionary

Search for a word

transcriptase

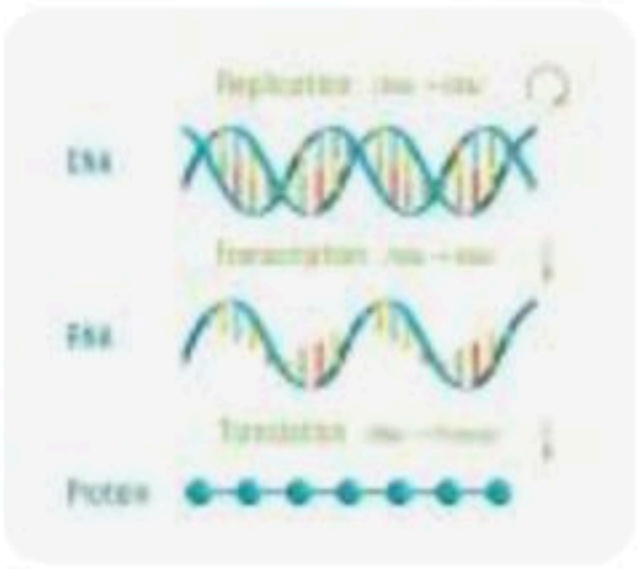
/tranˈskrɪpteɪz,trɑːn-/

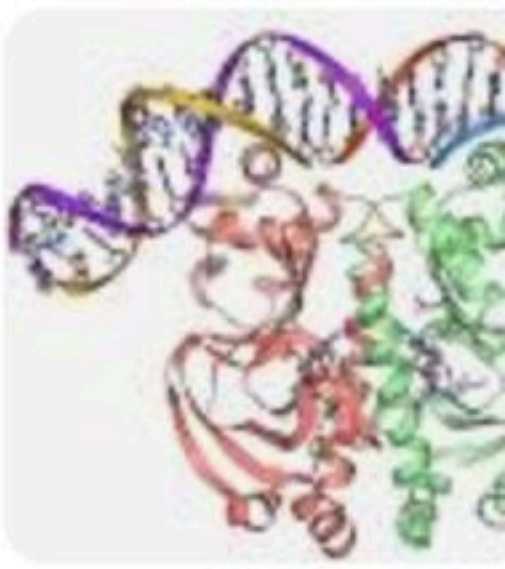
noun BIOCHEMISTRY

noun: **reverse transcriptase**

an enzyme which catalyses the formation of RNA from a DNA template during transcription, or (*reverse transcriptase*) the formation of DNA from an RNA template in reverse transcription.



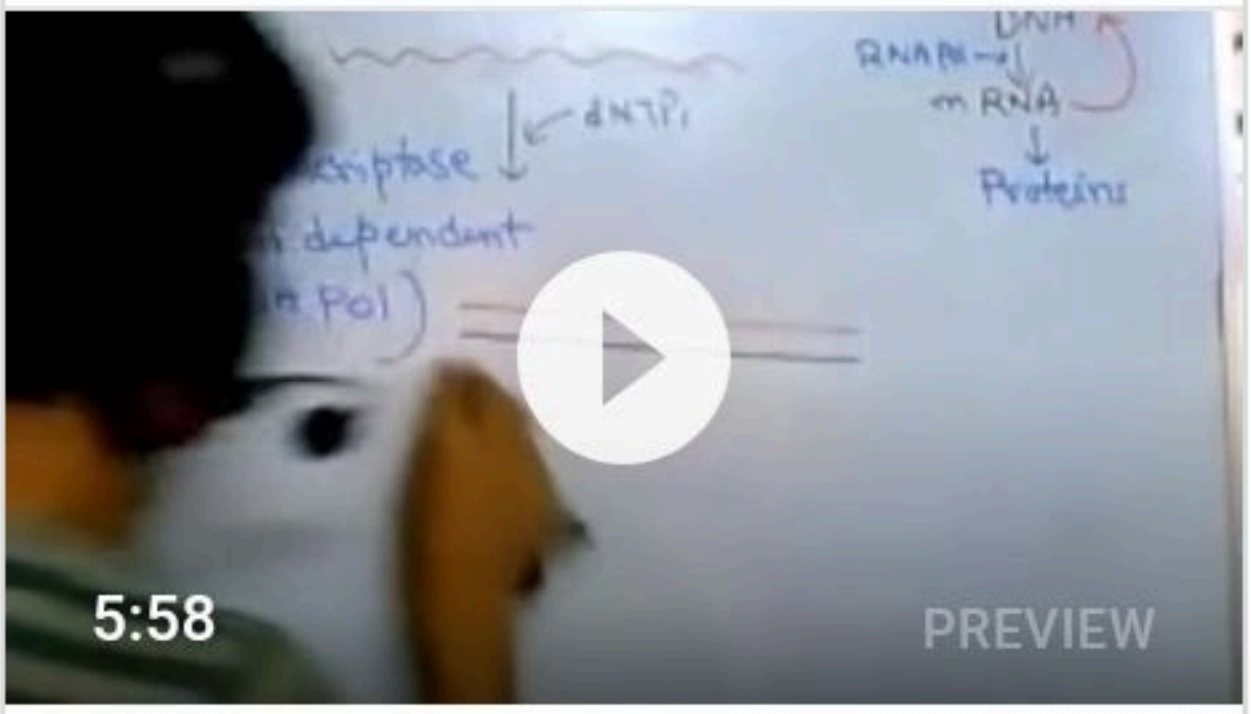




Definitions from Oxford Languages *Feedback*

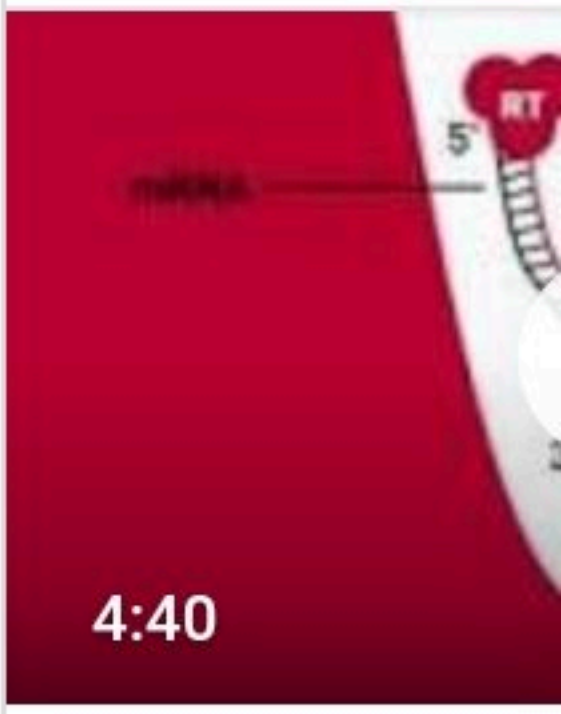
Translations and more definitions

22:59 4.00 KB/S Vo LTE2 4G 52%



Reverse transcription | by reverse transcriptase enzyme

29-Sep-2014



Simplified RT Transcription

06-Dec-2012

People also ask

What is the meaning of reverse transcriptase?

Reverse transcriptase, also called RNA-directed DNA polymerase, an enzyme encoded from the genetic material of retroviruses that catalyzes the **transcription** of retrovirus RNA (ribonucleic acid) into DNA (deoxyribonucleic acid).

https://www.britannica.com › science

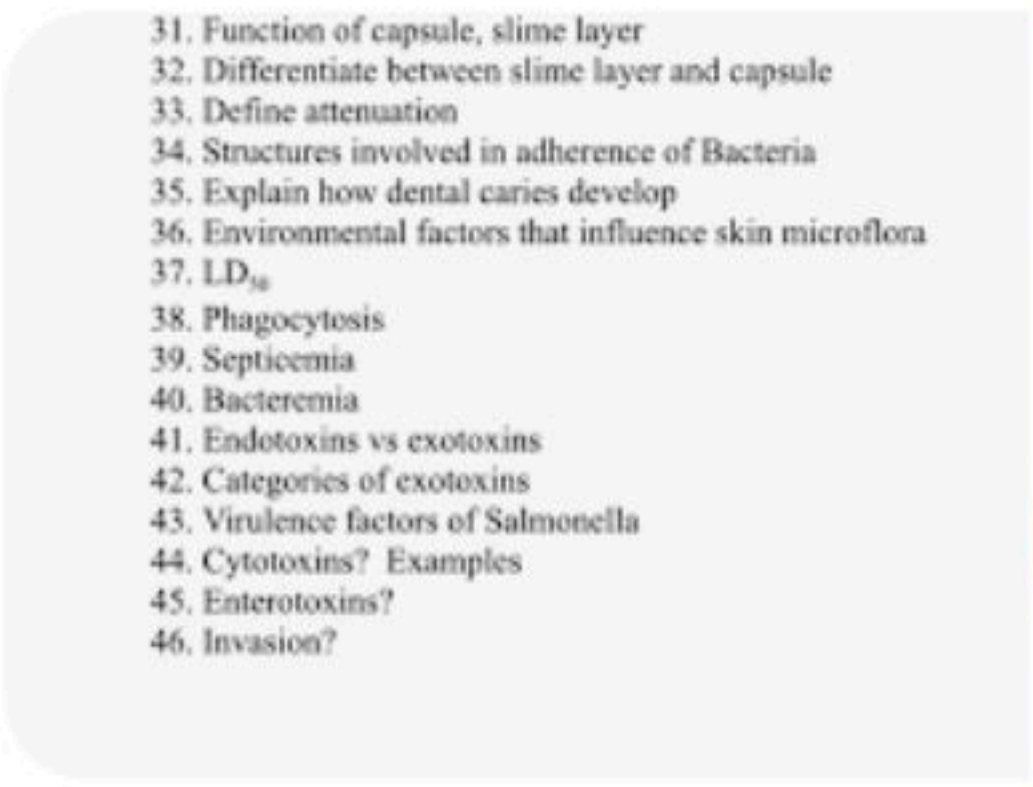
Reverse transcriptase | enzyme | Britannica

More results

What is the purpose of using reverse transcriptase?

What does transcriptase mean?

23:00 0.00 KB/S 52%
google.com/search?cli
Google
function of capsule
All Images News Videos Books


31. Function of capsule, slime layer
32. Differentiate between slime layer and capsule
33. Define attenuation
34. Structures involved in adherence of Bacteria
35. Explain how dental caries develop
36. Environmental factors that influence skin microflora
37. LD_{50}
38. Phagocytosis
39. Septicemia
40. Bacteremia
41. Endotoxins vs exotoxins
42. Categories of exotoxins
43. Virulence factors of Salmonella
44. Cytotoxins? Examples
45. Enterotoxins?
46. Invasion?



It has several **functions**: promote bacterial adhesion to surfaces or interaction with other organisms; act as a permeability barrier, as a defense mechanism against phagocytosis and/or as a nutrient reserve. Among pathogens, **capsule** formation often correlates with pathogenicity.

https://www.uniprot.org › keywords

Capsule - UniProt

About featured snippets Feedback

People also ask

23:00 2.00 KB/S 52%

as a nutrient reserve. Among pathogens, **capsule** formation often correlates with pathogenicity.

https://www.uniprot.org › keywords

Capsule - UniProt

About featured snippets Feedback

People also ask

What is the function of capsule in bacterial cell?

Capsules can protect a bacterial cell from **ingestion** and destruction by **white blood cells** (phagocytosis). While the exact mechanism for escaping phagocytosis is unclear, it may occur because capsules make bacterial surface components more slippery, helping the bacterium to escape engulfment by phagocytic cells.

https://www.britannica.com › science

Bacteria - Capsules and slime layers | Britannica

More results

What is the function of capsule in prokaryotic cell?

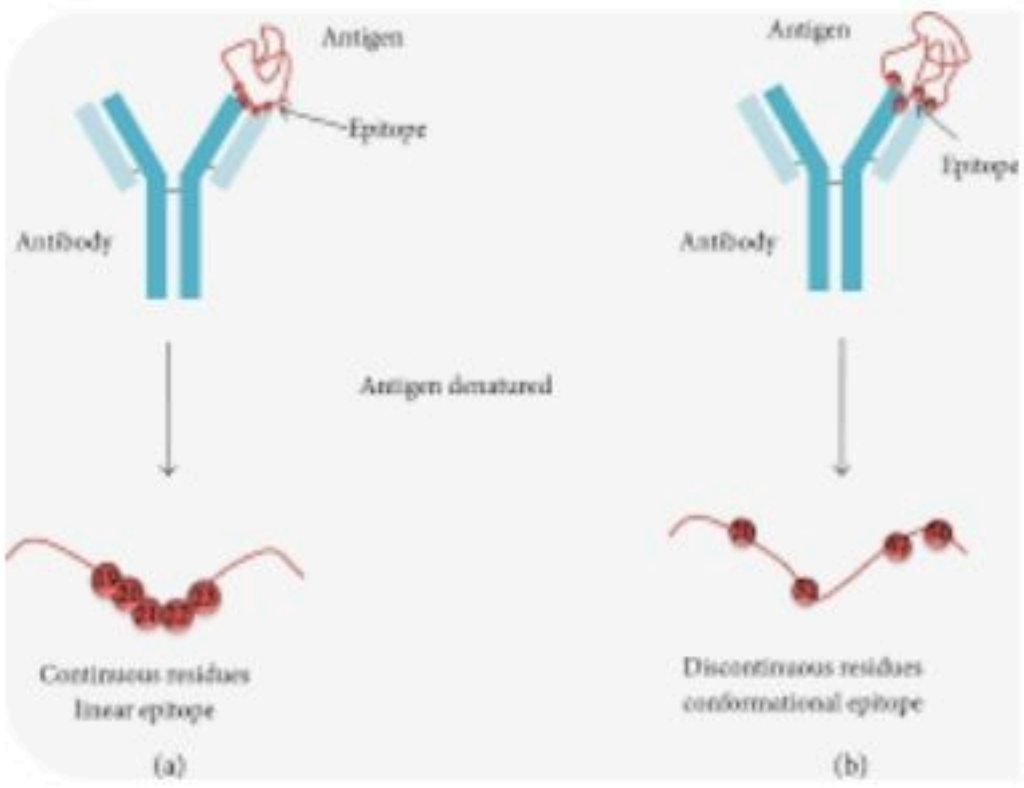
What is the function of a capsule quizlet?

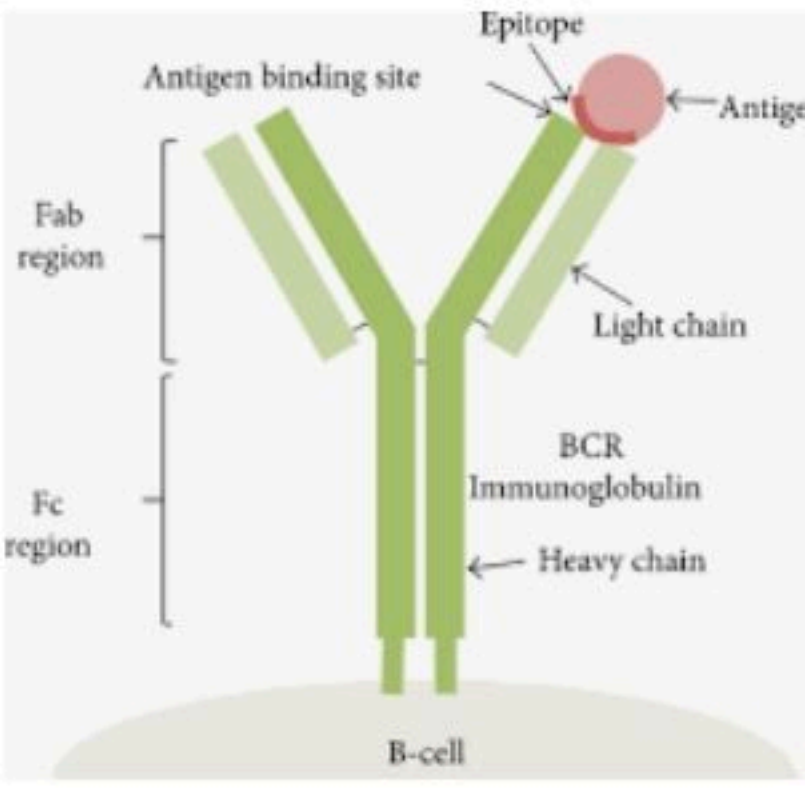
23:00 5.00 KB/S 52%
google.com/search?cli
5
All Images Videos News Books Sea

Showing results for ***epitope*** and antigen

Search instead for epitop and antigen

Epitope

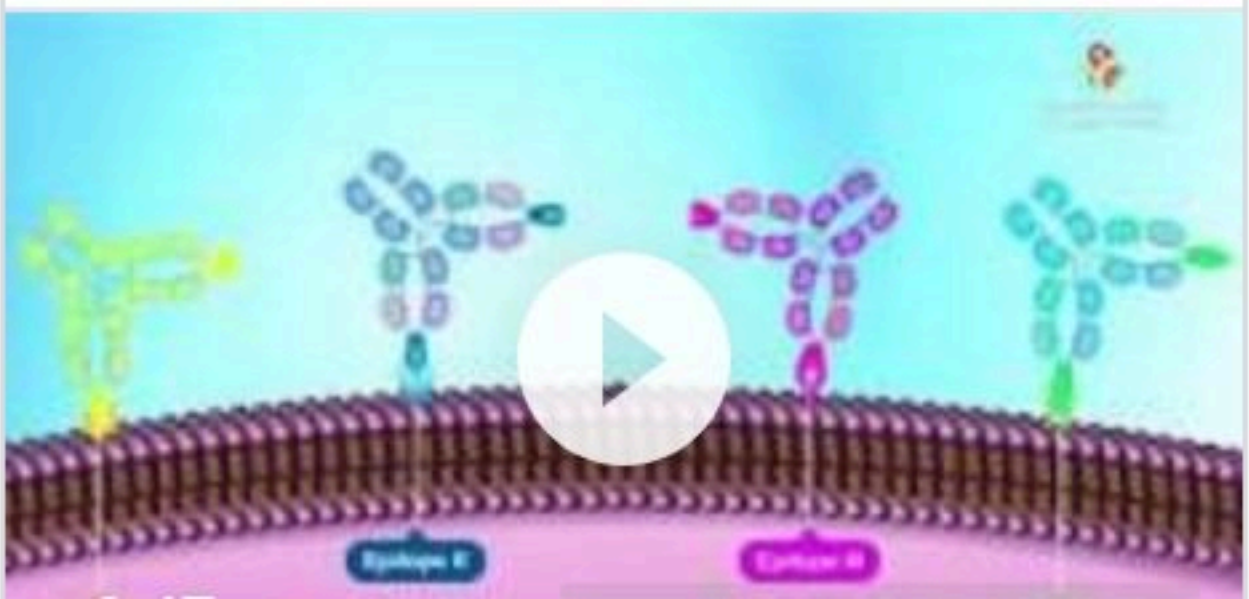




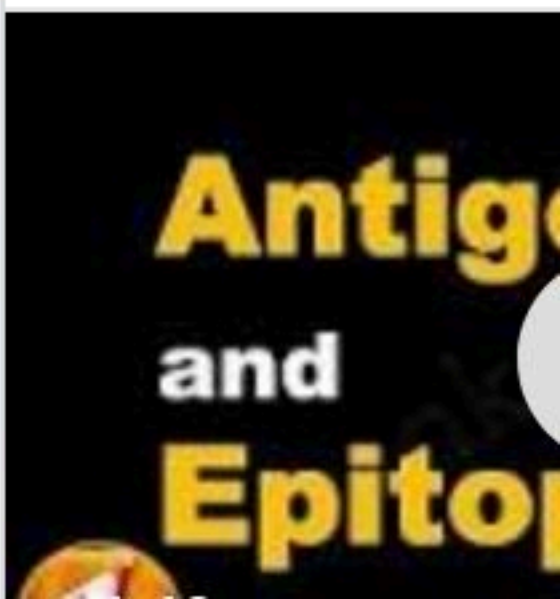
An epitope, also known as antigenic determinant, is the part of an antigen that is recognized by the immune system, specifically by antibodies, B cells, or T cells. For example, the epitope is the specific piece of the antigen to which an antibody binds. Wikipedia

Videos

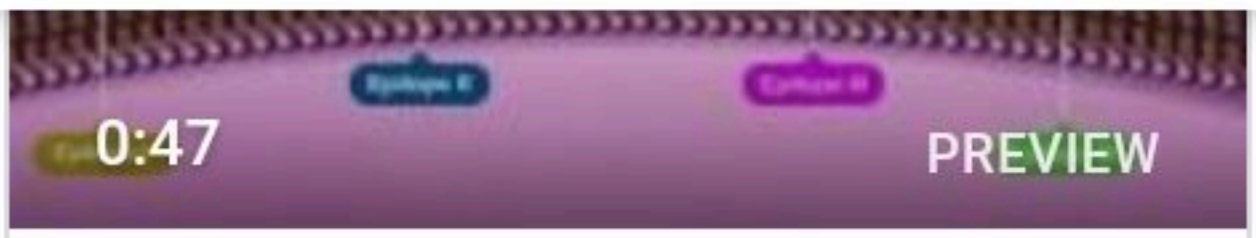
YouTube · Dr.G Bhanu Prakash Ani...



YouTube · Frank



23:01 2.00 KB/S 4G 52%
google.com/search?cli 5



Antigenic Determinants (Epitopes) - Animation

13-Feb-2019



Antigen and (Antigenic D (FL-Immuno

15-Apr-2017

People also ask

What is the difference between epitope and antigen?

An **epitope** (also known as the **antigenic** determinant) is that part of the **antigen** to which **antibodies** bind. While the **antigen** evokes the antibody response **in the** host, the antibody doesn't bind to the entire protein, but only to that segment called the **epitope**.

24-Nov-2016

https://socratic.org › questions › wh...

What is the difference between an epitope and an antigen? Does an ...

More results

How many epitopes can an antigen have?

23:01 22.0 KB/S 52%
google.com/search?q=
bacterial transmission
All Images Videos News Books

Scholarly articles for **bacterial transmission**

... risk factor for intraoperative **bacterial transmission** - Loftus - Cited by 185

... mutans and some aspects of the **bacterial transmission** - KÖHLER - Cited by 234

Bacterial transmission in periodontal diseases: a ... - Greenstein - Cited by 112

Bacteria are transmitted to humans through air, water, food, or living vectors. The principal modes of **transmission** of **bacterial** infection are contact, airborne, droplet, vectors, and vehicular. Preventive measures have a dramatic impact on morbidity and mortality.

https://www.ncbi.nlm.nih.gov › pmc

Bacterial Infections: Overview - NCBI - NIH

About featured snippets Feedback

People also ask

23:01 13.0 KB/S 52%
google.com/search?cli
5
Google
what is bacterial transmission
All Images Videos News Books Sea

Bacteria are transmitted to humans through air, water, food, or living vectors. The principal modes of **transmission** of **bacterial** infection are contact, airborne, droplet, vectors, and vehicular. Preventive measures have a dramatic impact on morbidity and mortality.

https://www.ncbi.nlm.nih.gov › pmc

Bacterial Infections: Overview - NCBI - NIH

About featured snippets Feedback

People also ask

How do bacteria spread in the body?

What are the 5 modes of transmission?

Can bacteria be transmitted from person to person?

23:02 13.0 KB/S 52%
google.com/search?cli
5
Google
define disc staining
All Images Videos News Books

https://en.m.wikipedia.org › wiki

Staining - Wikipedia

Stains may be used to define biological tissues (highlighting, for example, muscle fibers or connective tissue), cell populations (classifying ...

---

https://en.m.wikipedia.org › wiki

Disk diffusion test - Wikipedia

The disk diffusion test, or agar diffusion test, or Kirby–Bauer test is an antibiotic ... Hemolysis ·...

People also ask

What is mean by staining?

What are the types of staining?

What is staining and its types?

What is the purpose of staining?

23:02 3.00 KB/S 52%

https://en.m.wikipedia.org › wiki

Staining - Wikipedia

Stains may be used to define biological tissues (highlighting, for example, muscle fibers or connective tissue), cell populations (classifying ...

---

https://en.m.wikipedia.org › wiki

Disk diffusion test - Wikipedia

The disk diffusion test, or agar diffusion test, or Kirby–Bauer test is an antibiotic ... Hemolysis ·...

People also ask

What is mean by staining?

A **stain** is a mark or discoloration, and to **stain** something is to mark it with dirt or color it, either deliberately or accidentally. ... **Stain** comes, via the Middle English disteynen, or "discolor," from the Latin tingere, "to color or dye."

https://www.vocabulary.com › stain

stain - Dictionary Definition : Vocabulary.com

More results

What are the types of staining?

23:02 2.00 KB/S 52%

A **stain** is a mark or discoloration, and to **stain** something is to mark it with dirt or color it, either deliberately or accidentally. ... **Stain** comes, via the Middle English disteynen, or "discolor," from the Latin tingere, "to color or dye."

https://www.vocabulary.com › stain

stain - Dictionary Definition : Vocabulary.com

More results

What are the types of staining?

Seven Types of Stain

- Oil **Stain**. Oil **stains** are the most widely available and the **type of stain** most people think of when they think of **stain**. ...
- Varnish **Stain**. Varnish **stains** resemble oil **stains** in every way but one. ...
- Gel **Stain**. ...
- Lacquer **Stain**. ...
- Water-Soluble Dye **Stain**. ...
- Metal-Complex (Metalized) Dye **Stain**.

19-Jun-2020

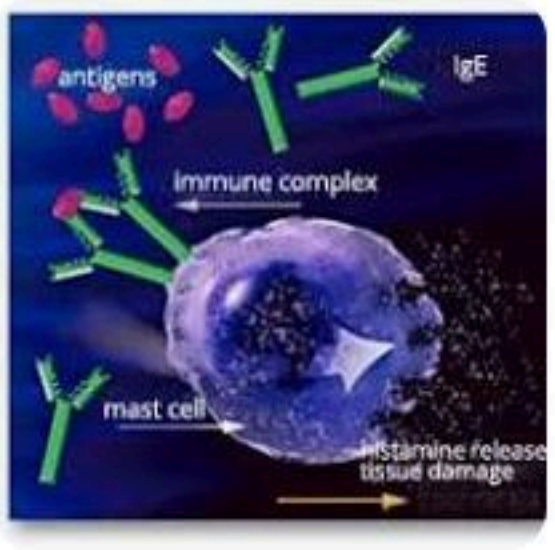
https://thefinishingstore.com › news

Seven Types of Stain – TheFinishingStore

23:03 23.0 KB/S 52%
google.com/search?cli 5

Reactions | Immunological and ...

Type I: Immediate Hypersensitivity (Anaphylactic Reaction) · Type II: Cytotoxic Reaction (Antibody-dependent) · Type III: Immune Complex ...



People also ask

What are the 4 types of hypersensitivity?

The four types of hypersensitivity are:

- **Type** I: reaction mediated by IgE antibodies.
- **Type** II: cytotoxic reaction mediated by IgG or IgM antibodies.
- **Type** III: reaction mediated by immune complexes.
- **Type** IV: delayed reaction mediated by cellular response.

10-Sep-2020

https://www.ncbi.nlm.nih.gov › books

Type IV Hypersensitivity Reaction - StatPearls - NCBI Bookshelf

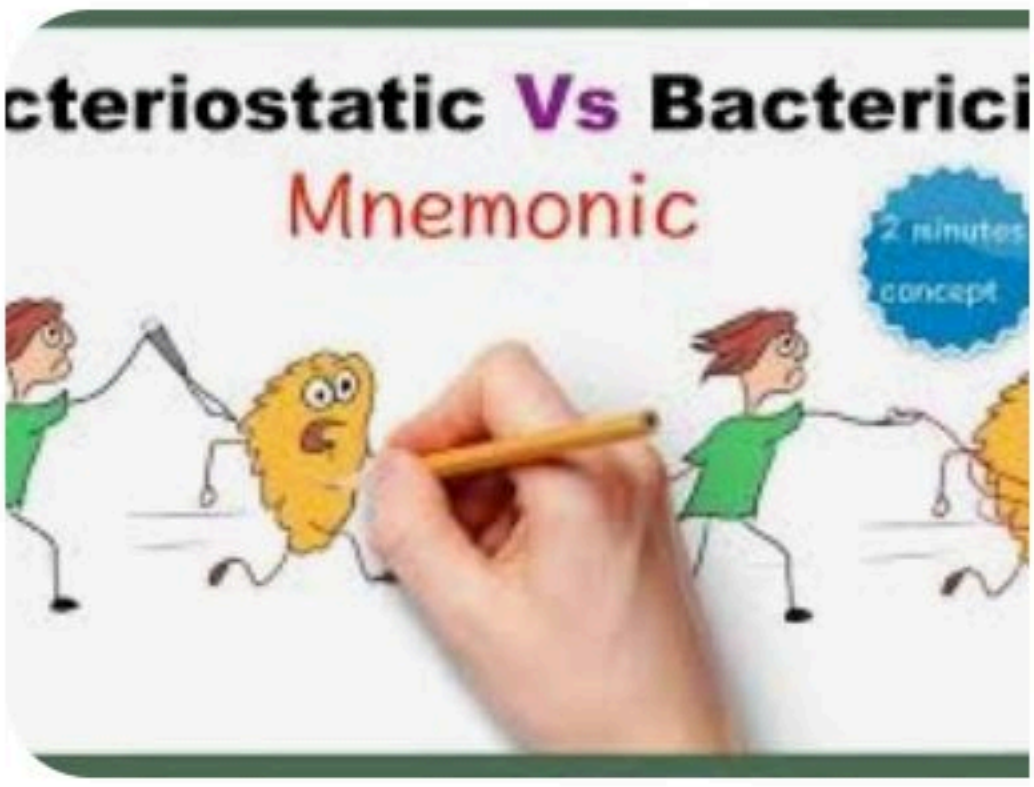
More results

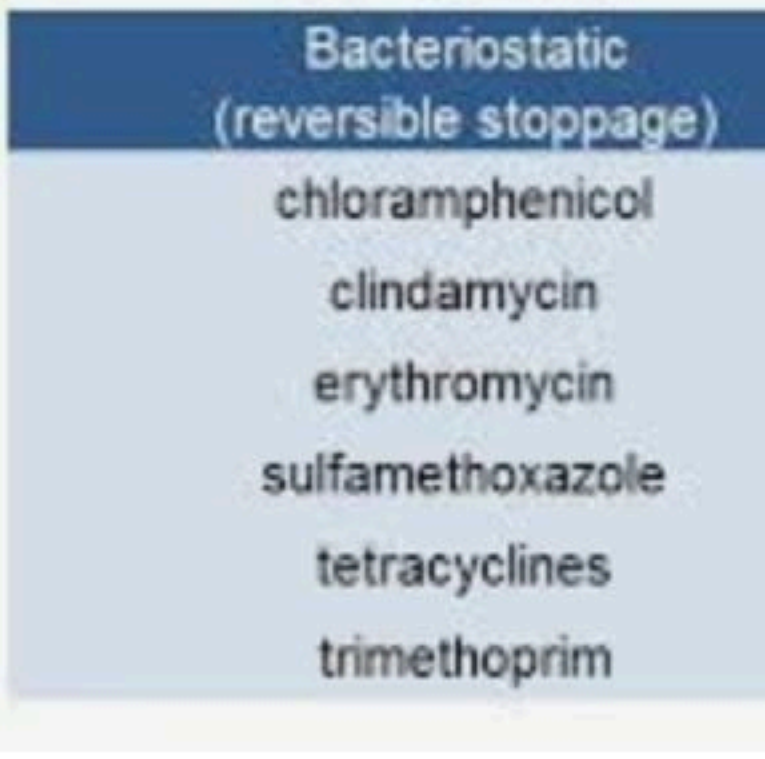
23:03 0.12 KB/S 51%

bactericidal

All Images News Videos Books Sear

Bactericide





A bactericide or bacteriocide, sometimes abbreviated Bcidal, is a substance which kills bacteria. Bactericides are disinfectants, antiseptics, or antibiotics. Wikipedia

People also search for: Bacteriostatic agent, Antiseptic, Antimicrobial, Antibiotics, Analgesic, more

People also ask

What does bactericidal mean?

Medical **Definition** of **bactericidal**

: destroying bacteria. Other Words from **bactericidal**. bactericidally \ -ᵊl-ē \ adverb.

23:04 3.00 KB/S 51%
google.com/search?cli
5
Google
bacteriostatic
All Books Images News Videos

Bacteriostatic agent

A bacteriostatic agent or bacteriostat, abbreviated Bstatic, is a biological or chemical agent that stops bacteria from reproducing, while not necessarily killing them otherwise. Depending on their application, bacteriostatic antibiotics, disinfectants, antiseptics and preservatives can be distinguished. Wikipedia

People also search for: Bactericide, Antiseptic, Antibiotics, Antimicrobial, Sulfonamide, **more**

Medical terminology

Treatment

Activities

People also ask

What is bacteriostatic and bactericidal?

23:04 0.00 KB/S 4G 51%

bacteriostatic

All Books Images News Videos Sea

Bacteriostatic agent

A bacteriostatic agent or bacteriostat, abbreviated Bstatic, is a biological or chemical agent that stops bacteria from reproducing, while not necessarily killing them otherwise. Depending on their application, bacteriostatic antibiotics, disinfectants, antiseptics and preservatives can be distinguished. Wikipedia

People also search for: Bactericide, Antiseptic, Antibiotics, Antimicrobial, Sulfonamide, **more**

Medical terminology

Treatment

Activities

People also ask

What is bacteriostatic and bactericidal?

The definitions of "**bacteriostatic**" and "**bactericidal**" appear to be straightforward: "**bacteriostatic**" means that the agent prevents the growth of bacteria (i.e., it keeps them in the stationary phase of growth), and "**bactericidal**" means that it kills bacteria.

23:04 19.0 KB/S 51%
google.com/search?cli
5
Google
virus
All Images News Books Videos

Virus

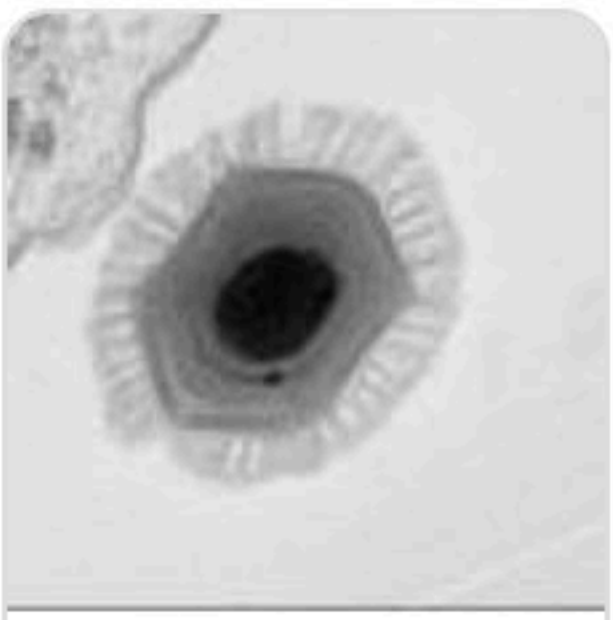
Infectious agent

Overview | Representative species

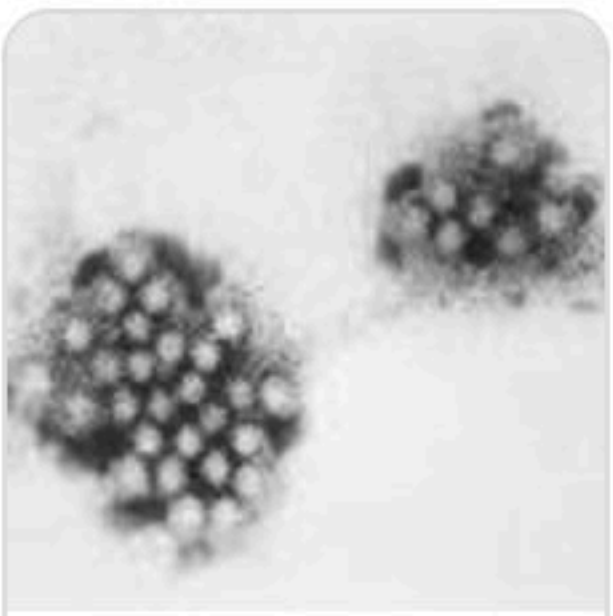
A virus is a submicroscopic infectious agent that replicates only inside the living cells of an organism. Viruses infect all types of life forms, from animals and plants to microorganisms, including bacteria and archaea. Wikipedia

Feedback

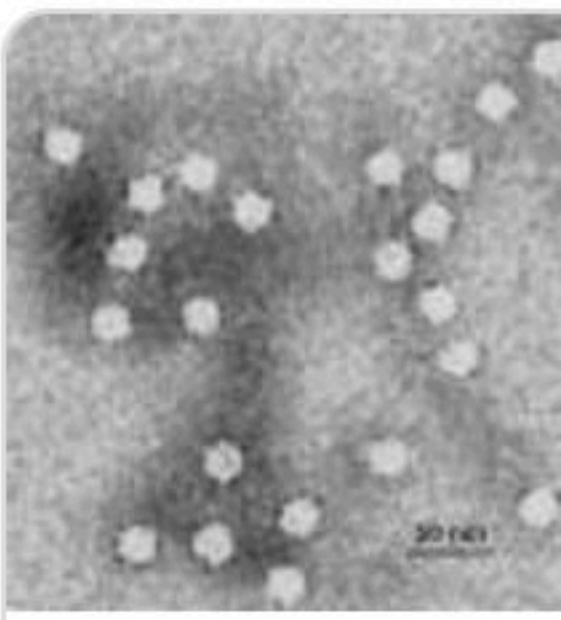
Representative species



Megavirus



Norwalk virus

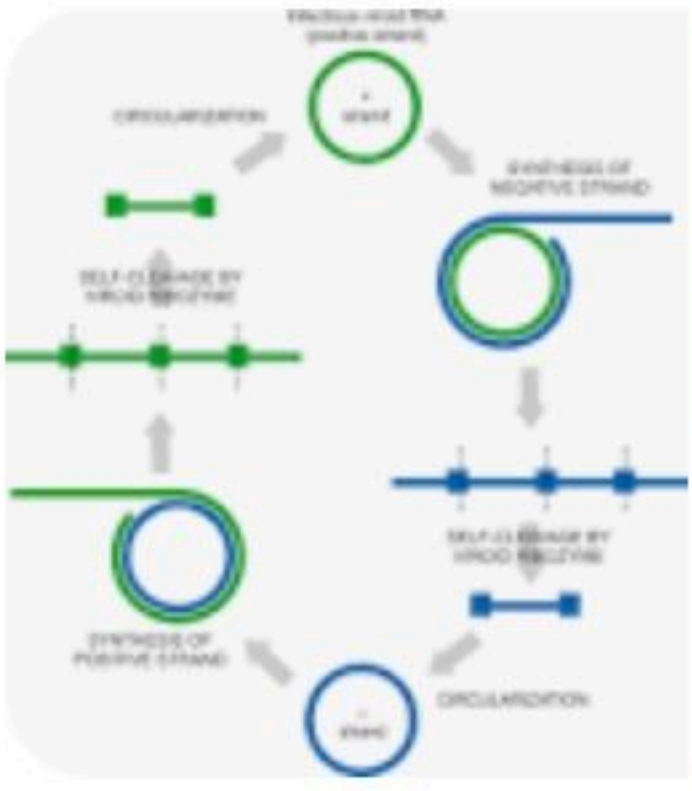


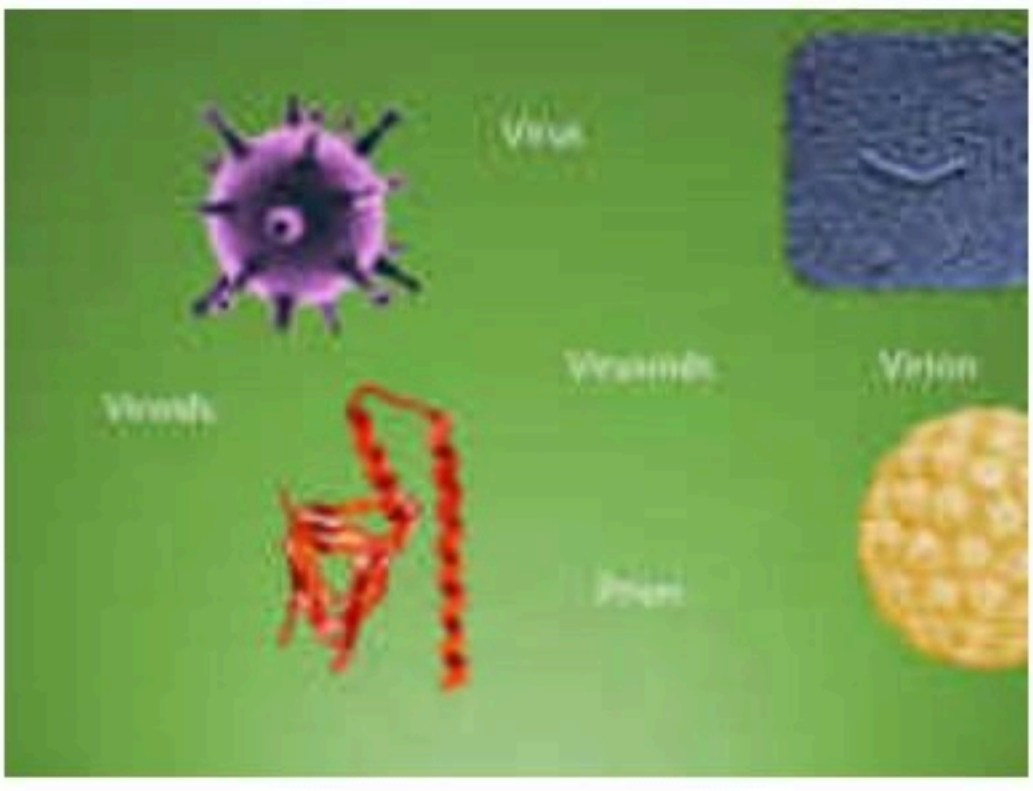
Canine parvovirus

23:04 6.00 KB/S 51%
google.com/search?cli
5
Google
viroids
All Images Videos News Books

Viroid

Overview Videos Representative specie





Viroids are the smallest infectious pathogens known. They are composed solely of a short strand of circular, single-stranded RNA that has no protein coating. All known **viroids** are inhabitants of angiosperms, and most cause diseases, whose respective economic importance on humans varies widely.

https://en.m.wikipedia.org › wiki

Viroid - Wikipedia

Feedback

23:05 3.00 KB/S 51%

People also ask

What is the difference between a virus and a virion?

Virus is a non-cellular, obligate parasite that is self-replicative inside a specific host cell. ... The main **difference between virus** and **virion** is that **virus** is the nucleoprotein particle whereas **virion** is the active, infectious form of the **virus**. A **virus** consists of genetic material covered by a protein capsid. 13-Feb-2018

https://pediaa.com › difference-betw...

Difference Between Virus and Virion | Definition, Role, Structure ...

More results

What is virion?

Virion, an entire **virus particle**, consisting of an outer protein shell called a capsid and an inner core of nucleic acid (either ribonucleic or deoxyribonucleic acid—RNA or DNA). The core confers infectivity, and the capsid provides specificity to the virus.

https://www.britannica.com › science

Virion | viral structure | Britannica

23:05 40.0 KB/S 51%
google.com/search?cli
5
Google
cytokines
All Images Books News Videos

Cytokine

Cytokines are a broad and loose category of small proteins important in cell signaling. Cytokines are peptides and cannot cross the lipid bilayer of cells to enter the cytoplasm. Cytokines have been shown to be involved in autocrine, paracrine and endocrine signaling as immunomodulating agents. Wikipedia

People also search for: Interferon, Antibody, Prostaglandin, Interleukin 6, **more**

Medical terminology

Number

Secretion

Function

People also ask

23:05 0.24 KB/S 51%

enter the cytoplasm. Cytokines have been shown to be involved in autocrine, paracrine and endocrine signaling as immunomodulating agents. Wikipedia

People also search for: Interferon, Antibody, Prostaglandin, Interleukin 6, **more**

Medical terminology

Number

Secretion

Function

People also ask

What is the function of cytokines?

What is a cytokine and what does it do?

Cytokines are small proteins that **are** crucial in controlling the growth and activity of other immune system cells and blood cells. When released, they signal the immune system to **do** its job. 27-Dec-2019

https://www.cancer.org › treatment

Cytokines and Their Side Effects - American Cancer Society

More results

23:06 3.00 KB/S 51%

Impact Articles | PPts | Journals ...

Bactericidal Drugs. Bactericidal agents is a one type of antibiotic which kills bacteria directly. The antibiotic polymyxin B injures the plasma ...

People also ask

What drugs are bactericidal?

Antibiotics that inhibit cell wall synthesis: the Beta-lactam antibiotics (penicillin derivatives (penams), cephalosporins (cephems), monobactams, and carbapenems) and vancomycin. Also **bactericidal** are daptomycin, fluoroquinolones, metronidazole, nitrofurantoin, co-trimoxazole, telithromycin.

https://www.hilarispublisher.com › b...

Bactericidal Drugs | List of High Impact Articles | PPts | Journals ...

More results

What does bactericidal mean?

Which drugs are bacteriostatic in nature?

Is Penicillin G bacteriostatic or bactericidal?

What is bactericidal effect?

23:06 17.0 KB/S 51%
google.com/search?cli
5
Google
bacteriostatic drug
All Images News Videos Books Sea

Bacteriostatic agent is a one type of antibiotic which inhibits the growth of bacteria. The bacteria don't die, but they can't grow or replicate either. Disinfectants, antiseptics, **bacteriostatic** antibiotics and preservatives can be distinguished depending upon their application.

https://www.hilarispublisher.com › b...

Bacteriostatic Drugs | List of High Impact Articles | PPts | Journals ...

About featured snippets Feedback

People also ask

Which antibiotics are bacteriostatic?

What do you mean by bacteriostatic?

Why do doctors prescribe bacteriostatic

23:07 24.0 KB/S 51%
google.com/search?cli
Google
type 3 of hypersensitivity
All Images Videos News Books

In **type** III **hypersensitivity** reaction, an abnormal immune response is mediated by the formation of antigen-antibody aggregates called "immune complexes." They can precipitate in various tissues such as skin, joints, vessels, or glomeruli, and trigger the classical complement pathway. 14-Dec-2020

https://www.ncbi.nlm.nih.gov › books

Type III Hypersensitivity Reaction - StatPearls - NCBI Bookshelf

About featured snippets Feedback

People also ask

What is an example of type 3 hypersensitivity?

What is the difference between Type 2 and Type 3 hypersensitivity?

23:07 24.0 KB/S 51%

type 3 of hypersensitivity

All Images Videos News Books Sea

In **type** III **hypersensitivity** reaction, an abnormal immune response is mediated by the formation of antigen-antibody aggregates called "immune complexes." They can precipitate in various tissues such as skin, joints, vessels, or glomeruli, and trigger the classical complement pathway. 14-Dec-2020

https://www.ncbi.nlm.nih.gov › books

Type III Hypersensitivity Reaction - StatPearls - NCBI Bookshelf

About featured snippets Feedback

People also ask

What is an example of type 3 hypersensitivity?

Examples of type III hypersensitivity reactions include drug-induced serum sickness, farmer's lung and systemic lupus erythematosus. 15-May-2013

https://onlinelibrary.wiley.com › abs

Hypersensitivity: Immune Complex Mediated (Type III) - Eggleton ...

23:07 18.0 KB/S 51%
google.com/search?cli
Google
infection disease by contact
All News Videos Images Books

Infectious diseases commonly spread through the direct transfer of bacteria, viruses or other germs from one person to another. This can happen when an individual with the bacterium or virus touches, kisses, or coughs or sneezes on someone who isn't **infected**. 17-Jul-2019

https://www.mayoclinic.org › syc-2...

Infectious diseases - Symptoms and causes - Mayo Clinic

About featured snippets Feedback

People also ask

Which disease spread through contact?

What are the 4 types of infectious diseases?

What are the 5 most common infectious diseases?

23:08 5.00 KB/S 51%
google.com/search?cli
Google
infection disease by vehicle
All News Images Videos Books

Vehicle transmission occurs when a substance, such as soil, water, or air, carries an **infectious** agent to a new host. Healthcare-associated **infections** (HAI), or nosocomial **infections**, are acquired in a clinical setting.



https://courses.lumenlearning.com › ...

Modes of Disease Transmission | Microbiology - Lumen Learning

About featured snippets Feedback

People also ask

What is the most common vehicle for disease transmission?

What are the 4 types of infectious diseases?

23:08 17.0 KB/S 51%
google.com/search?cli
5
Google
infection disease by vector
All Images News Videos Books

Vector-Borne **Disease**: **Disease** that results from an **infection** transmitted to humans and other animals by blood-feeding anthropods, such as mosquitoes, ticks, and fleas. Examples of **vector**-borne **diseases** include Dengue fever, West Nile Virus, Lyme **disease**, and malaria.

http://publichealth.lacounty.gov › acd

Vector-Borne Disease - Department of Public Health - Acute ...

About featured snippets Feedback

People also ask

What are the 4 major disease vectors?

What is the most common vector for human infection?

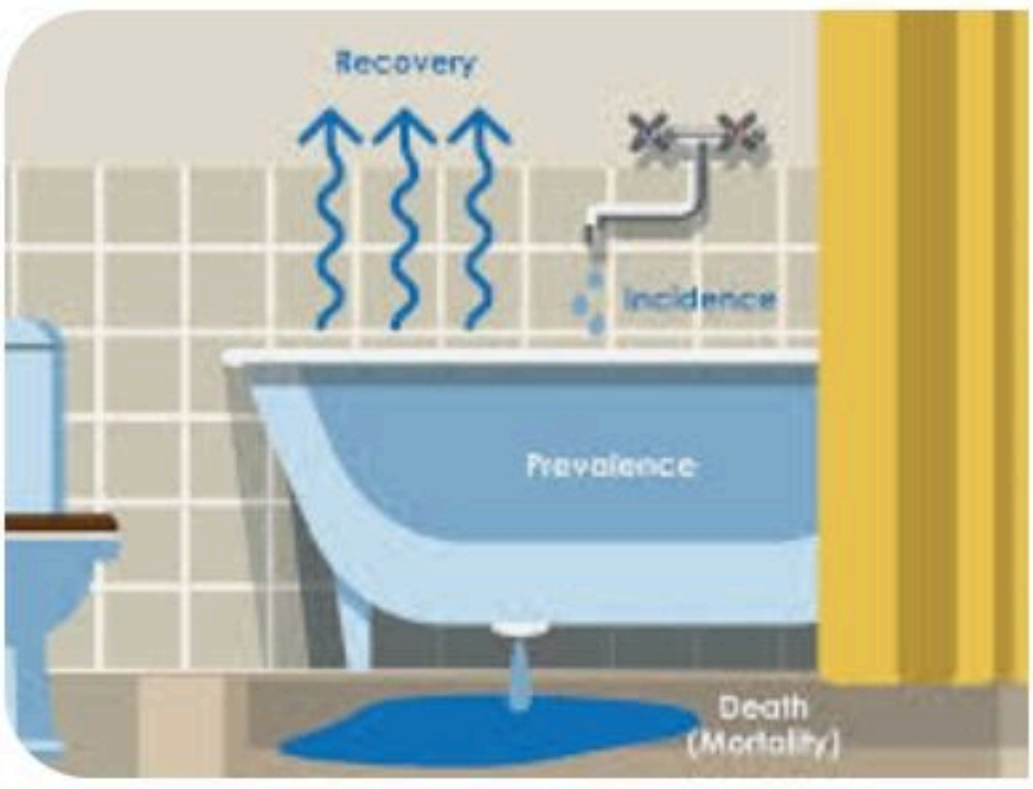
23:09 30.0 KB/S 51%
google.com/search?cli
5

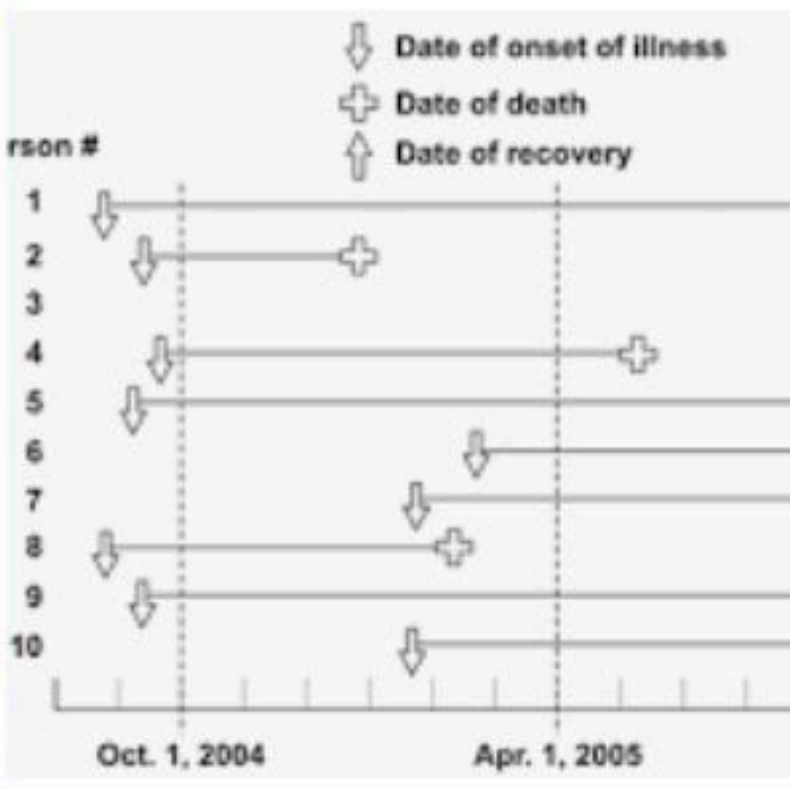
Scholarly articles for **incidence and prevalence**

Incidence and **prevalence** - Sauder - Cited by 441

Incidence and **prevalence** of idiopathic pulmonary ... - Raghu - Cited by 1264

Incidence and **prevalence** of hyponatremia - Upadhyay - Cited by 745





Prevalence and **incidence** are frequently confused. **Prevalence** refers to proportion of persons who have a condition at or during a particular time period, whereas **incidence** refers to the proportion or rate of persons who develop a condition during a particular time period.

CDC https://www.cdc.gov › csels › dsepd

Principles of Epidemiology | Lesson 3 - Section 2 - CDC

About featured snippets Feedback



SUGGESTED CLIP 57 seconds

YouTube · app

Incidence and Prevalence - Everything you need to know ...

Open on youtube.com

More results

What is an example of prevalence?

So, it can just be thought of as a wide "point in time". **Example**: During 1980 the Framingham Het Study examined 2,477 subjects for cataracts and found that 310 had them. So, the **prevalence** was 310/2,477 = 0.125. This can conveniently be expressed as 12.5 per 100 or 12.5% (per cent means 'per hundred').

https://sphweb.bumc.bu.edu › otlt

Prevalence - SPH - Boston University

More results

What prevalence means?

23:09 2.00 KB/S 51%

What is an example of prevalence?

So, it can just be thought of as a wide "point in time". **Example**: During 1980 the Framingham Het Study examined 2,477 subjects for cataracts and found that 310 had them. So, the **prevalence** was 310/2,477 = 0.125. This can conveniently be expressed as 12.5 per 100 or 12.5% (per cent means 'per hundred').

https://sphweb.bumc.bu.edu › otlt

Prevalence - SPH - Boston University

More results

What prevalence means?

Why is incidence and prevalence important?

The **prevalence** reflects the number of existing cases of a disease. In contrast to the **prevalence**, the **incidence** reflects the number of new cases of disease and can be reported as a risk or as an **incidence** rate. **Prevalence** and **incidence** are used for different purposes and to answer different research questions.
19-Feb-2010

https://www.karger.com › Fulltext

Measures of Disease Frequency: Prevalence and Incidence ...

23:10 50.0 KB/S 50%

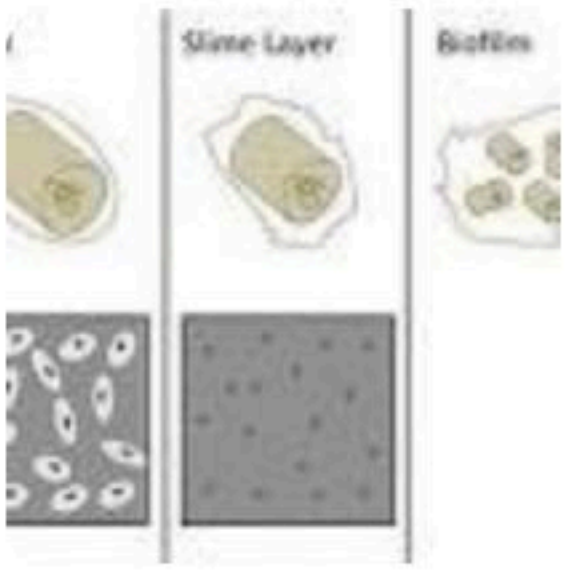
google.com/search?cli

Google

name the bacteria which form ca

All Images News Videos Books Sea

The capsule is found most commonly among gram-negative bacteria:



- Escherichia coli (in some strains)
- Neisseria meningitidis.
- Klebsiella pneumoniae.
- Haemophilus influenzae.
- Pseudomonas aeruginosa.
- Salmonella.

https://en.m.wikipedia.org › wiki

Bacterial capsule - Wikipedia

About featured snippets Feedback

People also ask

What are capsulated bacteria?

Can all bacteria form capsules?

23:10 12.0 KB/S 50%

strains)

- Neisseria meningitidis.
- Klebsiella pneumoniae.
- Haemophilus influenzae.
- Pseudomonas aeruginosa.
- Salmonella.

https://en.m.wikipedia.org › wiki

Bacterial capsule - Wikipedia

About featured snippets Feedback

People also ask

What are capsulated bacteria?

Capsule (also known as K antigen) is a major virulence factor of **bacteria**, e.g. all of the principal pathogens which cause pneumonia and meningitis, including Streptococcus pneumoniae, Haemophilus influenzae, Neisseria meningitidis, Klebsiella pneumoniae, Escherichia coli, and group B streptococci have polysaccharide ... 28-Apr-2013

https://microbeonline.com › bacteria...

Bacterial Capsule: Structure,Importance and ...

More results

Can all bacteria form capsules?

23:10 22.0 KB/S 50%
google.com/search?cli
Google
function of capsule
All Images News Videos Books Sea

31. Function of capsule, slime layer
32. Differentiate between slime layer and capsule
33. Define attenuation
34. Structures involved in adherence of Bacteria
35. Explain how dental caries develop
36. Environmental factors that influence skin microflora
37. LD_{50}
38. Phagocytosis
39. Septicemia
40. Bacteremia
41. Endotoxins vs exotoxins
42. Categories of exotoxins
43. Virulence factors of Salmonella
44. Cytotoxins? Examples
45. Enterotoxins?
46. Invasion?



It has several **functions**: promote bacterial adhesion to surfaces or interaction with other organisms; act as a permeability barrier, as a defense mechanism against phagocytosis and/or as a nutrient reserve. Among pathogens, **capsule** formation often correlates with pathogenicity.

https://www.uniprot.org › keywords

Capsule - UniProt

About featured snippets Feedback

People also ask

23:11 23.0 KB/S 50%
google.com/search?cli
5
Google
antifungal drugs
All Images Books Videos News Sea

Common names for antifungal medicines include:

- **clotrimazole**.
- **econazole**.
- **miconazole**.
- **terbinafine**.
- **fluconazole**.
- **ketoconazole**.
- **amphotericin**.

https://www.nhs.uk › conditions › a...

Antifungal medicines - NHS

About featured snippets Feedback

People also ask

What is the best oral antifungal medication?

What can naturally kill fungus?

23:11 28.0 KB/S 50%

All Images Books Videos News Sea

Common names for antifungal medicines include:

- **clotrimazole**.
- **econazole**.
- **miconazole**.
- **terbinafine**.
- **fluconazole**.
- **ketoconazole**.
- **amphotericin**.

https://www.nhs.uk › conditions › a...

Antifungal medicines - NHS

About featured snippets Feedback

People also ask

What is the best oral antifungal medication?

Oral **antifungal drugs** include **fluconazole** and **itraconazole**. These drugs can be used to treat more severe fungal **infections**. Terbinafine is an oral antifungal medicine that may be used to treat fungal **infections** of the nails.

https://www.webmd.com › how-are-...

How are oral antifungal agents used to treat skin conditions? -

23:11 20.0 KB/S 50%
google.com/search?cli
Google
inflammation and its consequenc
All Images News Videos Books Sea

When you have chronic **inflammation**, your body's **inflammatory** response can eventually start damaging healthy cells, tissues, and organs. Over time, this can lead to DNA damage, tissue death, and internal scarring. All of these are linked to **the** development of several diseases, including: cancer. 27-Jul-2018

https://www.healthline.com › health

Chronic Inflammation: Definition, Symptoms, Causes, and Treatment

About featured snippets Feedback

People also ask

What are the consequences of inflammation?

What happens to the body during inflammation?

What are 4 types of inflammation?

23:11 2.00 KB/S 50%
google.com/search?cli 5

When you have chronic **inflammation**, your body's **inflammatory** response can eventually start damaging healthy cells, tissues, and organs. Over time, this can lead to DNA damage, tissue death, and internal scarring. All of these are linked to **the** development of several diseases, including: cancer. 27-Jul-2018

https://www.healthline.com › health

Chronic Inflammation: Definition, Symptoms, Causes, and Treatment

About featured snippets Feedback

People also ask

What are the consequences of inflammation?

Over time, chronic inflammation can trigger your immune system to attack healthy tissue and organs in your body. When left untreated, prolonged chronic inflammation can increase your risk for diseases like diabetes, heart disease, **cancer** and rheumatoid arthritis.

https://www.elcaminohealth.org › ca...

The Causes and Effects of Inflammation | El Camino Health

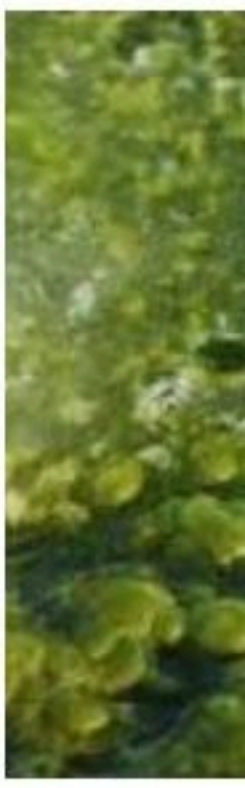
23:12 32.0 KB/S 50%
google.com/search?cli
5
All Images News Videos Books Sea

Did you mean: ***algae***

Algae

Overview | Representative species | Peop



Algae is an informal term for a large and diverse group of photosynthetic eukaryotic organisms. It is a polyphyletic grouping that includes species from multiple distinct clades. Wikipedia

Feeding

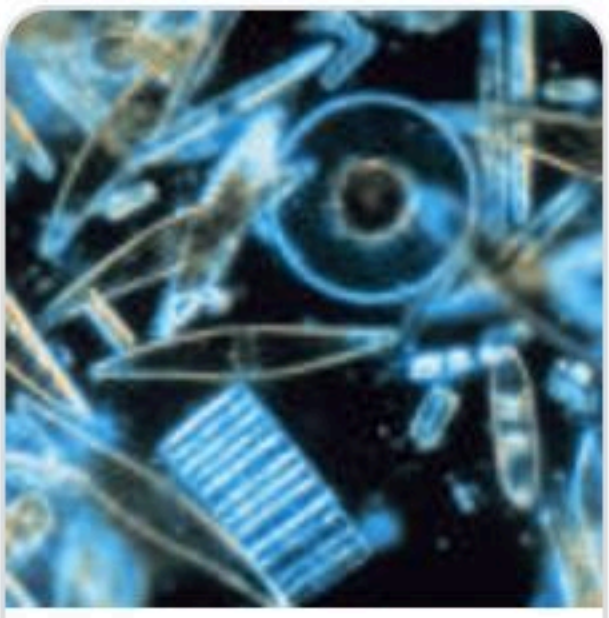
Energy source

Dictionary

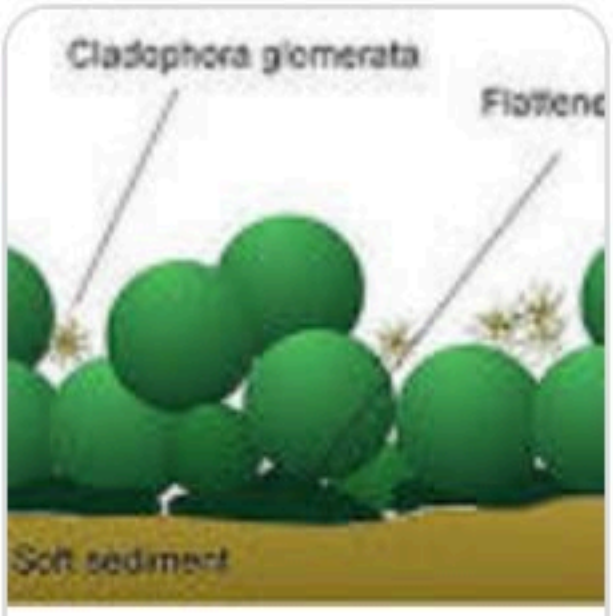
Domain

23:12 2.00 KB/S Vo LTE2 4G 50%

Representative species



Diatom



Marimo



Wakame

More about Algae

People also ask

Is algae bacteria or plant?

What algae means?

Algae are a diverse group of aquatic organisms that have the ability to conduct photosynthesis. Certain **algae** are familiar to most people; for instance, seaweeds (such as kelp or phytoplankton), pond scum or the **algal** blooms in lakes. 04-Jun-2016

https://www.livescience.com › 5497...

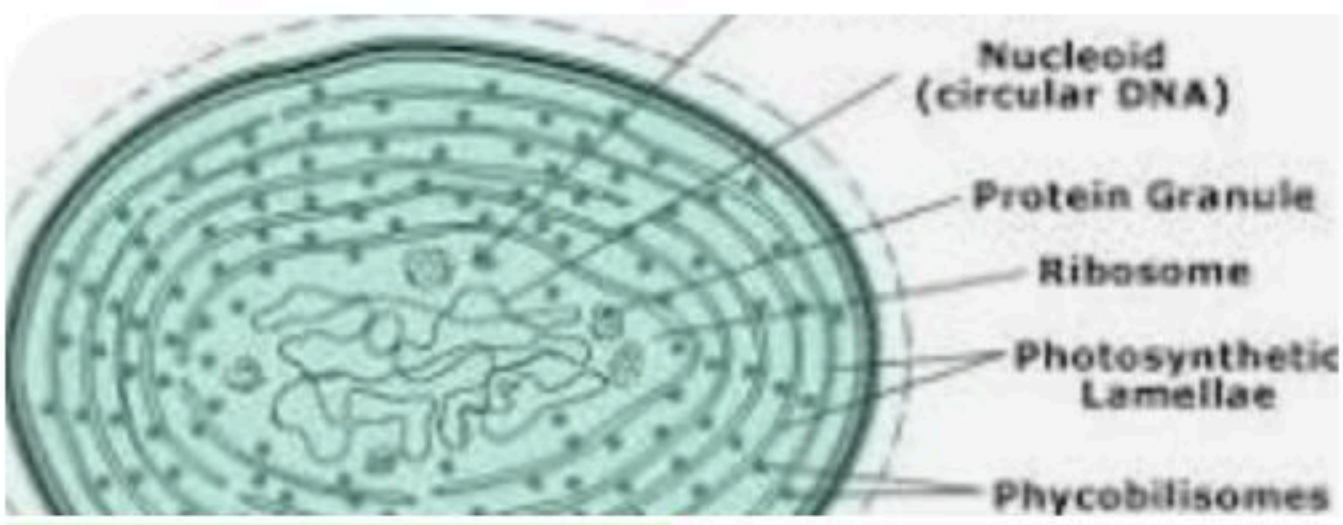
What Are Algae? | Live Science

More results

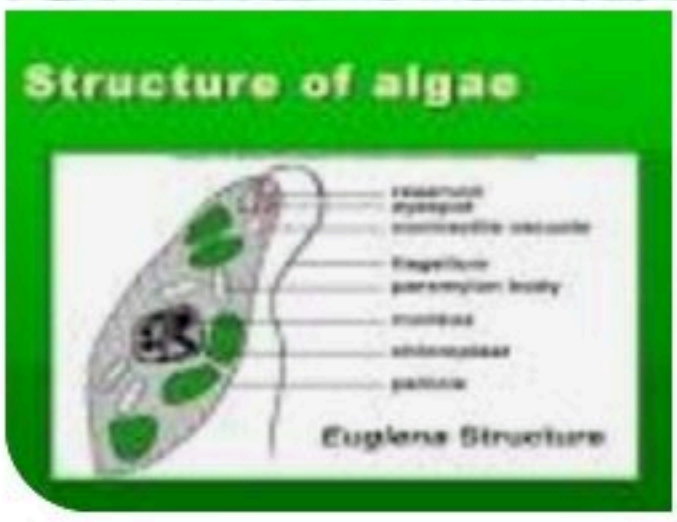
23:12 60.0 KB/S 50%
google.com/search?cli 5

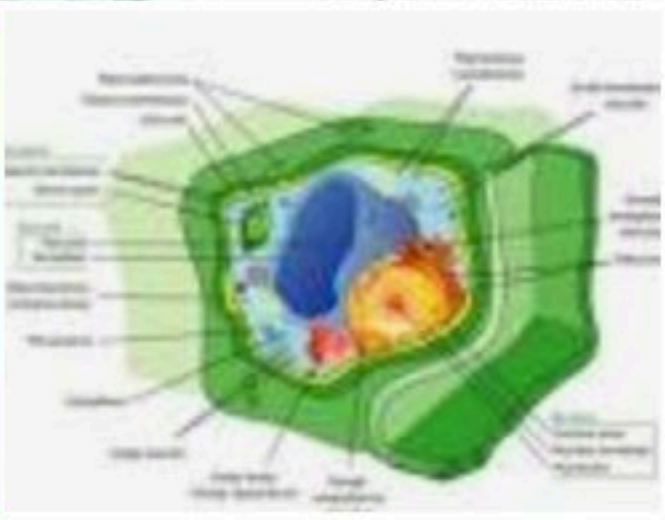
Search instead for algea structure

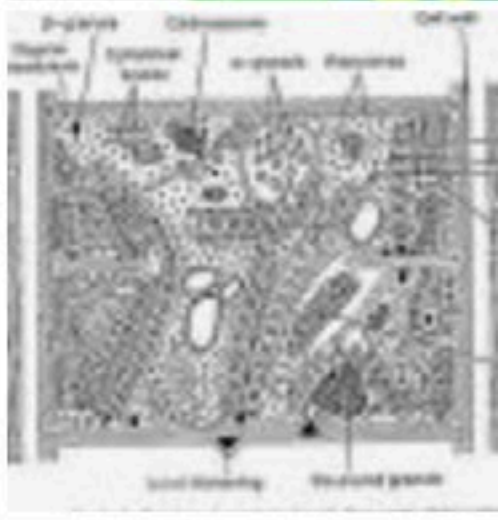
Structure of algae











Algal Cell **Structure**

Algae are eukaryotic cells, or cells that contain a nucleus, which makes them slightly more complex than bacteria. ... Some **algae** have silica exoskeletons, flagella for movement or other **structures**. The pigment used for photosynthesis can even vary, resulting in **algae** that appear green, red, or brown.

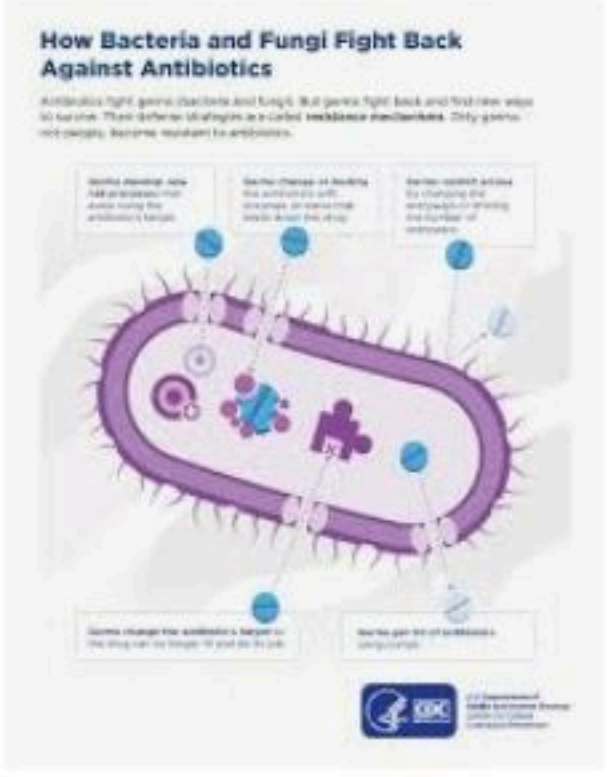
https://sciencing.com › the-structure...

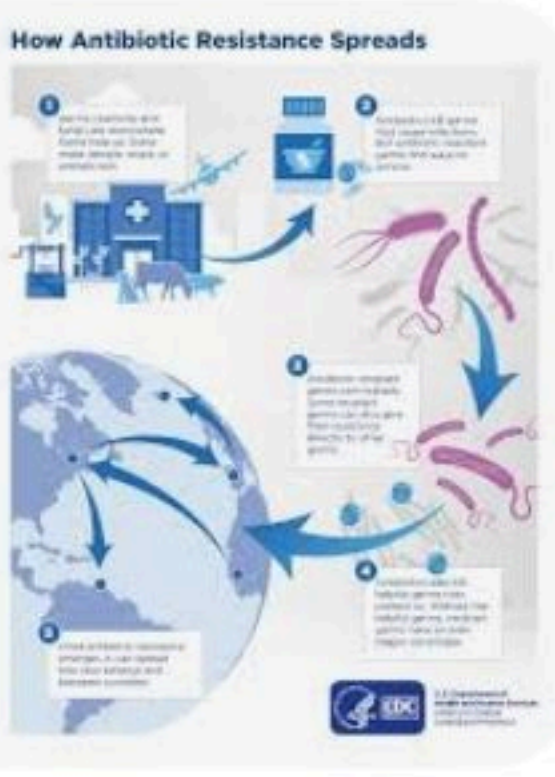
The Structure of Algae - Sciencing

About featured snippets Feedback

23:13 20.0 KB/S 50%
google.com/search?cli
Google
how microbes become resistant
All Videos News Images Books Sea







Bacteria develop **resistance** mechanisms by using instructions provided by their DNA. Often, **resistance** genes are found within plasmids, small pieces of DNA that carry genetic instructions from one germ to another. This means that some **bacteria** can share their DNA and make other germs **become resistant**. 10-Feb-2020

CDC https://www.cdc.gov › about › how-...

How Antibiotic Resistance Happens | Antibiotic/Antimicrobial ... - CDC

About featured snippets Feedback

23:14 45.0 KB/S 50%

Google

why haptin is a week antigen

All Images News Videos Books Sea

Did you mean: why haptin is a ***weak*** antigen

https://www.sciencedirect.com › topics

Hapten - an overview | ScienceDirect Topics

Antigens for B Cells. >> Immunogens contain. Epitopes that bind to the antigen-binding sites of antibodies. Class II epitopes ...

People also ask

Are haptens Antigenic?

A **hapten** is essentially an incomplete **antigen**. These small molecules can elicit an immune response only when attached to a large carrier such as a protein; the carrier typically does not elicit an immune response by itself.

24-Nov-2016

https://www.creative-diagnostics.com › ...

Immunogen, Antigen, Hapten, Epitope, and Adjuvant

More results

23:15 3.00 KB/S 50%

https://www.ncbi.nlm.nih.gov › pmc

The effect of haptens on protein-carrier immunogenicity

by T Gefen · 2015 · Cited by 39 · Related articles

Two weeks after each injection, blood was drawn and sera were separated and kept at − 20° until analysis. ... The presence of antibodies against carrier protein or hapten in mouse sera ...

Abstract | Introduction | Materials and methods

People also ask

What is the hapten carrier effect?

What is the difference between antigen and hapten?

What does a hapten do?

Are haptens immunogenic?

Hapten is a molecule that reacts with specific antibody but is not **immunogenic** by itself, it can be made **immunogenic** by conjugation to a suitable carrier. Many drugs like penicillins **are haptens**. A **hapten** is essentially an incomplete antigen. 24-Nov-2016

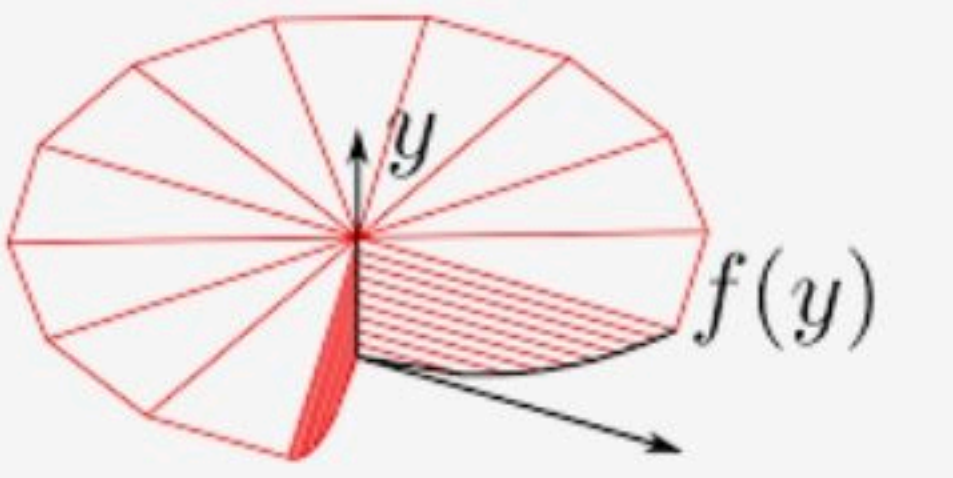
CD https://www.creative-diagnostics.com › ...

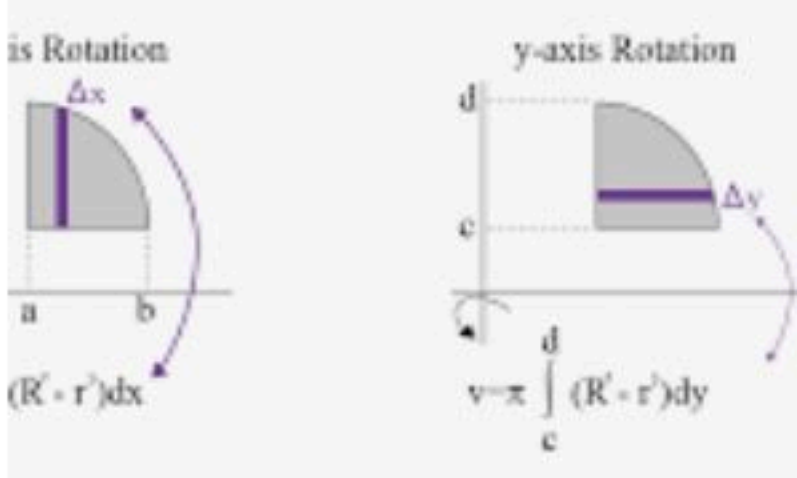
Immunogen, Antigen, Hapten, Epitope, and Adjuvant

23:15 33.0 KB/S 50%
google.com/search?cli
5
Google
disc methods
All Images Videos News Books Sea

Disc integration

Overview Formula





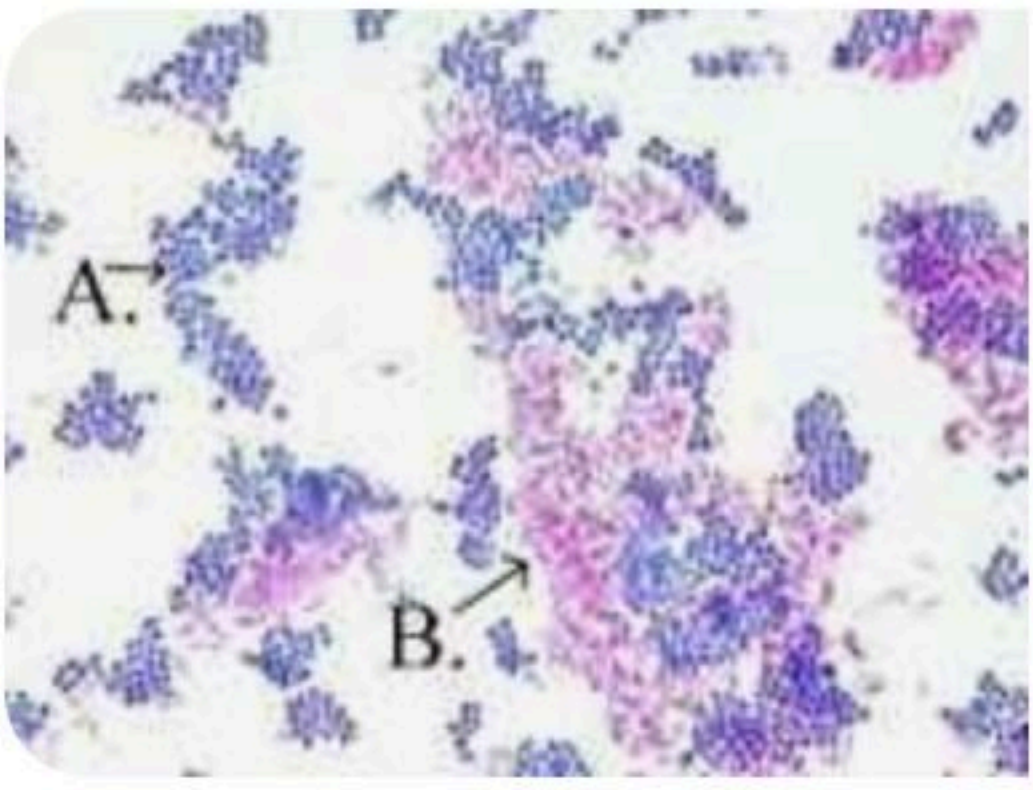
Disc integration, also known in integral calculus as the **disc method**, is a **method** for calculating the volume of a solid of revolution of a solid-state material when integrating along an axis "parallel" to the axis of revolution.

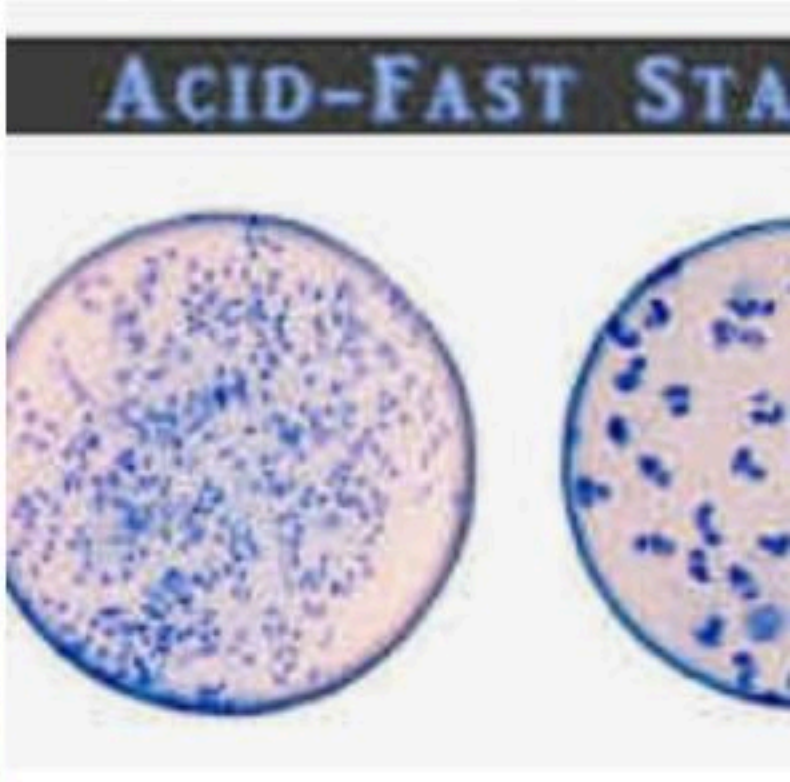
https://en.m.wikipedia.org › wiki

Disc integration - Wikipedia

Formula

23:15 27.0 KB/S 50%
google.com/search?cli
5
Google
acid fast staining
All Images Books Videos News Sea





Acid fast stains are used to differentiate **acid fast** organisms such mycobacteria. **Acid fast** bacteria have a high content of mycolic **acids** in their cell walls. **Acid fast** bacteria will be red, while nonacid **fast** bacteria will **stain** blue/green with the counterstain with the Kinyoun **stain**.

https://www.sciencedirect.com › topics

Acid-Fast - an overview | ScienceDirect Topics

About featured snippets Feedback

People also ask

23:16 4.00 KB/S 49%
google.com/search?cli
Google
how disease can be transmitted
All Images News Videos Books

Infectious **diseases** are **transmitted** from person to person by direct or indirect contact. Certain **types of** viruses, bacteria, parasites, and fungi **can all** cause infectious **disease**. Malaria, measles, and respiratory **illnesses** are **examples of** infectious **diseases**.

https://www.healthline.com › health

Disease Transmission: Direct Contact vs. Indirect Contact

About featured snippets Feedback

People also ask

How can diseases be transmitted?

What are the 4 types of disease transmission?

What are the 5 methods of disease transmission?

23:17 22.0 KB/S 49%

WORD COACH

Learn new words

Which word is *similar* to mortality?

Transience

or

Trace

SKIP

Feedback

People also ask

What is definition of mortality?

(mor-TA-lih-tee) Refers to the state of being mortal (destined to die). In medicine, a term also used for death rate, or the number of deaths in a certain group of people in a certain period of time.

https://www.cancer.gov › def › mort...

Definition of mortality - NCI Dictionary of Cancer Terms ...

More results

What is mortality and example?

23:17 5.00 KB/S 49%
google.com/search?cli
5
Feedback

People also ask

What is definition of mortality?

(mor-TA-lih-tee) Refers to the state of being mortal (destined to die). In medicine, a term also used for death rate, or the number of deaths in a certain group of people in a certain period of time.

https://www.cancer.gov › def › mort...

Definition of mortality - NCI Dictionary of Cancer Terms ...

More results

What is mortality and example?

Mortality is the condition of one day having to die or the rate of failure or loss. An **example** of **mortality** is that all animals eventually will die. An **example** of **mortality** is the number of high school students who don't graduate. noun.

https://www.yourdictionary.com › ...

Mortality Meaning | Best 13 Definitions of Mortality

23:17 41.0 KB/S 49%

Dictionary

Search for a word

pathogenicity

/ˌpaθədʒəˈnɪsɪti/

noun MEDICINE

the property of causing disease.
"tests on the pathogenicity of the virus"

Definitions from Oxford Languages *Feedback*

Translations and more definitions

People also ask

What is pathogenicity in biology?

Pathogenicity refers to the ability of an organism to cause disease (ie, harm the host). This ability represents a genetic component of the pathogen and the overt damage done to the host is a property of the host-pathogen interactions. Commensals and opportunistic pathogens lack this inherent ability to cause disease.

https://www.tulane.edu › notes › Path

Pathogenicity vs Virulence

More results

What Does pathogenicity and pathogen mean?

23:18 9.00 KB/S 49%

/mɔːˈbɪdɪti/

noun MEDICINE

the condition of suffering from a disease or medical condition.
"the therapy can substantially reduce respiratory morbidity in infants"

- the rate of disease in a population.
  "the levels of air pollution are associated with increased morbidity from respiratory diseases"

Definitions from Oxford Languages *Feedback*

Translations and more definitions

People also ask

What is difference between mortality and morbidity?

Morbidity and **mortality** are two terms that often get confused. **Morbidity** refers to disease states, while **mortality** refers to death. Both terms are commonly used in health- and death-related statistics.

https://www.verywellhealth.com › w...

Morbidity vs. Mortality: What's the Difference? - Verywell Health

More results

What is morbidity rate mean?

23:20 44.0 KB/S 49%

google.com/search?cli

incidence and prevalence

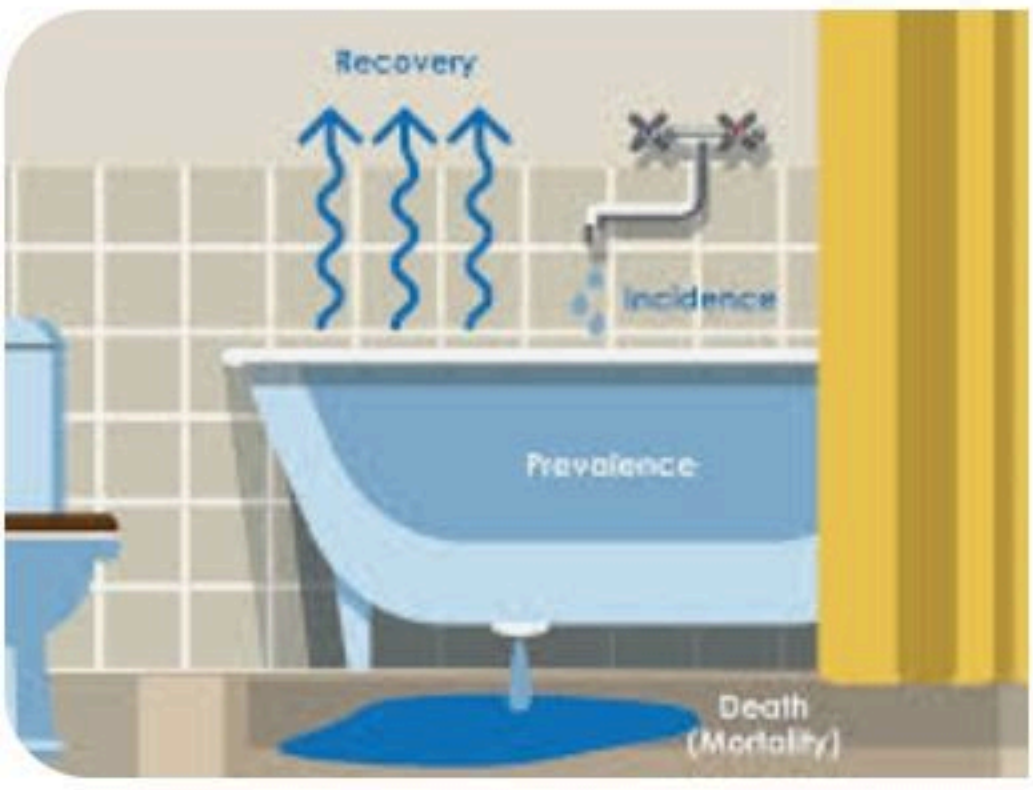
All Images Videos News Books Sea

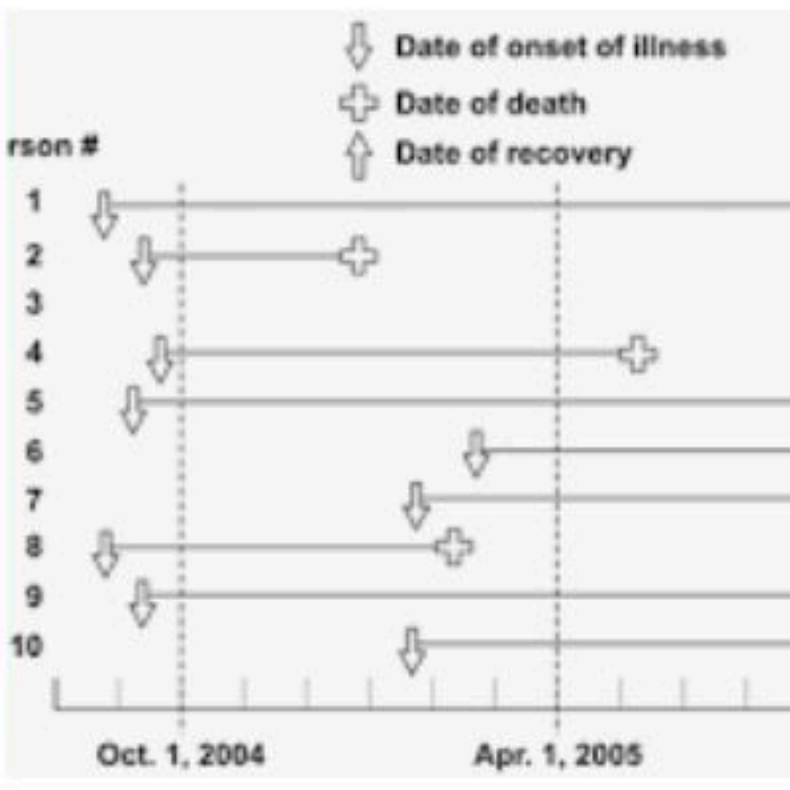
Scholarly articles for **incidence and prevalence**

Incidence and **prevalence** - Sauder - Cited by 441

Incidence and **prevalence** of idiopathic pulmonary ... - Raghu - Cited by 1264

Incidence and **prevalence** of hyponatremia - Upadhyay - Cited by 745



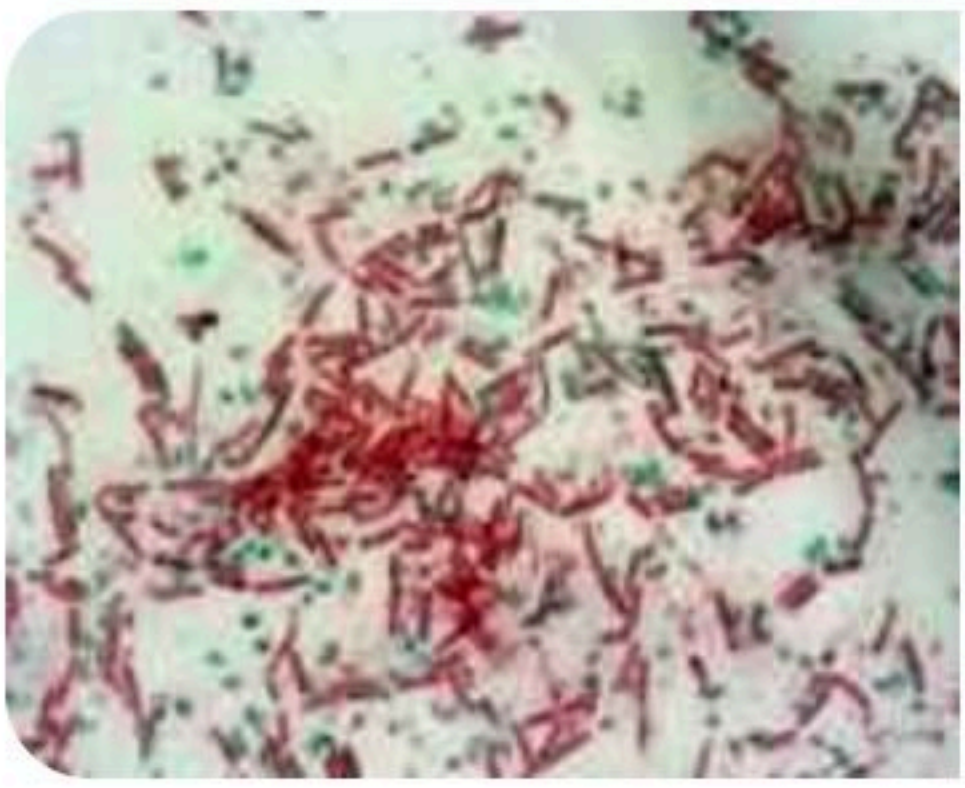


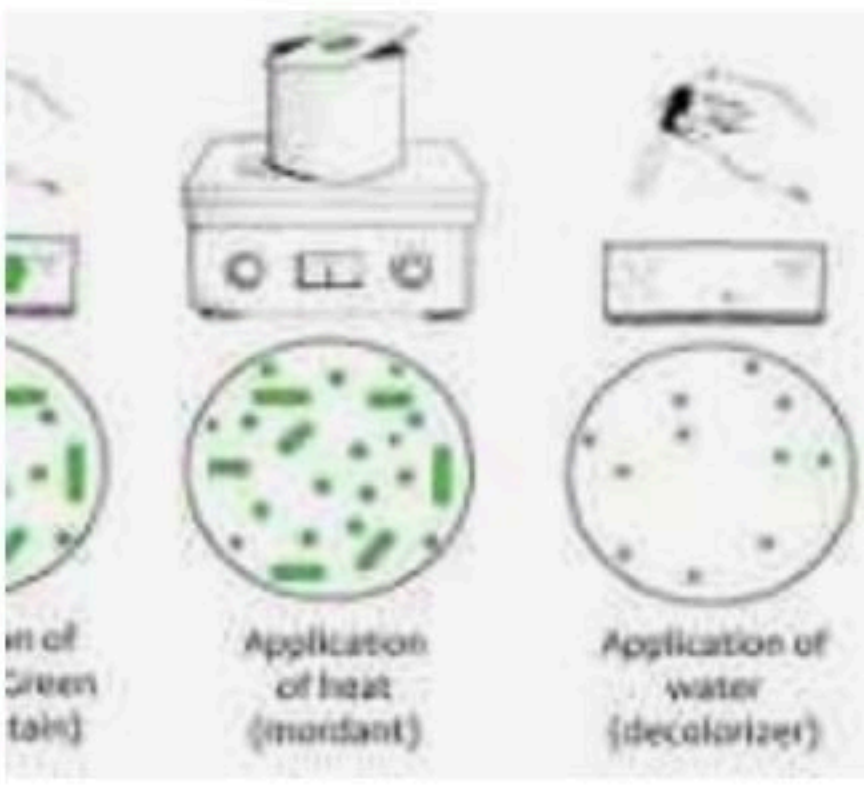
Prevalence and **incidence** are frequently confused. **Prevalence** refers to proportion of persons who have a condition at or during a particular time period, whereas **incidence** refers to the proportion or rate of persons who develop a condition during a particular time period.

CDC https://www.cdc.gov › csels › dsepd

23:21 23.0 KB/S 49%
google.com/search?cli
5
Google
spore staining
All Images Books Videos News Sea

Endospore staining





Endospore Staining is a technique used in bacteriology to identify the presence of endospores in a bacterial sample, which can be useful for classifying bacteria. Wikipedia

People also ask

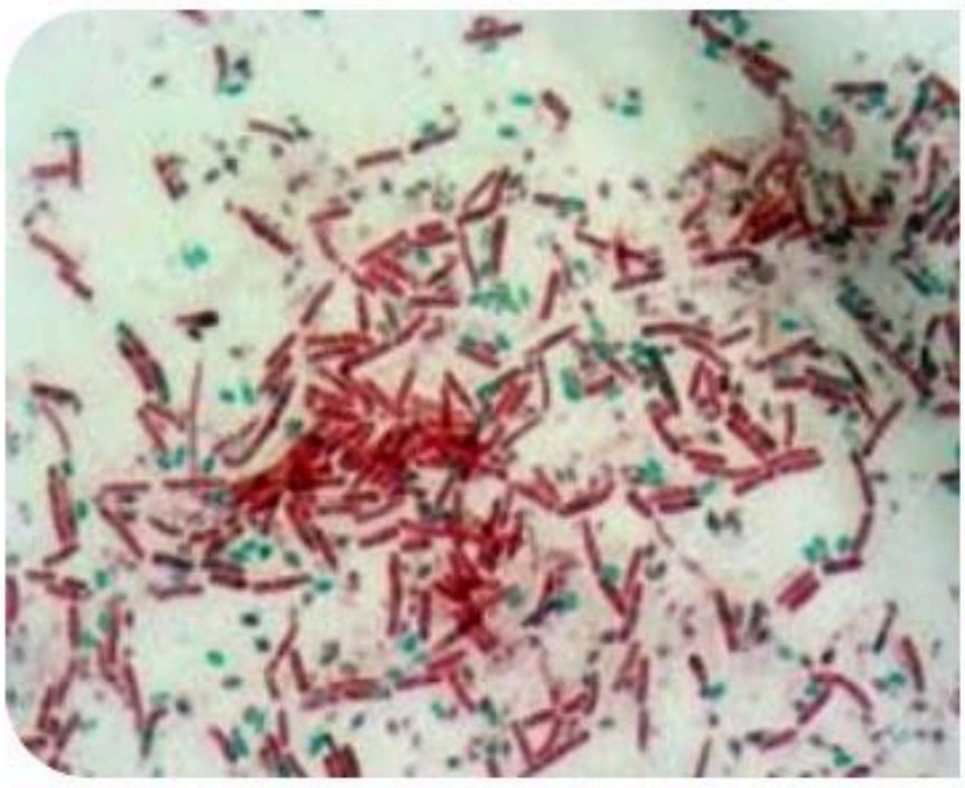
What is Spore staining?

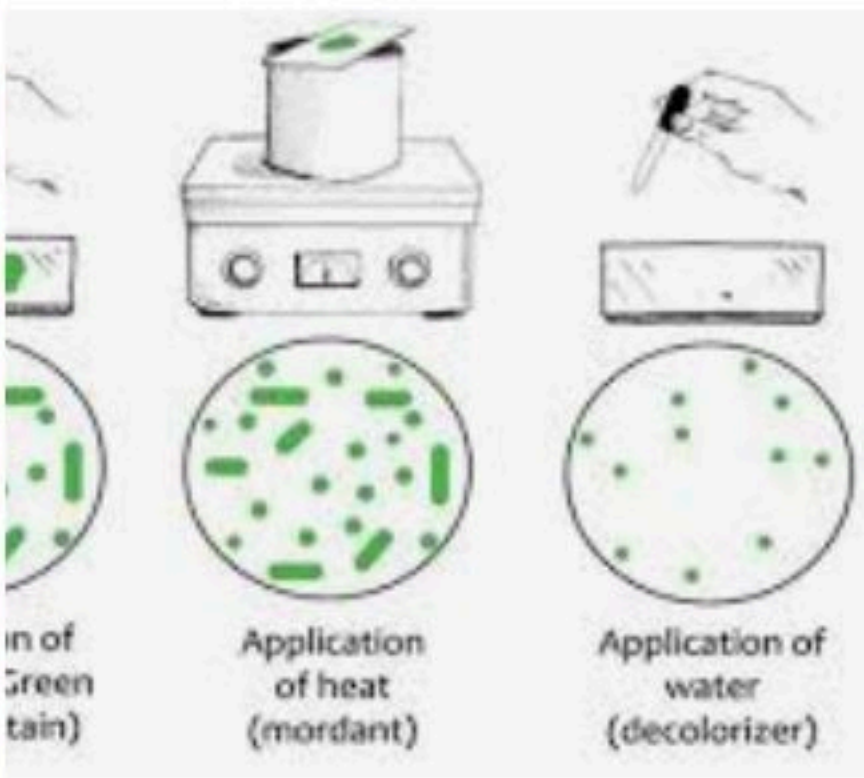
How does spore staining work?

What is the aim of spore staining?

23:21 45.0 KB/S 49%
All Images Books Videos News

Endospore staining





Endospore Staining is a technique used in bacteriology to identify the presence of endospores in a bacterial sample, which can be useful for classifying bacteria. Wikipedia

People also ask

What is Spore staining?

The **endospore stain** is a differential **stain** used to visualize bacterial endospores. Endospores are formed by a few genera of bacteria, such as Bacillus . By forming **spores**, bacteria can survive in hostile conditions. ... Endospores can form within different areas of the vegetative cell.

https://www.austincc.edu › microbugz

Welcome to Microbugz - Endospore Stain

23:21 22.0 KB/S 49%
google.com/search?cli
5
Google
microorganisms affecting body
All Images Videos News Books Sea

Scholarly articles for **microorganisms affecting body**

... and their regulation in oleaginous **microorganisms** - Dourou - Cited by 65

... detection of nucleic acids and **microorganisms** in **body** ... - Cheng - Cited by 7

Body weight selection **affects** quantitative genetic ... - Meng - Cited by 56

Impact of microbes on human life

Microbes cause infectious diseases such as flu and measles. There is also strong evidence that **microbes** may contribute to many non–infectious chronic diseases such as some forms of cancer and coronary heart disease. Different diseases are caused by different types of **micro-organisms**.

https://microbiologysociety.org › mi...

Microbes and disease | Microbes and the human body ...

impact of microbes on human life

23:21 2.00 KB/S 49%

Microbes cause infectious diseases such as flu and measles. There is also strong evidence that **microbes** may contribute to many non–infectious chronic diseases such as some forms of cancer and coronary heart disease. Different diseases are caused by different types of **micro-organisms**.

https://microbiologysociety.org › mi...

Microbes and disease | Microbes and the human body ...

impact of microbes on human life

About featured snippets Feedback

People also ask

What type of microorganism can affect our health?

Most microbes belong to four major groups: **bacteria**, viruses, **protozoa** or **fungi**. (To find out more, see the "**Bacteria**/Viruses/**Protozoa**" fact sheets). Disease-causing microbes can also be called pathogens, germs or bugs and are responsible for causing infectious diseases. 23-Jan-2017

https://www.safewater.org › disease-...

Disease Causing Micro-organisms — Safe Drinking Water Foundation

23:22 21.0 KB/S 49%
google.com/search?cli
Google
protein inhibiting antibiotics
All Images News Videos Books Sea

Scholarly articles for **protein inhibiting antibiotics**

... mode of translation **inhibition** by macrolide **antibiotics** - Kannan - Cited by 126

... concentrations of **antibiotics** that **inhibit protein** ... - Lin - Cited by 123

Toxicity of five **protein** synthesis **inhibiting antibiotics** ... - Brosche - Cited by 33

Antibiotics can **inhibit protein** synthesis by targeting either the 30S subunit, examples of which include spectinomycin, tetracycline, and the aminoglycosides kanamycin and streptomycin, or to the 50S subunit, examples of which include clindamycin, chloramphenicol, linezolid, and the macrolides erythromycin, ... 14-Aug-2016

https://link.springer.com › chapter

Antibiotics That Inhibit Protein Synthesis | SpringerLink

About featured snippets Feedback

23:22 2.00 KB/S 49%

examples of which include spectinomycin, tetracycline, and the aminoglycosides kanamycin and streptomycin, or to the 50S subunit, examples of which include clindamycin, chloramphenicol, linezolid, and the macrolides erythromycin, ... 14-Aug-2016

https://link.springer.com › chapter

Antibiotics That Inhibit Protein Synthesis | SpringerLink

About featured snippets Feedback

People also ask

What antibiotics work by inhibiting protein synthesis?

Which class of drugs inhibit protein synthesis?

Bacteriostatic protein-synthesis inhibitors that target the ribosome, such as tetracyclines and gly-cylcyclines, chloramphenicol, **macrolides** and ketolides, **lincosamides** (clindamycin), streptogramins (quinupristin/dalfopristin), oxazolidinones (linezolid), and aminocyclitols (spectinomycin).

https://accesspharmacy.mhmedical.com › ...

Protein Synthesis Inhibitors and Miscellaneous Antibacterial ...

23:22 5.00 KB/S 49%

People also ask

What antibiotics work by inhibiting protein synthesis?

Which class of drugs inhibit protein synthesis?

Bacteriostatic protein-synthesis inhibitors that target the ribosome, such as tetracyclines and gly-cylcyclines, chloramphenicol, **macrolides** and ketolides, **lincosamides** (clindamycin), streptogramins (quinupristin/dalfopristin), oxazolidinones (linezolid), and aminocyclitols (spectinomycin).

https://accesspharmacy.mhmedical.com › ...

Protein Synthesis Inhibitors and Miscellaneous Antibacterial ...

More results

Is Penicillin a protein synthesis inhibitor?

Penicillins have been shown to **inhibit** bacterial cell wall **synthesis**, and interact with **penicillin** binding **proteins**, leading to bacterial lysis. ... As per the present understanding **penicillin** acts principally by **inhibiting** cell wall **synthesis**.

https://pubmed.ncbi.nlm.nih.gov › ...

Inhibition of cell wall synthesis--is this the mechanism of action of ...

23:23 0.00 KB/S 48%
google.com/search?cli
Google
hypersensitivity diagnosis
All Images Videos News Books

Diagnosis of hypersensitivity

The **diagnostic** evaluation of **hypersensitivity** reactions consists of thorough history-taking, in vitro laboratory testing, and in vivo cutaneous tests and provocative tests (17). 23-Jul-2018

https://www.aerzteblatt.de › article

Drug Hypersensitivity: Diagnosis, Genetics, and Prevention (23.07 ...

diagnosis of hypersensitivity

About featured snippets Feedback

People also ask

How do you test for hypersensitivity?

What are the 4 types of hypersensitivity?

What are hypersensitivity diseases?

23:23 2.00 KB/S 48%

Diagnosis of hypersensitivity

The **diagnostic** evaluation of **hypersensitivity** reactions consists of thorough history-taking, in vitro laboratory testing, and in vivo cutaneous tests and provocative tests (17). 23-Jul-2018

https://www.aerzteblatt.de › article

Drug Hypersensitivity: Diagnosis, Genetics, and Prevention (23.07 ...

diagnosis of hypersensitivity

About featured snippets Feedback

People also ask

How do you test for hypersensitivity?

Delayed **hypersensitivity** skin and patch **tests** are the easiest methods of **testing** for type IV delayed **hypersensitivity**. A concentration of the suspected allergen is applied to the skin under a nonabsorbent adhesive patch and left for 48 hours. If burning or itching develops more rapidly, the patch is removed.
31-Aug-2018

https://labtestsonline.org › conditions

Allergies | Lab Tests Online

More results

23:23 7.00 KB/S 48%

People also ask

How do you test for hypersensitivity?

Delayed **hypersensitivity** skin and patch **tests** are the easiest methods of **testing** for type IV delayed **hypersensitivity**. A concentration of the suspected allergen is applied to the skin under a nonabsorbent adhesive patch and left for 48 hours. If burning or itching develops more rapidly, the patch is removed.

31-Aug-2018

https://labtestsonline.org › conditions

Allergies | Lab Tests Online

More results

What are the 4 types of hypersensitivity?

The four types of hypersensitivity are:

- **Type** I: reaction mediated by IgE antibodies.
- **Type** II: cytotoxic reaction mediated by IgG or IgM antibodies.
- **Type** III: reaction mediated by immune complexes.
- **Type** IV: delayed reaction mediated by cellular response.

10-Sep-2020

https://www.ncbi.nlm.nih.gov › books

23:23 9.00 KB/S 48%

- **Type** II: cytotoxic reaction mediated by IgG or IgM antibodies.
- **Type** III: reaction mediated by immune complexes.
- **Type** IV: delayed reaction mediated by cellular response.

10-Sep-2020

https://www.ncbi.nlm.nih.gov › books

Type IV Hypersensitivity Reaction - StatPearls - NCBI Bookshelf

More results

What are hypersensitivity diseases?

Summary. **Hypersensitivity diseases** reflect normal immune mechanisms directed against innocuous antigens. They can be mediated by IgG antibodies bound to modified cell surfaces, or by complexes of antibodies bound to poorly catabolized antigens, as occurs in serum sickness.

https://www.ncbi.nlm.nih.gov › books

Hypersensitivity diseases - Immunobiology - NCBI Bookshelf

More results

23:23 53.0 KB/S 48%

https://www.webmd.com › features

What Is a Pandemic? - WebMD

"A pandemic is basically a global epidemic -- an epidemic that spreads to more than one continent," says Dan Epstein, a spokesman for the ...

⚠ COVID-19 alert

Common questions

What is the definition of a pandemic?

A pandemic is defined as “an epidemic occurring worldwide, or over a very wide area, crossing international boundaries and usually affecting a large number of people”. The classical definition includes nothing about population immunity, virology or disease severity.

By this definition, pandemics can be said to occur annually in each of the temperate southern and northern hemispheres, given that seasonal epidemics cross international boundaries and affect a large number of people. However, seasonal epidemics are not considered pandemics.

https://www.who.int › volumes

WHO | The classical definition of a pandemic is not elusive

More results

23:24 50.0 KB/S 48%

adjective

1. (of a disease or condition) regularly found among particular people or in a certain area.
   "complacency is **endemic in** industry today"
2. (of a plant or animal) native and restricted to a certain place.
   "a marsupial **endemic to** north-eastern Australia"

noun

an endemic plant or animal.
"there are three types of island endemics"

Definitions from Oxford Languages *Feedback*

Translations and more definitions

People also ask

What do you mean by endemic?

1a : belonging or native to a particular people or country. b : characteristic of or prevalent in a particular field, area, or environment problems **endemic** to translation the self-indulgence **endemic** in the film industry. 2 : restricted or peculiar to a locality or region **endemic** diseases an **endemic** species.

https://www.merriam-webster.com › ...

Endemic | Definition of Endemic by Merriam-Webster

23:24 5.00 KB/S 48%
google.com/search?cli
5
endemic
All News Books Images Videos

Dictionary

Search for a word

endemic

/ɛnˈdɛmɪk/

See definitions in:

All Medical Biology

adjective

1. (of a disease or condition) regularly found among particular people or in a certain area.
   "complacency is **endemic in** industry today"
2. (of a plant or animal) native and restricted to a certain place.
   "a marsupial **endemic to** north-eastern Australia"

noun

an endemic plant or animal.
"there are three types of island endemics"

Definitions from Oxford Languages *Feedback*

Translations and more definitions

23:24 0.71 KB/S 48%
google.com/search?cli
5

People also ask

What do you mean by endemic?

1a : belonging or native to a particular people or country. b : characteristic of or prevalent in a particular field, area, or environment problems **endemic** to translation the self-indulgence **endemic** in the film industry. 2 : restricted or peculiar to a locality or region **endemic** diseases an **endemic** species.

https://www.merriam-webster.com › ...

Endemic | Definition of Endemic by Merriam-Webster

More results

What is an example of an endemic?

Endemic: A characteristic of a particular population, environment, or region. **Examples** of **endemic** diseases include chicken pox that occurs at a predictable rate among young school children in the United States and malaria in some areas of Africa.

https://www.rxlist.com › endemic

Definition of Endemic - RxList

23:24 8.00 KB/S 48%
google.com/search?cli
5
Google
epidemic
All Images News Books Videos Sear

Dictionary

Search for a word

epidemic

/ɛpɪˈdɛmɪk/

noun

a widespread occurrence of an infectious disease in a community at a particular time.
"a flu epidemic"

Similar: outbreak plague




adjective

of the nature of an epidemic.
"epidemic diseases"

Similar: rife rampant

Definitions from Oxford Languages *Feedback*

23:24 6.00 KB/S 48%

or

Streak

SKIP

Feedback

COVID-19 alert

Common questions

What is the definition of a pandemic?

A pandemic is defined as “an epidemic occurring worldwide, or over a very wide area, crossing international boundaries and usually affecting a large number of people”. The classical definition includes nothing about population immunity, virology or disease severity.

By this definition, pandemics can be said to occur annually in each of the temperate southern and northern hemispheres, given that seasonal epidemics cross international boundaries and affect a large number of people. However, seasonal epidemics are not considered pandemics.

https://www.who.int › volumes

WHO | The classical definition of a pandemic is not elusive

More results

Define :

The definition of algae is a single or multi-cellular organism that has no roots, stems or leaves and is often found in water

These are simple eukaryotic cells. Some are unicellular, others are multicellular (thallus); however, they lack tissues such as roots, stem and leaves typically seen in plants. Algae absorb nutrients from water through their surfaces and are mostly photoautotrophs; however, a few are chemoheterotrophs. They are responsible for 80% atmospheric O2 on the face of the earth.

Types of algae:

1. **Phaeophyta (Kelp):** Commonly known as brown algae, brownish in color, contains cellulose and alginic acid cell walls, are multicellular and contain chlorophyll *a* and *c*, xanthophylls. These store carbohydrates and are used for algin which is a thickener in ice creams. Can grow upto 20 cm in one day.

2. **Rhodophyta:** Red Alage, these are reddish in color and contain cellulose in their cell walls. Most are multicellular and contain chlorophyll a and d, and phycobiliproteins. They store glucose polymer and used for agar and carrageenan (both are used as thickeners of foods).

3. **Chlorophyta:** Green Algae: have all the features of red algae except that chlorophyll is of a and b types only. These algae are believed to have given rise to terrestrial plants.

4. **Bacillariophyta:** Cell wall is composed of pectin and silica. They are unicellular and possess chlorophyll a and c, and carotene xanthophylls. These algae store oil and produce domoic acid with which human can be intoxicated by eating contaminated mussels. **Diatoms**: Come in beautiful colors and shapes.

5. **Dinoflagellata:** These are unicellular algae collectively called as plankton (free floating organisms). These cells have cellulose in their cytoplasmic membrane which gives the membrane rigidity. They have chlorophyll *a* and *c*, and carotene. These algae store starch in them. Dinoflagellates in the genus Alexandrium produce neurotoxins (called saxitoxins) which can cause paralytic shellfish poisoning in humans when they consume contaminated mussels and clams. Blooms of Alexndrium gives ocean a deep red color commonly known as red tide.

6. **Oomycota (Water molds):** They are basically decomposers of dead matter. Superficially, these resemble with fungi; however, these are motile with flagella and their cell wall is composed of cellulose not chitin. Their spores are called zoospores. Terrestrial members of water molds are also plant parasite. *Phytophthora infestans* was responsible for Irish potato blight during mid-1800s. *P. ramorum* causes sudden oak death

2. HYPERSENSITIVITY AND ITS TYPE 5 marks

What is hypersensitivity?

It is a normal Immune response but in a damaging way. In other words, immune system is acting in a way for which it has made to. So when an antigen gets into the body, immune system recognizes it as foreign and start making antibodies against the antigen. However, the consequences are bad and damaging to the body. It may be called an abnormal antigenic response. Allergy is another name for hypersensitivity. Those antigens that act in a bad way are called allergens

Types of hypersensitivity:

Hypersensitivities can be grouped into 4 types based on the types of antibodies or cells involved.

1. **Type I (Anaphylactic):** Also called anaphylaxis or immediate hypersensitivity. This is IgE mediated.
2. **Type II (Cytotoxic):** IgG and IgM mediated as these antibodies attach to cells and destroy them by activating complement.
3. **Type III (Immune complex mediated):** Immune complexes are formed by IgM or IgG antibodies and these complexes lodge in the capillary beds and cause damage by complement activation.
4. **Type IV (delayed type cell-mediated):** CD4+ helper cells along with macrophages are involved.

3. Define antigen and epitope

1. **Antigen**: A foreign substance that provokes immunity is called an antigen.
   The definition of antigen is a harmful substance which enters the body which causes the body to make antibodies as a response to fight off disease.
   More specifically called as an immunogen. So immunogen and antigen although slightly different but used interchangeably. o Proteins are best antigen, though glycoproteins and lipoproteins can also act as antigens. Carbohydrates and lipid alone are poor antigens. o Antigens are foreign to the body. o **Degradability**: Inert molecules are not antigenic.
2. **Epitopes**: Specific regions on an antigen. In other words, any antigen has multiple regions each one of those act as a small antigen. These are called epitopes.These epitopes are also called as antigenic determinants. o Most antigens have MW >10,000 D. Molecules that are less than 10,000 D are generally poor antigen.

   A small molecular region of an antigen that binds to a particular antibody or antigen receptor on a T cell; an antigenic determinant. A single antigen can have multiple epitopes

4. Describe two transported system in the blood and why it is necessary 10 marks

Cardiovascular and **lymphatics** are the two transport systems of the body by which cells of the immune system patrol the whole body.

Cardiovascular system consists of the heart and associated blood vessels (arteries and veins) in which blood circulates

lymphatics are channels that start from tissue spaces and collect tissue fluid (called lymph) from tissue spaces and bring them back to the heart. Along these lymphatic channels, there are strategically placed lymph nodes which filter lymph for microbes much like blood is filtered in the spleen. Lymph nodes contain aggregations of B and T cells along with macrophages and dendritic cells which interact with each other inducing adaptive immune responses.

Why Cardiovascular and lymphatics are needed?

Well, in unicellular organisms, nutrients can be absorbed directly from the environment in which a unicellular organism

3. Define antigen and epitope

1. **Antigen**: A foreign substance that provokes immunity is called an antigen.
   The definition of antigen is a harmful substance which enters the body which causes the body to make antibodies as a response to fight off disease.
   More specifically called as an immunogen. So immunogen and antigen although slightly different but used interchangeably. o Proteins are best antigen, though glycoproteins and lipoproteins can also act as antigens. Carbohydrates and lipid alone are poor antigens. o Antigens are foreign to the body. o **Degradability**: Inert molecules are not antigenic.
2. **Epitopes**: Specific regions on an antigen. In other words, any antigen has multiple regions each one of those act as a small antigen. These are called epitopes.These epitopes are also called as antigenic determinants. o Most antigens have MW >10,000 D. Molecules that are less than 10,000 D are generally poor antigen.

   A small molecular region of an antigen that binds to a particular antibody or antigen receptor on a T cell; an antigenic determinant. A single antigen can have multiple epitopes

4. Describe two transported system in the blood and why it is necessary 10 marks

Cardiovascular and **lymphatics** are the two transport systems of the body by which cells of the immune system patrol the whole body.

Cardiovascular system consists of the heart and associated blood vessels (arteries and veins) in which blood circulates

lymphatics are channels that start from tissue spaces and collect tissue fluid (called lymph) from tissue spaces and bring them back to the heart. Along these lymphatic channels, there are strategically placed lymph nodes which filter lymph for microbes much like blood is filtered in the spleen. Lymph nodes contain aggregations of B and T cells along with macrophages and dendritic cells which interact with each other inducing adaptive immune responses.

Why Cardiovascular and lymphatics are needed?

Well, in unicellular organisms, nutrients can be absorbed directly from the environment in which a unicellular organism is living. However, a multicellular organism does not have this easy access to the nutrients. So, nutrients must surround the cells in a multicellular organism. These nutrients are provided by cardiovascular system which pumps blood to all organs. Remember that only fluid part of blood leaves blood vessels. RBCs never leave blood vessels. Sometimes proteins also leak from these blood vessels; however, once they leak from the blood vessels, they never can enter back into blood vessels because of their bigger size. If proteins remain in the tissue spaces, they exert osmotic pressure which attracts water into these space resulting in edema. However, nature has provided another mechanism in the form of lymphatic channels which start from tissue spaces and can drain tissue fluid and proteins back to the heart. This is the role of blood vessels and lympahtics that they ensure

circulation of nutrients throughout the body without causing any edema or abnormality. B cells and T cells also keep circulating between these two channels (cardiovascular and lymphatics). Unfortunately, microbes can also use these two systems to spread themselves in the body. That is the reason, spleen and lymph nodes are placed in these channels by the nature to deal with these microbes that find their ways into blood and lymph. Lymphatic System consists of the following: o Bone marrow o Lymph nodes o Spleen o Tonsils o Thymus o Mucosa associated o Bursa of Fabricius in birds

5. Disk diffusion method

A standardized inoculum of the organisms is spread on agar plate. Filter paper disks soaked with anti bacterials are then placed on the lawn of the organism. Plates are incubated at 37oC and results are recorded. Drugs diffuse from the paper disks radially and if the organisms are sensitive to the drug, a clear area with no growth around the disks develops. The size of the zone directly relates to the degree of sensitivity of the drug. This zone is compared to a standard table for recording the results of sensitivity test. Results are reported as sensitive or resistant or intermediate (between the two extremes).

6. How microbes develop resistance in antimicrobial?

7. Define immunity and difference between innate immunity and adaptive immunity. 10 marks

What is immunity?

Immunity: Ability of the body to protect itself from harmful effects of diseases.

Susceptibility: lack of immunity which leads to easy establishment of disease.

Types of immunity

Innate (Natural):

Present at birth: One is born with these defenses, so it is present without any exposure to the microbes. That is the reason, it is non-specific.

Rapid to respond: Because it is already there, it interacts with the microbes as soon as they enter the body.

Nonspecific: It means that it does not differentiate if it is an E. coli or a Staph or Salmonella. Body defense system against these entire pathogens act in a similar fashion. In other words, the same macrophage can phagocytize E. coli,

is living. However, a multicellular organism does not have this easy access to the nutrients. So, nutrients must surround the cells in a multicellular organism. These nutrients are provided by cardiovascular system which pumps blood to all organs. Remember that only fluid part of blood leaves blood vessels. RBCs never leave blood vessels. Sometimes proteins also leak from these blood vessels; however, once they leak from the blood vessels, they never can enter back into blood vessels because of their bigger size. If proteins remain in the tissue spaces, they exert osmotic pressure which attracts water into these space resulting in edema. However, nature has provided another mechanism in the form of lymphatic channels which start from tissue spaces and can drain tissue fluid and proteins back to the heart. This is the role of blood vessels and lympahtics that they ensure

circulation of nutrients throughout the body without causing any edema or abnormality. B cells and T cells also keep circulating between these two channels (cardiovascular and lymphatics). Unfortunately, microbes can also use these two systems to spread themselves in the body. That is the reason, spleen and lymph nodes are placed in these channels by the nature to deal with these microbes that find their ways into blood and lymph. Lymphatic System consists of the following: o Bone marrow o Lymph nodes o Spleen o Tonsils o Thymus o Mucosa associated o Bursa of Fabricius in birds

5. Disk diffusion method

A standardized inoculum of the organisms is spread on agar plate. Filter paper disks soaked with anti bacterials are then placed on the lawn of the organism. Plates are incubated at 37oC and results are recorded. Drugs diffuse from the paper disks radially and if the organisms are sensitive to the drug, a clear area with no growth around the disks develops. The size of the zone directly relates to the degree of sensitivity of the drug. This zone is compared to a standard table for recording the results of sensitivity test. Results are reported as sensitive or resistant or interm he tw extremes).

6. How microbes develop resistance in antimicrobial?

7. Define immunity and difference between innate immunity and adaptive immunity. 10 marks

What is immunity?

Immunity: Ability of the body to protect itself from harmful effects of diseases.

Susceptibility: lack of immunity which leads to easy establishment of disease.

Types of immunity

Innate (Natural):

Present at birth: One is born with these defenses, so it is present without any exposure to the microbes. That is the reason, it is non-specific.

Rapid to respond: Because it is already there, it interacts with the microbes as soon as they enter the body.

Nonspecific: It means that it does not differentiate if it is an E. coli or a Staph or Salmonella. Body defense system against these entire pathogens act in a similar fashion. In other words, the same macrophage can phagocytize E. coli, Staph or Salmonella at the same time (if these pathogens are present there).

Does not increase in response with repeated exposures to the same agent. Therefore, it has no memory.

First line of defense: Physical barriers (skin, mucus membrane) provide the first line of defense as they prevent the entry of the organisms by providing a physical barrier which if broken will let the microbes go inside.

Second line of defense: Phagocytic cells and their secretions provide a second line of barriers for microbes. So, if microbes succeed in breaching or breaking the first barrier, macrophages, neutrophils and such other cells phagocytize these microbes and kill them before these can establish infection in the body.

Adaptive Immunity: This will be studied in detail in a separate lecture soon.

Adaptive (Acquired): It is the third line of defense. Acquired by experience, specific to an organism.

Specific: It means that antibodies made against E. coli will destroy E. coli only, and will not do any harm to a Staph for example.

Have memory: It means if antibodies have once been made against E. coli, this exposure will be remembered by the body, and if a second encounter is made with the same type of E. coli again, the body would mount a heightened immune response against E. coli because it has memory cells developed in the body which upon second stimulation proliferate quickly and mount a greater defensive response.

O Because of memory, body's immune response increase with every repeated exposure. It has a limit also!

Slower to respond: For the first time exposure, the body's response is slower and takes a long time compared with the innate immunity which is immediate.

Adaptive Immune Response can further be divided into:

Humoral immune response (antibody response by B cells or lymphocytes)

Cellular immune response (T cells are made which kills microbial harboring cells).

8. Agglutination and Neutralization 2 marks

Agglutination: Antibodies can agglutinate antigens which are then cleared as a group by macrophages

Neutralization: Toxins bind with antibodies and they become incapable of binding to their target receptors thus the toxins become harmless.

Types of immunity

Innate (Natural):

Present at birth: One is born with these defenses, so it is present without any exposure to the microbes. That is the reason, it is non-specific.

Rapid to respond: Because it is already there, it interacts with the microbes as soon as they enter the body.

Nonspecific: It means that it does not differentiate if it is an E. coli or a Staph or Salmonella. Body defense system against these entire pathogens act in a similar fashion. In other words, the same macrophage can phagocytize E. coli, Staph or Salmonella at the same time (if these pathogens are present there).

Does not increase in response with repeated exposures to the same agent. Therefore, it has no memory.

First line of defense: Physical barriers (skin, mucus membrane) provide the first line of defense as they prevent the entry of the organisms by providing a physical barrier which if broken will let the microbes go inside.

Second line of defense: Phagocytic cells and their secretions provide a second line of barriers for microbes. So, if microbes succeed in breaching or breaking the first barrier, macrophages, neutrophils and such other cells phagocytize these microbes and kill them before these can establish infection in the body.

Adaptive Immunity: This will be studied in detail in a separate lecture soon.

Adaptive (Acquired): It is the third line of defense. Acquired by experience, specific to an organ...

Specific: It means that antibodies made against E. coli will destroy E. coli only, and will not do any harm to a Staph for example.

Have memory: It means if antibodies have once been made against E. coli, this exposure will be remembered by the body, and if a second encounter is made with the same type of E. coli again, the body would mount a heightened immune response against E. coli because it has memory cells developed in the body which upon second stimulation proliferate quickly and mount a greater defensive response.

O Because of memory, body's immune response increase with every repeated exposure. It has a limit also!

Slower to respond: For the first time exposure, the body's response is slower and takes a long time compared with the innate immunity which is immediate.

Adaptive Immune Response can further be divided into:

Humoral immune response (antibody response by B cells or lymphocytes)

Cellular immune response (T cells are made which kills microbial harboring cells).

8. Agglutination and Neutralization 2 marks

Agglutination: Antibodies can agglutinate antigens which are then cleared as a group by macrophages

Neutralization: Toxins bind with antibodies and they become incapable of binding to their target receptors thus the toxins become harmless.

9. What is plaque 2 marks

Plaques are equivalent to bacterial colonies. Basically a plaque is an area where the virus infection has lysed the cells. These plaques are formed on cultured cells in a culture dish. A monolayer of the cells is infected with the virus to be enumerated and then the culture is overlaid with the semisolid agar so that infected and lysed cells do not move away from the site in the dish. Plaques can be counted easily much like bacterial colonies

10. Difference between capsule and slime

Capsule: If glycocalyx is organized and firmly attached to the cell, it is called a capsule.

– Plays important role in virulence (degree of pathogenicity)

– Can also be a part of the vaccine against the bacteria to which it belongs.

– Can be demonstrated by negative staining technique.

Slime: If glycocalyx is loosely attached to the cell, it is called slime.

11.PCR method 3 marks

PCR goes through 3 step cycles for about 30 to 35 times. Each cycle consists of a denaturing stage (about 95oC), primer annealing (50 to 56oC) and extending stage (72oC). A primer is a fragment of DNA which is specific to the signature sequence. Two primers span the signature sequence actually. At the end of 30 or so such cycles, enough DNA

accumulates that one can visualize on an agarose gel easily.

12. Type of T cell

T helper cells

9. What is plaque 2 marks

Plaques are equivalent to bacterial colonies. Basically a plaque is an area where the virus infection has lysed the cells. These plaques are formed on cultured cells in a culture dish. A monolayer of the cells is infected with the virus to be enumerated and then the culture is overlaid with the semisolid agar so that infected and lysed cells do not move away from the site in the dish. Plaques can be counted easily much like bacterial colonies

10. Difference between capsule and slime

Capsule: If glycocalyx is organized and firmly attached to the cell, it is called a capsule.

– Plays important role in virulence (degree of pathogenicity)

– Can also be a part of the vaccine against the bacteria to which it belongs.

– Can be demonstrated by negative staining technique.

Slime: If glycocalyx is loosely attached to the cell, it is called slime.

11.PCR method 3 marks

PCR goes through 3 step cycles for about 30 to 35 times. Each cycle consists of a denaturing stage (about 95oC), primer annealing (50 to 56oC) and extending stage (72oC). A primer is a fragment of DNA which is specific to the signature sequence. Two primers span the signature sequence actually. At the end of 30 or so such cyc

accumulates that one can visualize on an agarose gel easily.

12. Type of T cell

T helper cells

TH1 (CD4+)

TH2 (CD4+)

T cytotoxic cells (CD8+)

13. Inhibits growth cell wall drug name

Penicillins and cephalosporins are the major antibiotics that **inhibit** bacterial **cell wall** synthesis. They are called beta-lactams because of the unusual 4-member ring that is common to all their members.

14. Microbiota factor and distribution composition

Nutrients: Availability of various nutrients determines the type of organisms present in a system. Similarly, physical and chemical factors also influence the type of microbiota. Other factors include pH, O2, and CO2. Host defenses, level of stress and age of an individual also matters for allowing microbiota of an individual.

15. Capsule function 2 marks

The capsule impairs phagocytosis and help the organisms evade the immune system; however, antibodies can help cells of the immune system in phagocytosis of such encapsulated organisms. Only a few bacteria produce capsule which include Strep pneumonia, Klbesiella pneumonia.

16. Viral replication stages

Viral multiplication goes through various stages in the cell. These include viral attachment, entry into the cell by pinocytosis or fusion with the plasma membrane, uncoating of the virus, biosynthesis of capsids and nucleic acids, maturation (assembly of the virus) and final release of the virus from the cell.

17. Economic benefits of fungi 3 marks

Saccharomyces are used for alcoholic beverages under anaerobic conditions for alcohol production (fermentatively). However, if Saccharomyces are incubated aerobically (for bread making), they metabolize glucose to produce CO2 (some ethanol is also produced but gets evaporated during baking) which expands the dough causing it to rise. This yeast is also extensively used for molecular biology work and vaccine production. Vitamin C is obtained from Aspergillus niger; cellulase from Trichoderma, antibiotics from many fungi and anticancer Taxol from Taxomyces are some other examples of useful fungal products. Entomophaga (kills gypsy moth), hence is used as a biological pest control. Mushrooms can be cooked and eaten as a source of proteins for humans.

18.Types of symbiosis 3 marks

Interaction of microbiota with the host leads to a relationship called symbiosis. There are a few outcomes of this symbiosis which we must know. The two participants of this relationship are microbiota and the host. At least one of these participants is benefited and sometimes both can be benefited from this mutual relationship. Different names or terms have been given to these mutual relationships as under:

Mutualism: both are benefited

Parasitism: one is harmed.

Commensalism: One is benefited, other is unharmed.

19. Inflammation and write its consequence 5 marks

Damage to body tissues triggers a local defensive response which is called inflammation. It causes redness, pain, swelling, heat, and loss of function of that organ

During inflammation, various mediators of inflammation are released, some of which include:

20. Cause and diagnose of anaphylaxis

The patient also experiences difficulty in breathing. It is an emergency as it may kill the person. Anaphylaxis may be local or systemic.

o Shock from drug reactions, venoms, and common allergens causing asthma are some examples of type I hypersensitivity.

Hypersensitivity can be diagnosed by demonstration of IgE by intradermal injections of allergens one by one. If specific IgE are present in the serum, a reaction will be visible on the skin locally in the form of a wheel of inflammation. Moreover, IgE levels can be measured by ELISA. Measurement of IgE is more practical as nature of allergens is not known most of the time.

o Rx: Epinephrine & antihistaminic drugs provide a means of treatment.

21. Two name of bacteria which is capsulated

Bacillus anthracic, streptococcus pneumoniae and Klebsiella are examples of capsulated organisms.

22. What is Epidemiology and its types

The study of where and when diseases occur and how they are transmitted in a population.

There are three basic types of investigations relating to epidemiology

1. Descriptive Epidemiology
2. Analytical Epidemiology
3. Experimental Epidemiology:

23. Target site of antibiotics

24. Capsomere

Capsid is made up of protein subunits called capsomeres that are arranged in 3 possible ways to give a helical, or icosahedral or complex symmetry to the viruses.

 Capsomeres: Subunit called protomers aggregate to form capsomeres.

25. Character of antigen

Antigen: A foreign substance that provokes immunity is called an antigen.

More specifically called as an immunogen. So immunogen and antigen although slightly different but used interchangeably.

Proteins are best antigen, though glycoproteins and lipoproteins can also act as antigens. Carbohydrates and lipid alone are poor antigens.

Antigens are foreign to the body.

Degradability: Inert molecules are not antigenic.

26. Algae structure

Body of multicellular alga such as seaweed is called a thallus which consists of branched holdfasts (anchor alga to rock) stemlike hollow stipes and leaflike blades. There is no vascular tissue in these algae. Also, the stipe is not lignified or woody, so it does not provide support to the weed. Surrounding water provides the support for the thallus. Some algae have a gas filled body inside them which keeps them floating in the water. This gas filled structure is called pneumatocyst or float

27. Acid fast stanning 5 marks

Some bacteria such as Mycobacteria have a waxy material in their cell wall.

Specific name for this waxy substance is mycolic acid

Mycobacterium tuberculosis: Causative agent for tuberculosis

M. leprae: Causative agent for leprosy

- **Principle of Acid-fast staining**: Mycobacteria are lipophilic, not easy to stain

But once stained, they are resistant to acidic alcohol decolorization process.

- **Overall Procedure for Acid-fast staining**:

Carbol fuchsin: It is a primary stain Heat is used to enhance penetration of carbol fuchsin.

Acid alcohol is a decolorizer.

Methylene blue is a counterstain.

Please note, Mycobacteria are gram positive, however, they look red in acidfast staining.

28. Capsule staining 10 marks

Capsule: If glycocalyx is organized and firmly attached to the cell, it is called a capsule.

– Plays important role in virulence (degree of pathogenicity)

– Can also be a part of the vaccine against the bacteria to which it belongs.

Capsid is made up of protein subunits called capsomeres that are arranged in 3 possible ways to give a helical, or icosahedral or complex symmetry to the viruses.

 Capsomeres: Subunit called protomers aggregate to form capsomeres.

25. Character of antigen

Antigen: A foreign substance that provokes immunity is called an antigen.

More specifically called as an immunogen. So immunogen and antigen although slightly different but used interchangeably.

Proteins are best antigen, though glycoproteins and lipoproteins can also act as antigens. Carbohydrates and lipid alone are poor antigens.

Antigens are foreign to the body.

Degradability: Inert molecules are not antigenic.

26. Algae structure

Body of multicellular alga such as seaweed is called a thallus which consists of branched holdfasts (anchor alga to rock) stemlike hollow stipes and leaflike blades. There is no vascular tissue in these algae. Also, the stipe is not lignified or woody, so it does not provide support to the weed. Surrounding water provides the support for the thallus. Some algae have a gas filled body inside them which keeps them floating in the water. This gas filled structure is called pneumatocyst or float

27. Acid fast stanning 5 marks

Some bacteria such as Mycobacteria have a waxy material in their cell wall.

Specific name for this waxy substance is mycolic acid

Mycobacterium tuberculosis: Causative agent for tuberculosis

M. leprae: Causative agent for leprosy

- **Principle of Acid-fast staining**: Mycobacteria are lipophilic, not easy to stain

But once stained, they are resistant to acidic alcohol decolorization process.

- **Overall Procedure for Acid-fast staining**:

Carbol fuchsin: It is a primary stain Heat is used to enhance penetration of carbol fuchsin.

Acid alcohol is a decolorizer.

Methylene blue is a counterstain.

Please note, Mycobacteria are gram positive, however, they look red in acidfast staining.

28. Capsule staining 10 marks

Capsule: If glycocalyx is organized and firmly attached to the cell, it is called a capsule.

– Plays important role in virulence (degree of pathogenicity)

– Can also be a part of the vaccine against the bacteria to which it belongs.

– Can be demonstrated by negative staining technique.

– Capsulated organism make glistening colonies as seen in the accompanying diagram below:

 Bacillus anthracic, streptococcus pneumoniae and Klebsiella are examples of capsulated organisms.

 Thickness of the capsule depends upon the culture conditions.

Capsules are mostly are water soluble. Also remember that capsular organisms usually make the broth viscous and stringy

Capsule staining procedure

1. Prepare a thick smear in a loopful of congo red (1%) stain
2. Fixation in acid alcohol for 15 sec
3. Wash with dH2O
4. Cover it with acid fuchsin for 1 min.
5. Wash with water
6. A bacterium stains red, capsule remains colorless and is seen in the dark blue background as seen in the accompanying diagram.

29. Define vaccine and its types 10 marks

What is vaccine:

A product that creates immunity against an infection is called a vaccine. It could be the whole organism or a part of the organism.

The observation that people or animals once recover from an infection do not get sick easily to that organism again led to the discovery of vaccines and vaccination.

Types of vaccine:

Live Attenuated Vaccines: These are weakened organisms so that they do not harm the body, but divide or replicate in the same way as pathogenic version of the organism does in the body. So, when a live virus or bacteria multiplies in

proteins that can induce immunity. In other words, this is another version of DNA vaccines. In subunit vaccines, antigenic epitopes are included in the genome which is inserted into an expression vector that makes those subunit proteins. These subunit proteins are collected and used as vaccines to stimulate the immune system. Again, these act like a killed vaccine. There is another mechanism being explored in these days for recombinant vaccines. A subunit vaccine for Foot and Mouth disease virus is available.

Such genes are being incorporated in plants now that are acting like an expression vector. Animals can eat those plants and subunit vaccines can induce immune response through M cells of the intestine. Success is limited though so far.

30. Describe features of fungi 5 marks

These are simple eukaryotic cells. Some are unicellular, others are multicellular (thallus); however, they lack tissues such as roots, stem and leaves typically seen in plants. Algae absorb nutrients from water through their surfaces and are mostly photoautotrophs; however, a few are chemoheterotrophs. They are responsible for 80% atmospheric O2 on the face of the earth. Microscopic exam is needed to identify unicellular and filamentous algae. However, multicellular algae that are commonly known as seaweeds are macroscopic in nature and can be identified morphologically without the help of a microscope. Four groups of such algae include blue-green algae, green algae, brown algae and red algae. These algae are located in the sea at various locations and absorb light of various wavelengths, hence red algae are located r from the surface and can use blue light from the sun as blue light is of shorter wavelength and can penetrate deep in the sea. Please see the accompanying figure (above) for a better understanding of how wavelength relates with various algae. Also remember, blue-green algae need magnification in order to be correctly identified, although they are not microscopic.

31. Treatment of hypersensitivity ii 3 marks

This type of hypersensitivity can be diagnosed by history and symptoms. Glomerulonephritis and cellulitis is very common. Deposition of immune complexes in the kidneys can be diagnosed by presence of IgG in kidney cells by using anti-IgG antibodies.

32. What are antibiotics 3 marks

antibiotics are used to kill organisms in disease states, some antibiotics are not very effective for this purpose; however, these antibiotics could be used in food to prevent food spoilage. Nisin and natamycin prevent spoilage of cheese

33. Define virion and viroid 2 marks

A **virion** is an infectious particle composed of nucleic acid that is surrounded by a protein coat (capsid). Viruses are classified based on their nucleic acid and protein coat. Viruses have an extracellular and an intracellular state. Outside of a cell, in the extracellular state, a virus is called a virion.

VIROIDS: They are composed only of circular single stranded (ss) RNA.

34. Write entry point of microorganism 2 marks

Mucus membranes: These include linings of digestive tract, respiratory tract, urogenital system, and conjunctiva

Skin: Surface area wise, skin is the largest organ of the body. Unbroken skin is impermeable to organisms

Parenteral Route: Injury to the skin or mucous membrane leads to easy access for the organisms in the body. Any cut received on the skin for example becomes a portal of entry

35. Role of thin clonal selection 2 marks

According to this theory, B cells are generated in the bone marrow randomly without seeing the antigen. When such a B cell encounters its specific antigen in the body, it starts proliferating into a clone which ultimately becomes plasma cells and start secreting antibodies. A beautiful example to understand clonal selection theory is like going to a shoe store and select a shoe size that fits you very well. Note, that the shoe that you selected was made randomly without any regard to your foot size.

35.Antigen presenting cell 3 marks

These cells are able to present antigens to B cells for making antibodies by B cells.

There are three cells that fall into this category: Dendritic cells, macrophages, and B cells

36.How antibodies provide defense against microbes 5 marks

37.Reverse transcription 2 marks

A reverse transcriptase is an enzyme used to generate complementary DNA from an RNA template, a process termed reverse transcription

38.Bactericidal and bacteriostatic drug 2 marks

Bactericidal: Those antibiotics that kill and lyse the cells are called bactericidal.

Bacteriostatic: Such antibiotics stop the growth of the organisms, they do not kill or lyse the cells. Cells of the immune system then clear them from the body

39. Define plasmogamy karyogamy and meiosis 3 marks

1.Plasmogamy: A haploid nucleus of a donor cell (+) penetrates the cytoplasm of a recipient cell (-). Both cells are haploid (one set of chromosomes)

2. **Karyogamy** The (+) and (-) nuclei fuse to form a diploid zygote nucleus.

hypersensitivity. Reaction to hair dyes is another example. Poison Ivy is another example in which contact with this plant initiate type IV hypersensitivity response. Chemicals from poison ivy interact with skin proteins and change them into foreign-looking protein molecules. Body's immune system mounts an immune response against these proteins resulting in a damaging type of intense inflammatory response

41. Cytokines 2 marks

Cytokines are released as a result of defense cell stimulation by PAMPs and TLRs interactions. Cytokines recruits more immune cells and also stimulate adaptive immune response

42. Note on infectious disease transmission with vector vehicle and contacts?

Direct Transmission: person to person physical contact is a direct contact. Influenza spread is a typical example.

Vehicle Transmission: Transmission of disease agents by a medium, such as water, food, or air. Other media include blood and other body fluids, drugs, and intravenous fluids.

A vector is a vehicle (animate or inanimate) that carries a pathogen from one host to another. o Arthropods are vectors for many diseases. There are two ways these arthropods can transmit organisms to susceptible hosts:

Mechanical Transmission: In this type of transmission, organisms stick to the body parts of insects and get transmitted passively from one location to another.

Biological Transmission: Microbes multiply in the insect and is transmitted actively by insect bites from one individual to another.

43. Incidence and Prevalence 2 marks

Incidence: # of new cases in a given time period in a population

Prevalence: New + existing cases

44. PRIMARY AND SECONDARY IMMUNE RESPONSE 5 marks

When a person is exposed to an antigen (a microbe for example), the antigen is presented to B cells which start making antibodies. Initially, there are no detectable antibodies in the serum. However, after 3-5 days, IgM antibodies are secreted and then 7-10 days after, the same B cell starts making IgG. The level of IgG rises to a level and then it starts declining. This is a typical response of an antigen exposure and is called a primary immune response. It is interesting to note that a second exposure with the same antigen results in a heightened response in terms of very high levels of antibodies that are achieved in a very short time (within 2-7 days). This is called a secondary immune response. It is also called a memory or anamnestic response because memory cells are involved that ensure a quick response to the second exposure by the same antigen (to which the immune system has already been exposed). The accompanying figure illustrates this concept (next page). Blue line in the figure depicts a primary and a secondary response.

Adaptive Immune Response can be acquired naturally or artificially. Each of these categories can further be divided either into active or passive type.

Naturally Acquired Active Immunity: This type of adaptive immunity develops in a person who gets infected with a microbe but then recovers from that illness making antibodies.

Naturally Acquired Passive Immunity: A typical example of this type of immunity is transfer of antibodies from colostrum or milk of a mother to the offspring.

Artificially Acquired Active Immunity: When a person receives a vaccine against a microbe and the body makes antibodies in response to that vaccine, it is called artificial acquired active immunity because body makes these antibodies actively by itself.

Artificially Acquired Passive Immunity: When premade antibodies are given to a person in order to protect a person from an illness or a snake bite, this constitutes an artificially acquired passive immunity type. The following figure illustrates these four types.

45. Vegetative Hyphae and Aerial hyphae 2 marks

Vegetative Hyphae: Hyphae that are embedded in the medium and are used to obtain food are called vegetative hyphae.

Aerial or Reproductive Hyphae: The portion of hyphae that is concerned with reproduction. This portion is projected into the air. They bear reproductive spores.

46. Signature sequence and primer

signature sequences

There are many tests that involve nucleic acid sequence detection for the presence of microbes in a clinical sample. Please remember, as antibodies are specific to the organisms, DNA or RNA sequences must be uniquely specific for the microbe that we are going to detect. Such sequences that are specific to an organism are called **signature sequences**

Primer

A primer is a fragment of DNA which is specific to the signature sequence

Feature of yeast cells

illustrates these four types.

45. Vegetative Hyphae and Aerial hyphae 2 marks

Vegetative Hyphae: Hyphae that are embedded in the medium and are used to obtain food are called vegetative hyphae.

Aerial or Reproductive Hyphae: The portion of hyphae that is concerned with reproduction. This portion is projected into the air. They bear reproductive spores.

46. Signature sequence and primer

signature sequences

There are many tests that involve nucleic acid sequence detection for the presence of microbes in a clinical sample. Please remember, as antibodies are specific to the organisms, DNA or RNA sequences must be uniquely specific for the microbe that we are going to detect. Such sequences that are specific to an organism are called **signature sequences**

Primer

A primer is a fragment of DNA which is specific to the signature sequence

Feature of yeast cells

47. Active and passive transport

active Movement

Movement against the concentration gradient

From low concentration to high concentration of a substance

Requires energy (ATP)

Substances that are actively transported into the cells undergo either no change or they are chemically modified to keep them there in the cell. Remember, active transport brings substances inside the cell against the gradient; hence molecules brought in can go out. That is why these molecules may be modified to keep them inside the cells. In other words, this modification avoids passive diffusion of those molecules from the cells outwards.

Passive Movement

Movement of substances with the concentration gradient. This means that substances will move from higher concentration of a substance to lower concentration. Gradient provides the force for movements of molecules and this happens with no energy expenditures.

What are the 2 benefits algae get from oceans 2 marks

Algae are important for aquatic life. They are at the bottom of the food chain because they fix CO_2 into organic molecules (Carbs) which could be used by other organisms.

They produce 80% of molecular oxygen on earth.

Algal blooms which result from fluctuation of temperature sometimes may cause problems for nature but that happens occasionally.

Algal blooms tell us about the status of pollution in the environment. Planktonic organisms (dead and decaying) are responsible for oil production. Petroleum and natural gas are the residues (hydrocarbons) of diatoms and other planktonic organisms.

Define lichens 2 marks

Lichens are a combination of green alga (or a cyanobacterium) and a fungus. Green alga or cyanobacterium is also called a phycobiont and the fungus is called a mycobiont in this relationship. Both partners have a symbiotic relationship with each other

Hypersensitivity type 1 3 marks

It involves IgE and it occurs within 30 minutes of exposure to the allergen. Antigens called allergens preferentially induce IgE production instead of IgG against the antigens. These IgEs get bound to mast cells in the tissues and a second exposure to the same allergen activates mast cells to secrete histamine and other mediators of inflammation. Shock from drug reactions, venoms, and common allergens causing asthma are some examples of type I hypersensitivity.

Hypersensitivity can be diagnosed by demonstration of IgE by intradermal injections of all spec IgE are present in the serum, a reaction will be visible on the skin locally in the form of a w.

10 / 11

Transmission of disease 5 marks

Direct Transmission: person to person physical contact is a direct contact. Influenza spread is a typical example.

Indirect Transmission: A nonliving object is involved. Objects in the use of a patient such as handkerchief, utensils, pillow and bedding are called fomites. These can transmit organisms to susceptible individuals.

Droplet Transmission: Mucus droplets are created when you sneeze for example. These droplets carry

organisms that may infect another individual who comes in contact with the droplets. One sneeze may produce 20,000 droplets. Influenza can be spread by this route.